

Step By Step Guide to Install Oracle 19c RAC installation Using Virtual-Box

Description:-

In this article we are going to see step by step oracle Rac 19c installation on OEL 7.8

Download Software

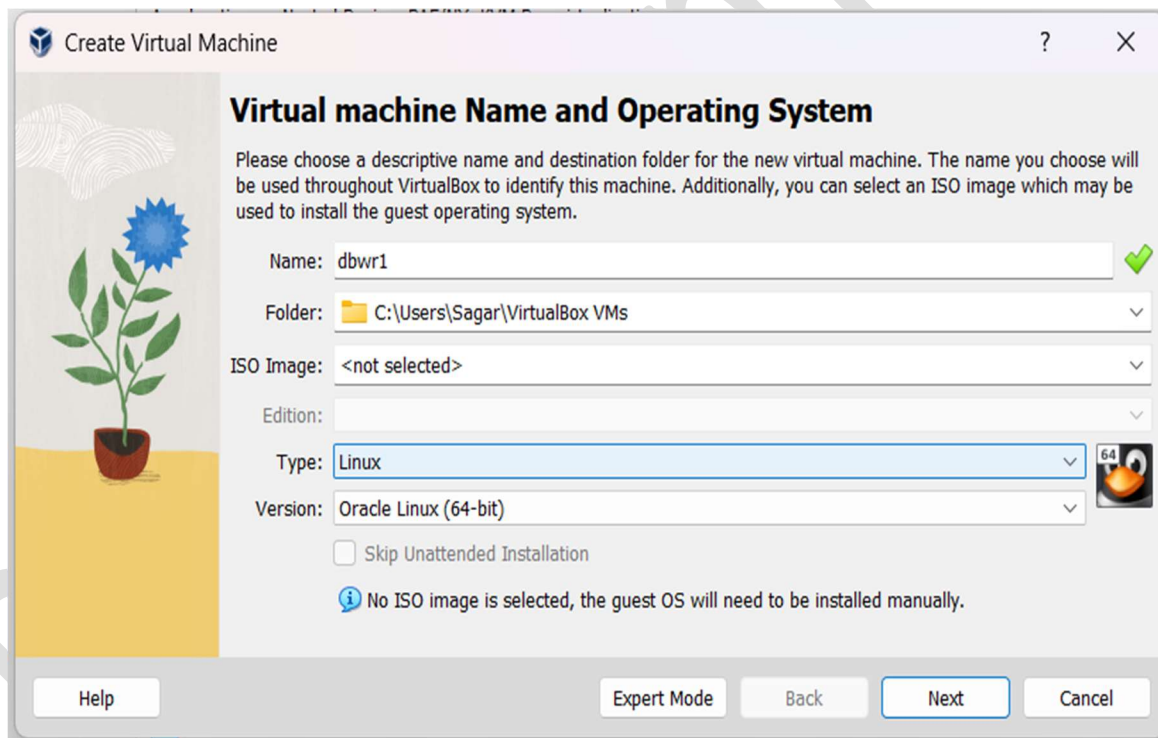
Download the following Software.

- [Oracle Linux 7](#)
- [VirtualBox for Window hosts](#)
- [Oracle Database 19c Release 3 – Linux x86-64](#)

Oracle VirtualBox Setup

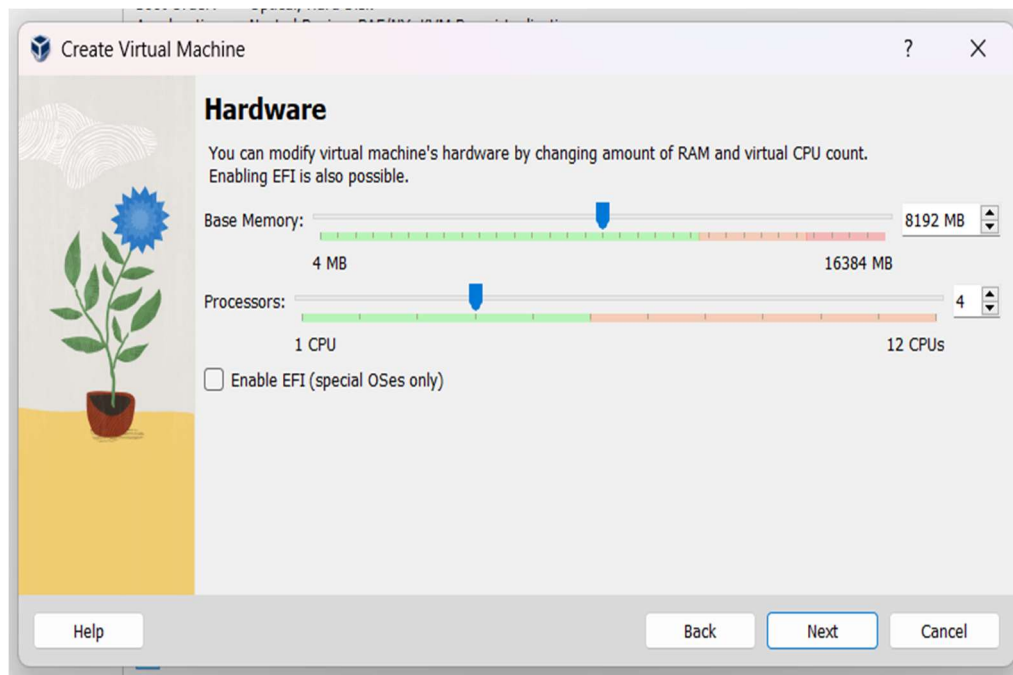
Click on new and Click Next

Provide a Name as dbwr1 type as Linux and version as Oracle 64bit. Click Next

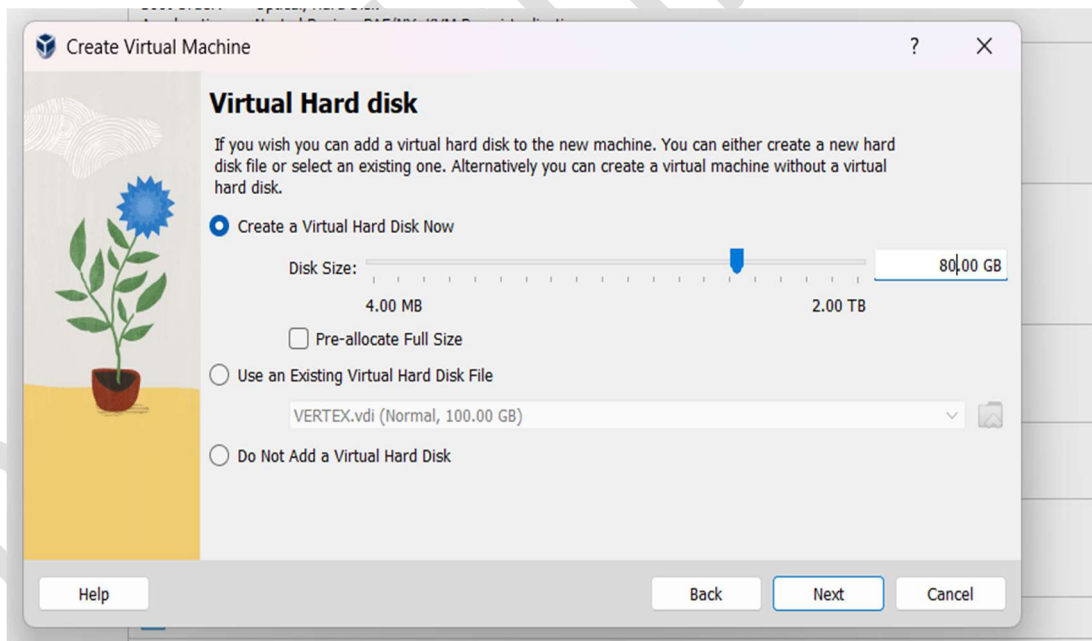


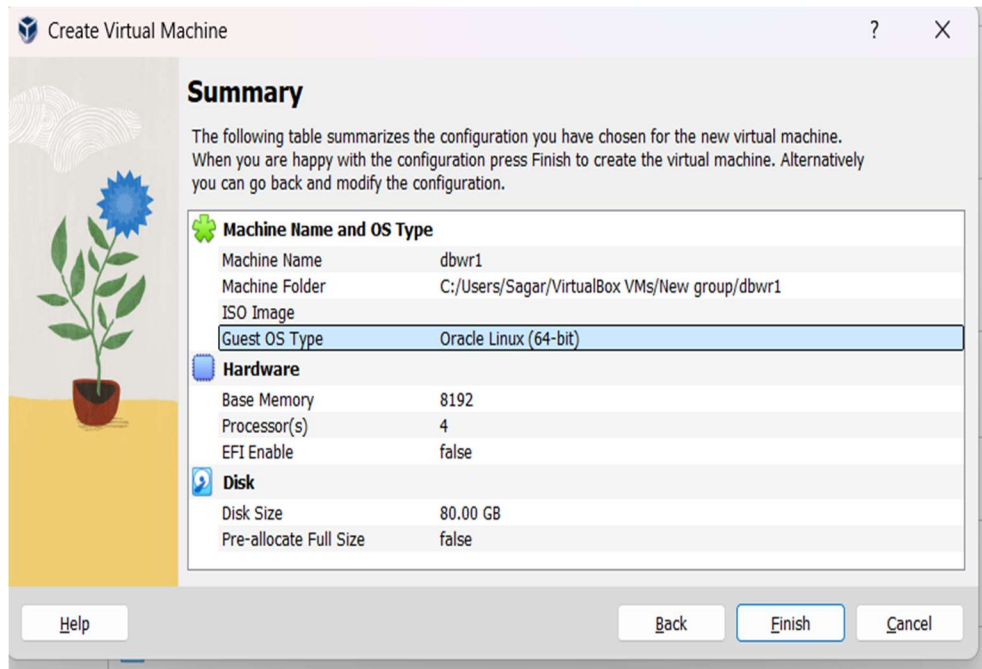
Enter Memory value to 8GB and then click the Next button

Note:-Oracle recommends at least 8 GB RAM for Oracle Grid Infrastructure installations. If the production system does not have enough RAM, you will see instance evictions and node reboot that may not be very explainable very often.

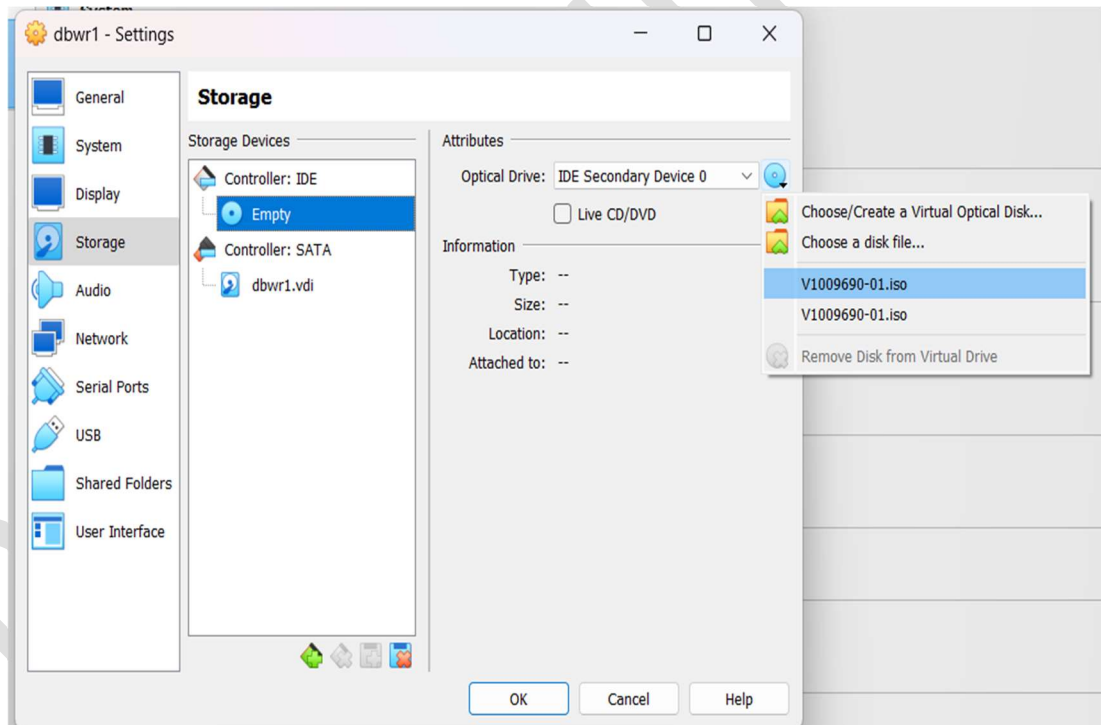


Accept the default option to create a new virtual hard disk by clicking the Create button
Provide the location where you want to save the VM files. Also enter the size of your vm from default to say 80GB and Click Create

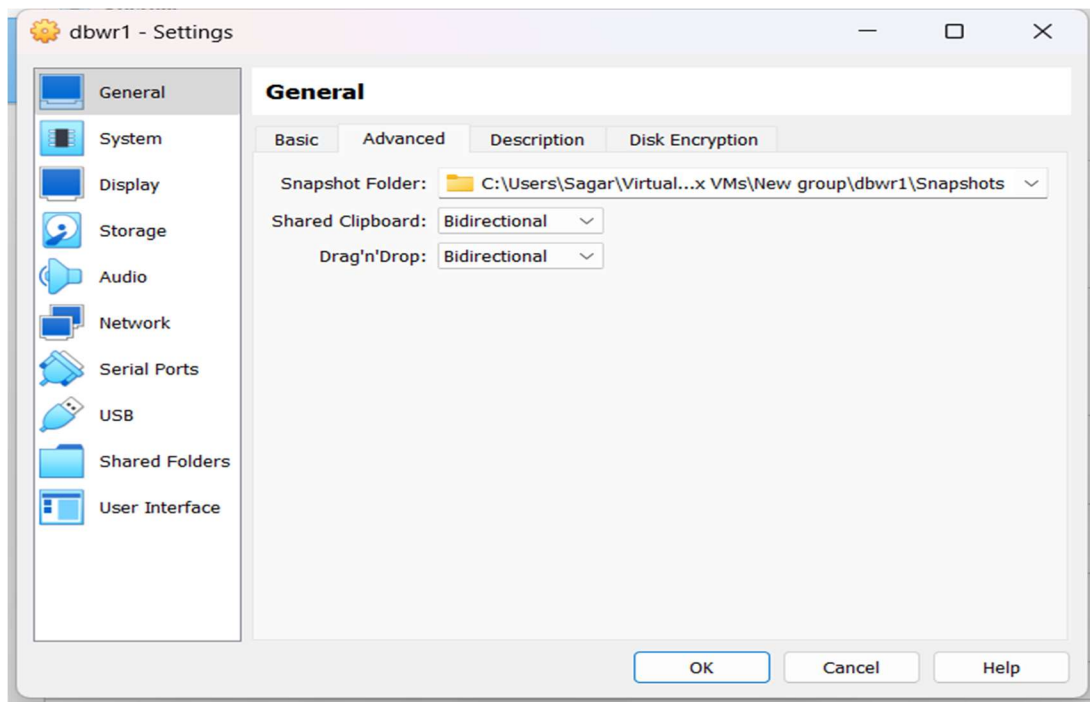




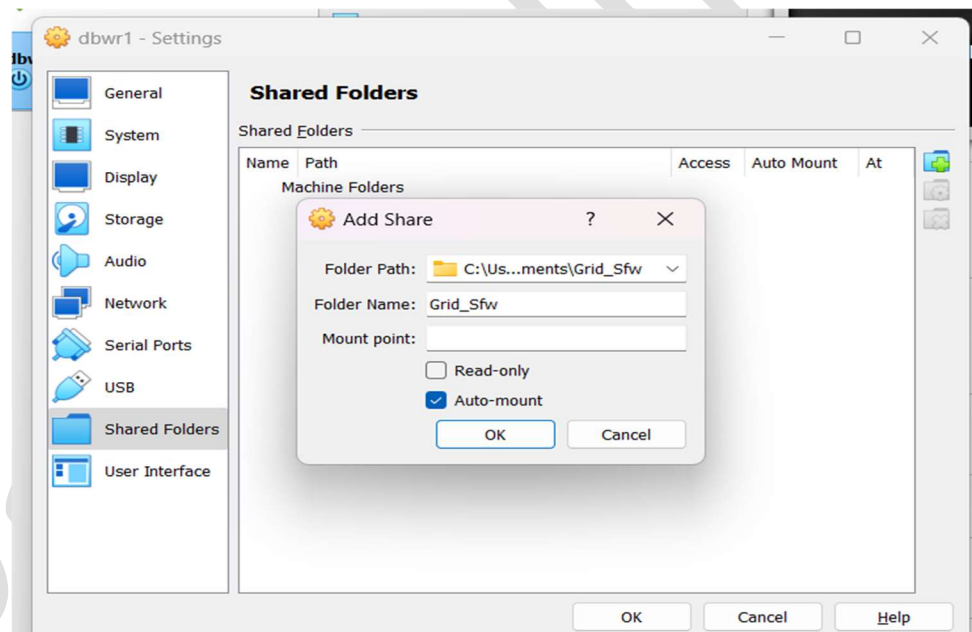
Click on General->Storage Select OEL7 ISO and click ok



Click on General->Advanced and change shared clipboard to Bidirectional and Change Drag 'n' Drop to Bidirectional and click Ok



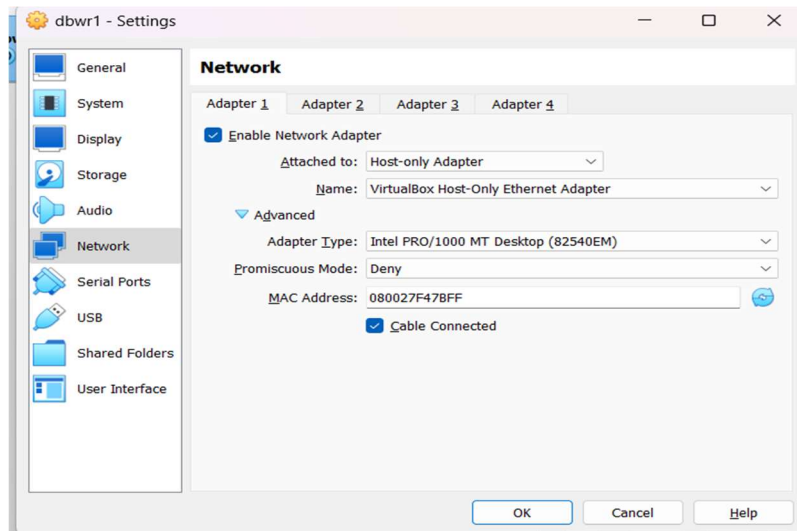
Click on General->Shared Folder this will be helpful to get any windows folder mounted on OEL7.8



Add three Adaptors.

- 1)public
- 2)private
- 3)Internet for installing RPM's

Click on General->Network-Adapter 1: Make sure Adapter 1 is enabled, attached to Internal Network or Host-only Adapter. This interface will be used for public network, for example, for connection to the RAC database from other applications.



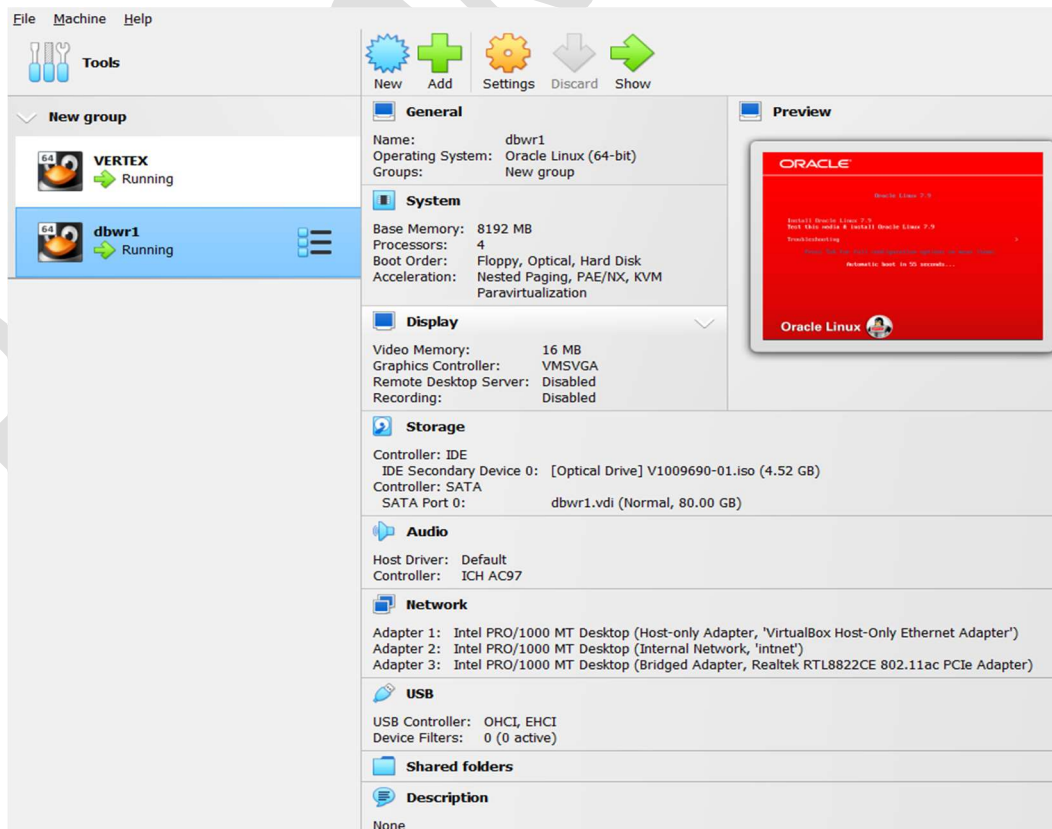
Click on General->Network-Adapter 2

Make sure Adapter 2 is enabled and attach to Internal Network. Name this network private network

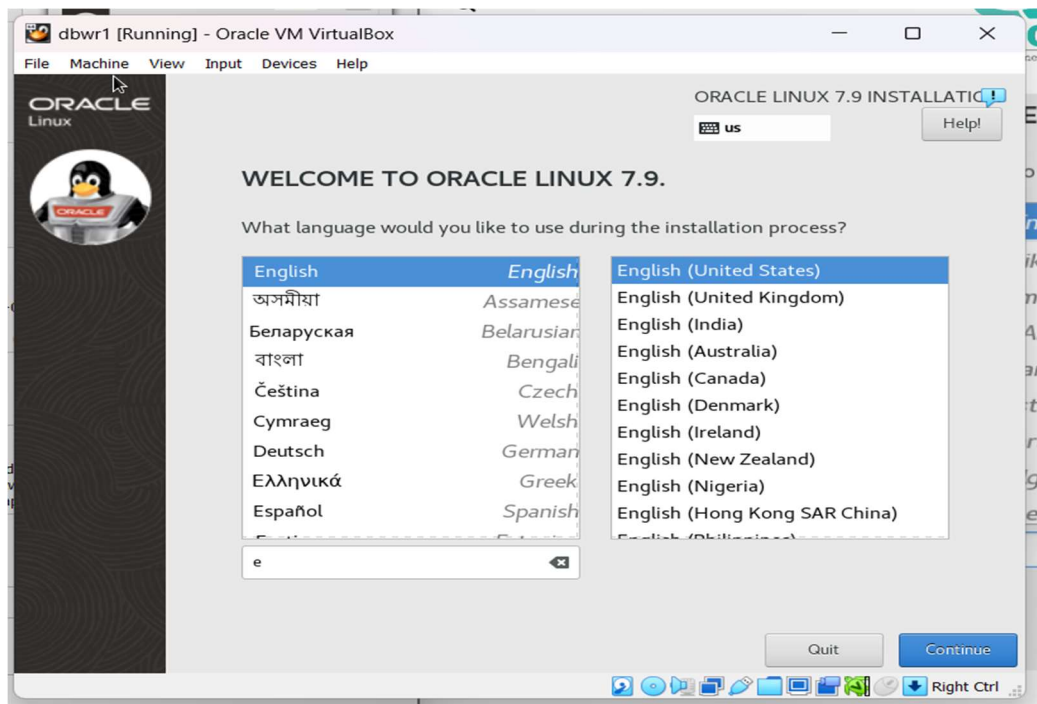
Click on General->Network-Adapter 3

Enable Adapter 3 and attached to Bridged Adapter. This adapter will be used for internet. Then press ok button

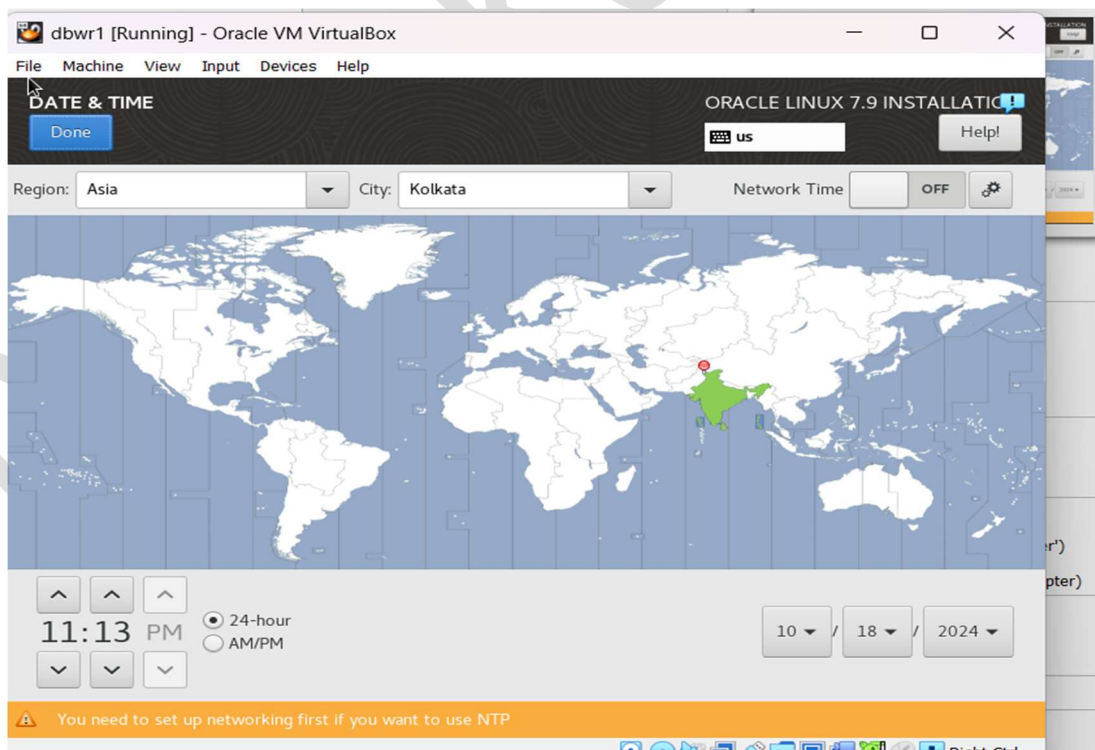
Double Check before starting Linux installation and Click on the Start button



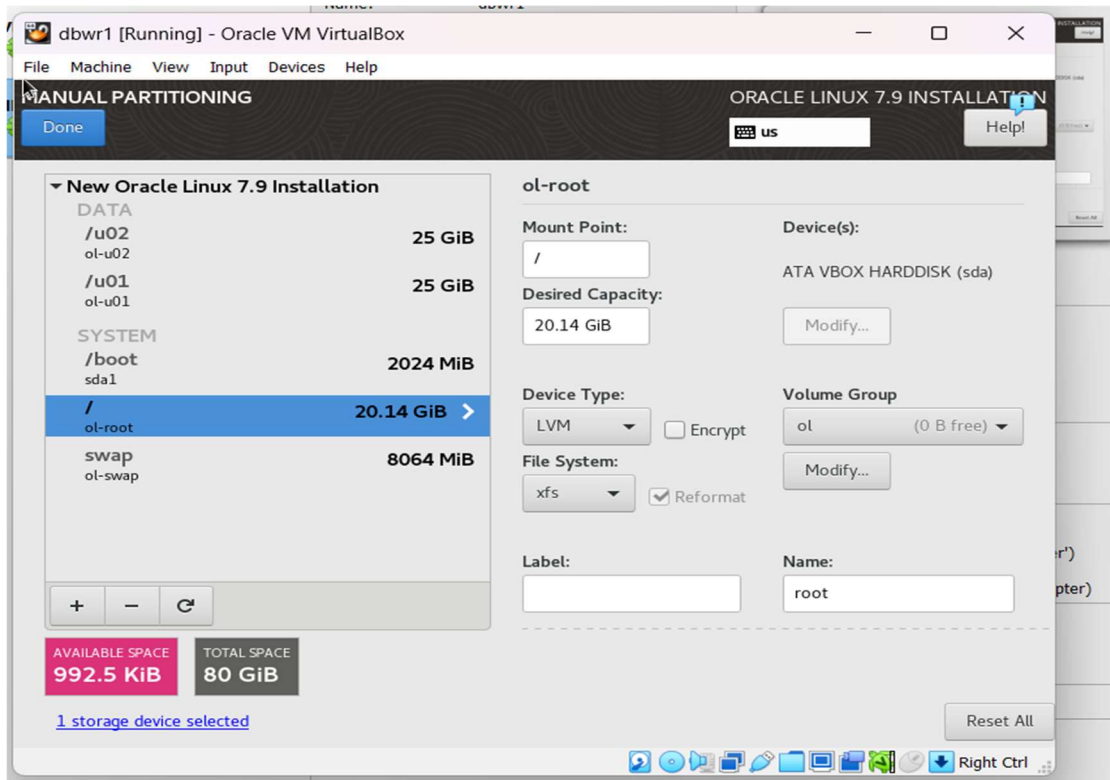
Click Continue



Select Region and city and press Done button



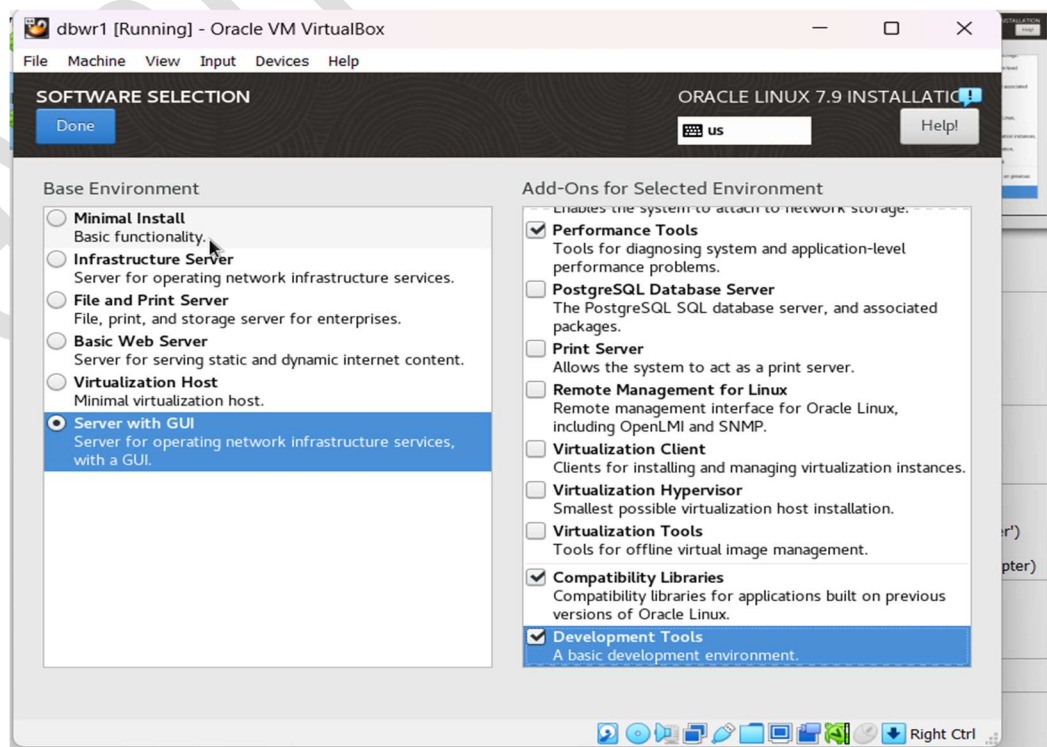
Next installation Destination-> select 80GB->I will configure partitioning and press Done button



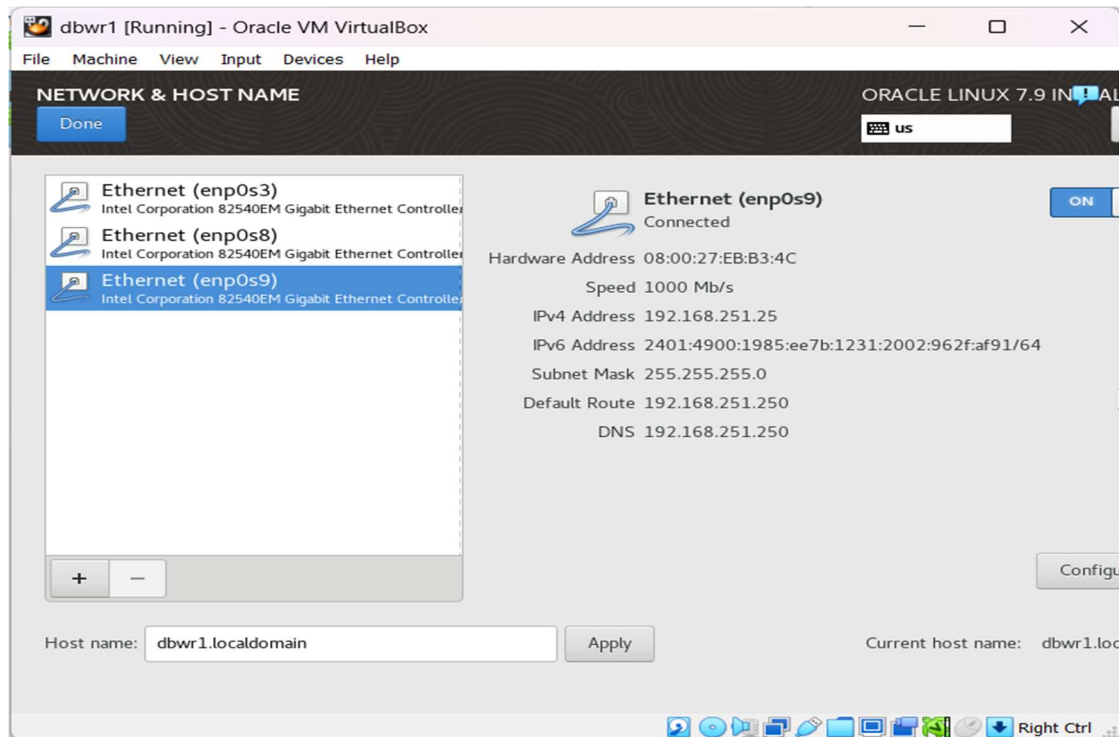
This is main screen one by one we need to configure

Select following packages and click done button

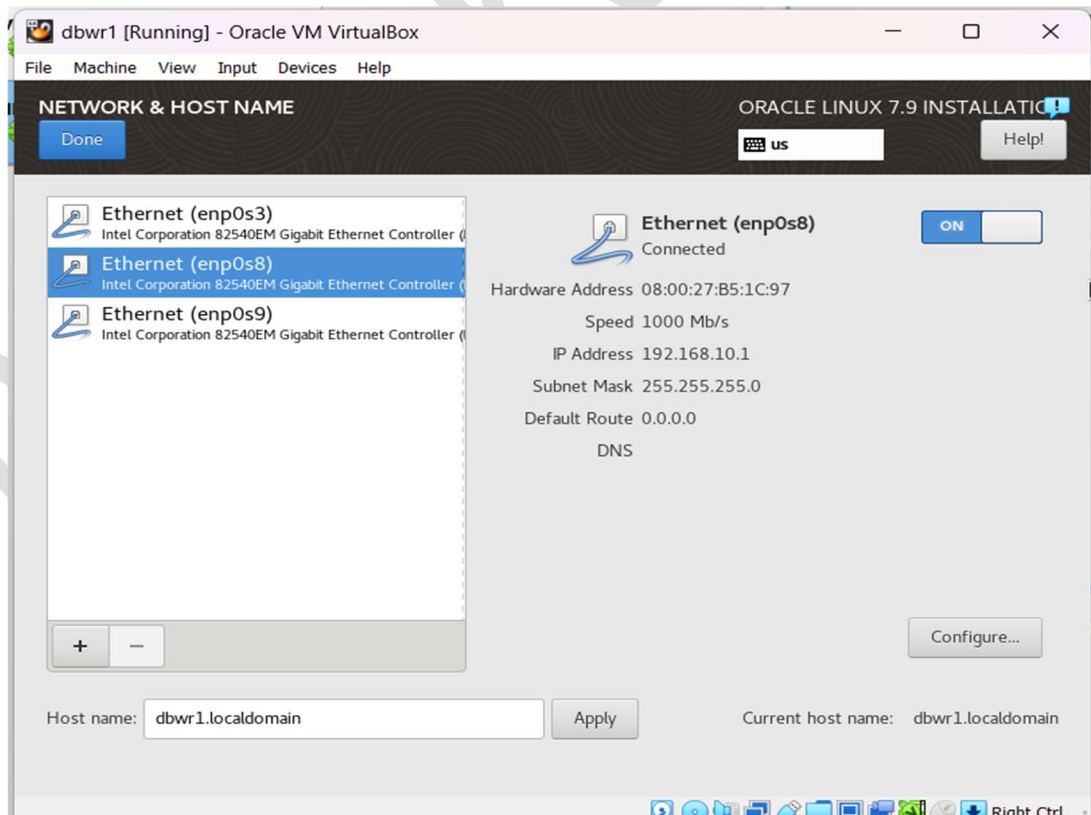
Server with GUI
Hardware Monitoring
Utilities
Large Systems
Performance
Network file system
client
Performance Tools
Compatibility
Libraries
Development Tools



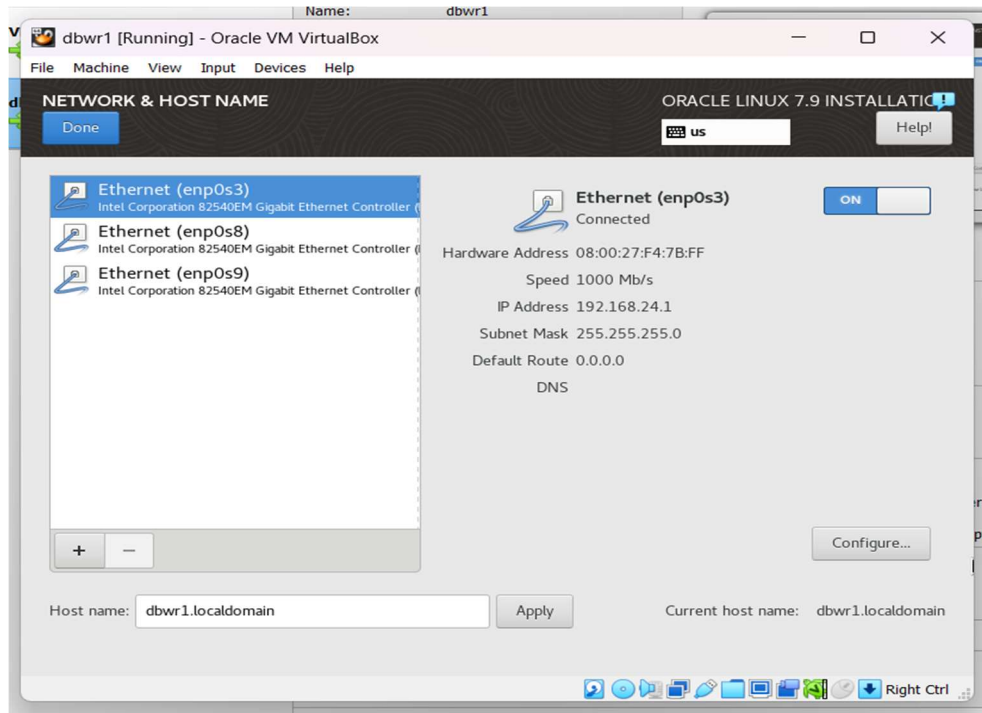
Next Configure Network



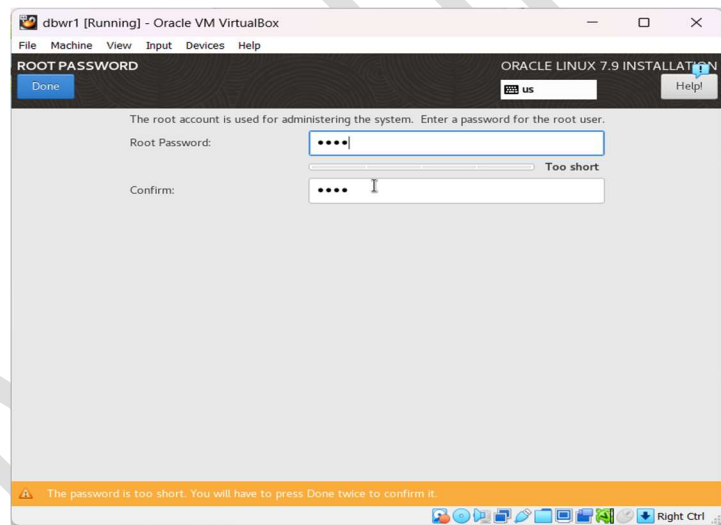
Select IPv4 Settings tab and change Method to Manual; Press Add and fill Address: 192.168.24.1; Netmask: 255.255.255.0; Gateway: 0.0.0.0. Press save then done



Select IPv4 Settings tab; change Method to Manual. Press Add and fill Address: 192.168.10.1; Netmask: 255.255.255.0; Gateway: 0.0.0.0. save and press done button



Enter Root Password



Install the system Linux prerequisites

First check internet is working fine or not
ping google.com

The package oracle-database-preinstall-19c contains all the prerequisites on Oracle Linux using the Oracle Unbreakable Enterprise Kernel (UEK).

```
# yum update -y
# yum install -y oracle-database-preinstall-19c.x86_64
```

```
# yum install oracleasm-support
# yum install bind* -y
# systemctl -p
```

Configure Hosts File

```
[root@dbwr1 ~]# cat /etc/hosts
127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1 localhost localhost.localdomain localhost6 localhost6.localdomain6
```

Public

```
192.168.24.1 dbwr1.localdomain dbwr1
192.168.24.2 dbwr2.localdomain dbwr2
```

Private

```
192.168.10.1 dbwr1-priv.localdomain dbwr1-priv
192.168.10.2 dbwr2-priv.localdomain dbwr2-priv
```

Virtual

```
192.168.24.31 dbwr1-vip.localdomain dbwr1-vip
192.168.24.32 dbwr2-vip.localdomain dbwr2-vip
```

SCAN

```
192.168.24.41 dbwr-scan.localdomain dbwr-scan
192.168.24.42 dbwr-scan.localdomain dbwr-scan
192.168.24.43 dbwr-scan.localdomain dbwr-scan
```

Create Groups and permissions

```
[root@dbwr1 ~]# groupadd -g 54327 asmdba
[root@dbwr1 ~]# groupadd -g 54328 asmoper
[root@dbwr1 ~]# groupadd -g 54329 asmadmin
```

You might need to run usermod to change **oracle user groups as it's being created** by oracle-database-preinstall-19c

```
usermod -G asmdba,asmoper,asmadmin oracle
```

Create Oracle user password


```
[root@dbwr1 ~]# passwd oracle
Changing password for user oracle.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
```

Create the directories in which the Oracle software will be installed.

```
mkdir -p /u01/app/19c/grid
mkdir -p /u01/app/oracle/product/19c/db_1
chown -R oracle:oinstall /u01
chmod -R 775 /u01/
```

Configure Bash Profile of Users.

We are going to install Grid and Oracle as oracle user.
Update the bash profile file for grid and oracle respectively as below.

Log in as the “oracle” user and add the following lines at the end of the “/home/oracle/.bash_profile” file.

```
# Oracle Settings
export TMP=/tmp
export TMPDIR=$TMP
export ORACLE_BASE=/u01/app/oracle
export GRID_HOME=/u01/app/19c/grid
export DB_HOME=$ORACLE_BASE/product/19c/db_1
export ORACLE_HOME=$DB_HOME
export ORACLE_SID=oradbwr
export ORACLE_TERM=xterm
export BASE_PATH=/usr/sbin:$PATH
export PATH=$ORACLE_HOME/bin:$BASE_PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/lib:/usr/lib
export CLASSPATH=$ORACLE_HOME/JRE:$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib

alias grid='. /home/oracle/grid.env'
alias db='. /home/oracle/db.env'
```

Grid Home bash profile

Create a file called “/home/oracle/grid.env” with the following contents.

```
export ORACLE_SID=+ASM1
export ORACLE_HOME=$GRID_HOME
export PATH=$ORACLE_HOME/bin:$BASE_PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/lib:/usr/lib
export CLASSPATH=$ORACLE_HOME/JRE:$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib
```


Oracle Home bash profile

Create a file called “/home/oracle/db.env” with the following contents.

```
export ORACLE_SID=oradbwr
export ORACLE_HOME=$DB_HOME
export PATH=$ORACLE_HOME/bin:$BASE_PATH
export LD_LIBRARY_PATH=$ORACLE_HOME/lib:/lib:/usr/lib
export CLASSPATH=$ORACLE_HOME/JRE:$ORACLE_HOME/jlib:$ORACLE_HOME/rdbms/jlib
```

Firewall Stop and Disable

```
[root@dbwr1 ~]#
[root@dbwr1 ~]# systemctl stop firewalld.service
[root@dbwr1 ~]#
[root@dbwr1 ~]# systemctl disable firewalld.service
Removed symlink /etc/systemd/system/multi-user.target.wants/firewalld.service.
Removed symlink /etc/systemd/system/dbus-org.fedoraproject.FirewallD1.service.
[root@dbwr1 ~]#
[root@dbwr1 ~]#
```

Chrony NTP Configuration

```
[root@dbwr1 ~]# systemctl enable chronyd.service
Created symlink from /etc/systemd/system/multi-user.target.wants/chronyd.service to /usr/lib/systemd/system/chronyd.service.
[root@dbwr1 ~]#
[root@dbwr1 ~]# systemctl restart chronyd.service
[root@dbwr1 ~]#
[root@dbwr1 ~]#
[root@dbwr1 ~]# chronyc -a 'burst 4/4'
200 OK
[root@dbwr1 ~]# chronyc -a makestep
```

Make the DNS server's own IP address in the below file:

```
[root@dbwr1 ~]# systemctl status dnsmasq
```

```
[root@dbwr1 ~]# systemctl start dnsmasq
```

```
[root@dbwr1 ~]#
[root@dbwr1 ~]# systemctl start dnsmasq
[root@dbwr1 ~]# systemctl status dnsmasq
● dnsmasq.service - DNS caching server.
   Loaded: loaded (/usr/lib/systemd/system/dnsmasq.service; disabled; vendor preset: disabled)
   Active: active (running) since Fri 2024-10-18 12:49:21 IST; 7s ago
     Main PID: 8609 (dnsmasq)
       Tasks: 1
      CGroup: /system.slice/dnsmasq.service
              └─8609 /usr/sbin/dnsmasq -k

Oct 18 12:49:21 dbwr1.localdomain systemd[1]: Started DNS caching server..
Oct 18 12:49:21 dbwr1.localdomain dnsmasq[8609]: started, version 2.76 cachesize 150
Oct 18 12:49:21 dbwr1.localdomain dnsmasq[8609]: compile time options: IPv6 GNU-getopt D...fy
Oct 18 12:49:21 dbwr1.localdomain dnsmasq[8609]: reading /etc/resolv.conf
Oct 18 12:49:21 dbwr1.localdomain dnsmasq[8609]: using nameserver 192.168.251.250#53
Oct 18 12:49:21 dbwr1.localdomain dnsmasq[8609]: using nameserver 2401:4900:1d32:1e34::c#53
Oct 18 12:49:21 dbwr1.localdomain dnsmasq[8609]: read /etc/hosts - 11 addresses
Hint: Some lines were ellipsized, use -l to show in full.
[root@dbwr1 ~]#
[root@dbwr1 ~]#
```

```
[root@dbwr1 ~]# cat /etc/resolv.conf
```

```
# Generated by NetworkManager
search localdomain
nameserver 192.168.24.1
options timeout:1
options attempts:5
```

```
[root@dbwr2 ~]# cat /etc/resolv.conf
```

```
# Generated by NetworkManager
search localdomain
nameserver 192.168.24.1
options timeout:1
options attempts:5
```

Now Check for the Available Hosts in DNS:

```
[root@dbwr1 ~]# cat /etc/resolv.conf
# Generated by NetworkManager
search localdomain
nameserver 192.168.24.1
options timeout:1
options attempts:5

[root@dbwr1 ~]#
[root@dbwr1 ~]# nslookup dbwr-scan
Server:      192.168.24.1
Address:     192.168.24.1#53

Name:   dbwr-scan.localdomain
Address: 192.168.24.43
Name:   dbwr-scan.localdomain
Address: 192.168.24.41
Name:   dbwr-scan.localdomain
Address: 192.168.24.42
** server can't find dbwr-scan.localdomain: REFUSED

[root@dbwr1 ~]#
```

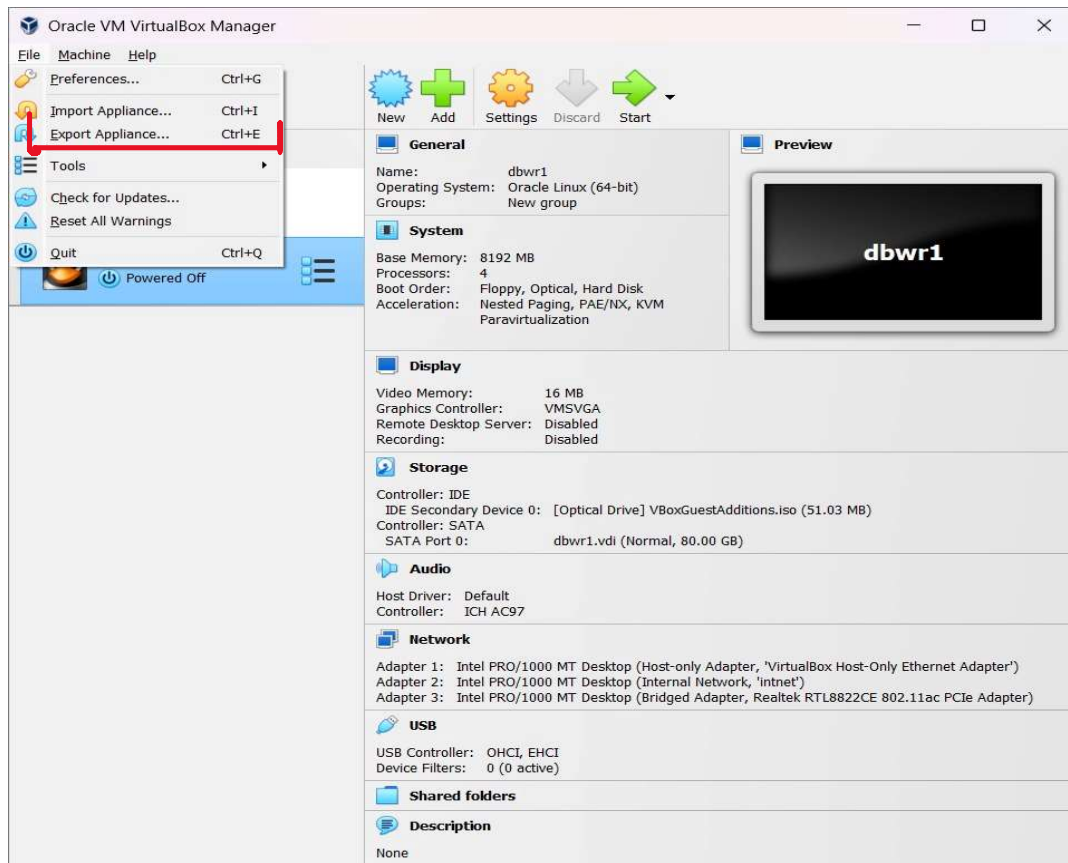
Backup VMbox for build 2nd Node and adding Diskgroup

Now we will shut down the server as we need to take backup and adding asm disks.

```
[root@dbwr1 19c]# init 0
```

Backup VMBox

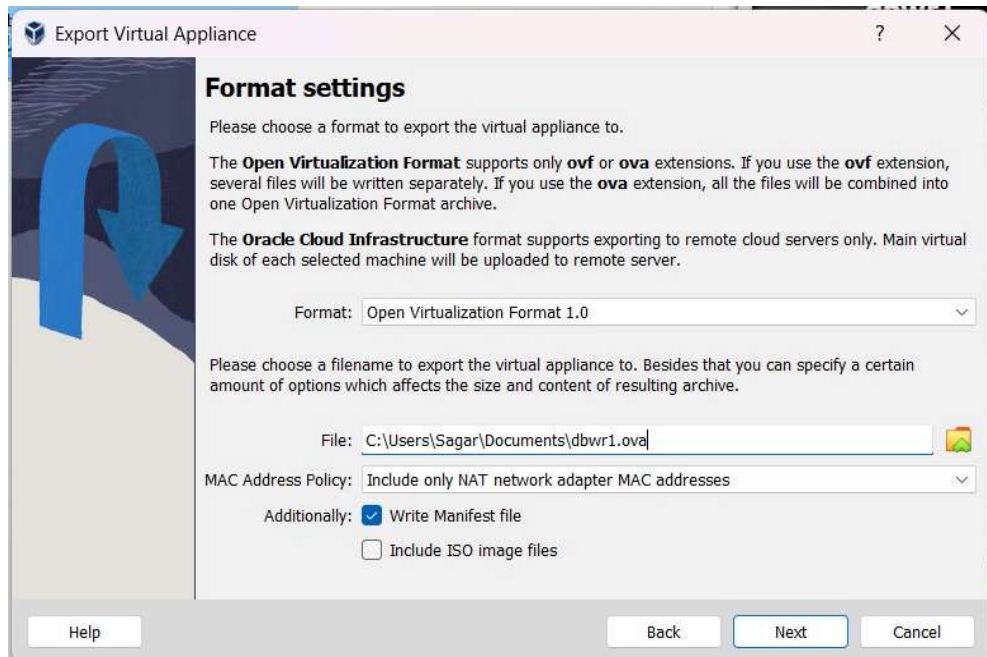
Click File export Appliance



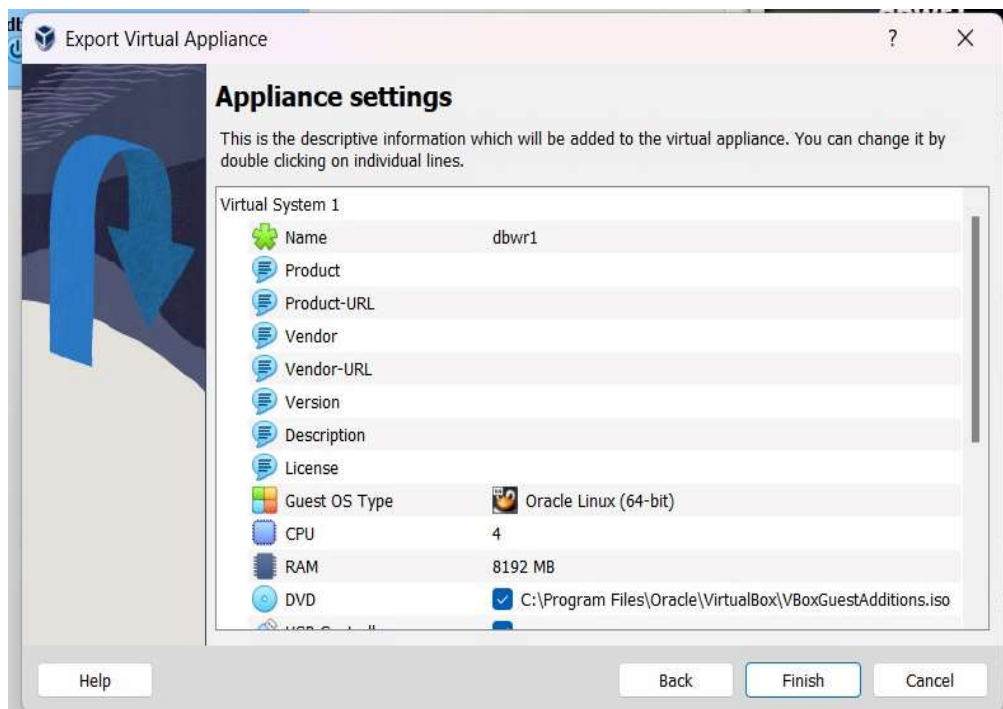
Select our VMBox name and press next



Select the location of backup



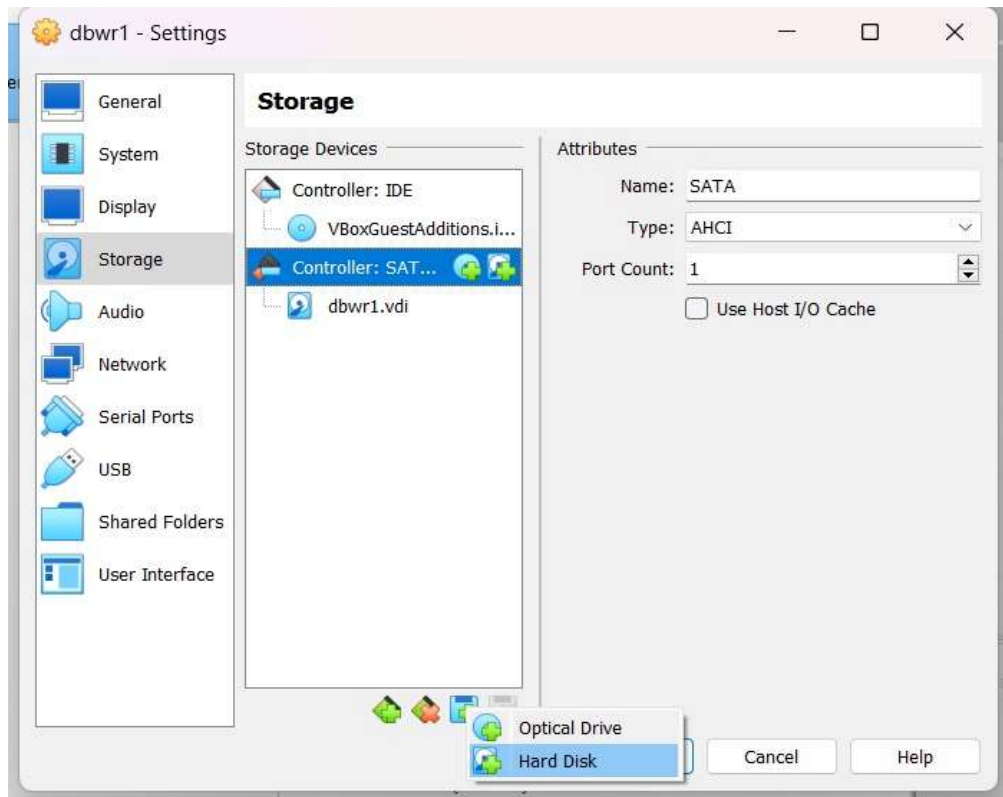
Click export



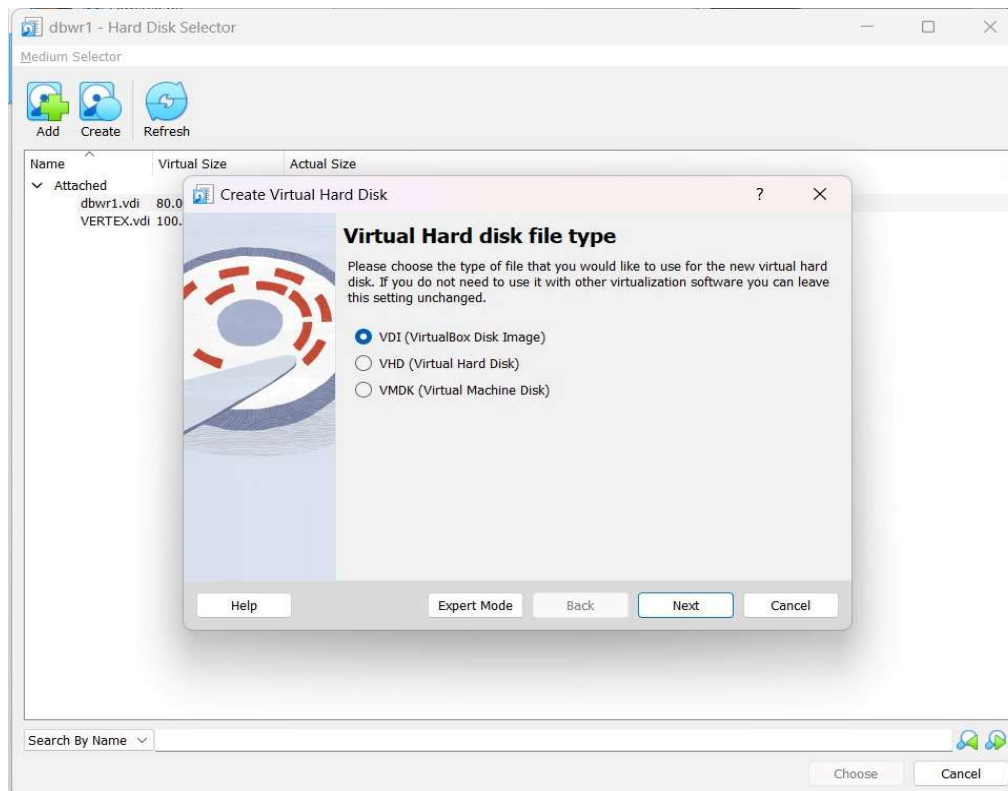
ASM Disk Group Preparation

After completed the backup. We need to add disks for ASM storage.

Click General-> Storage->Hard Disk

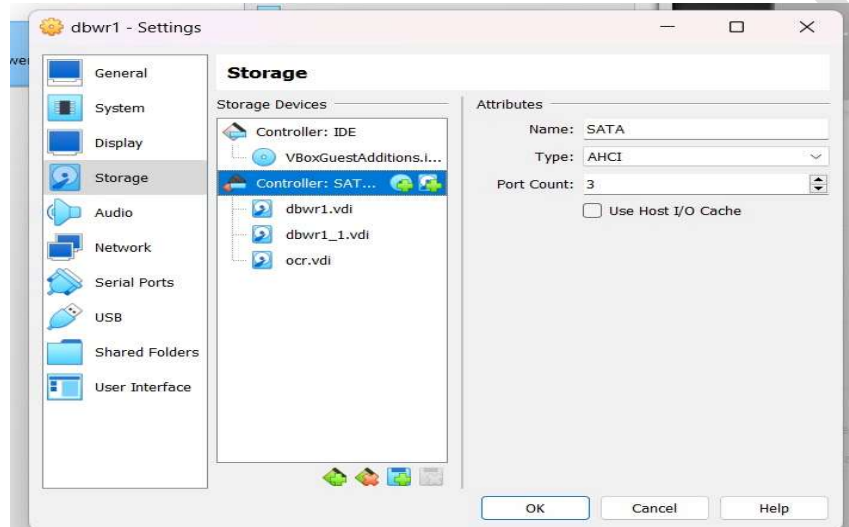
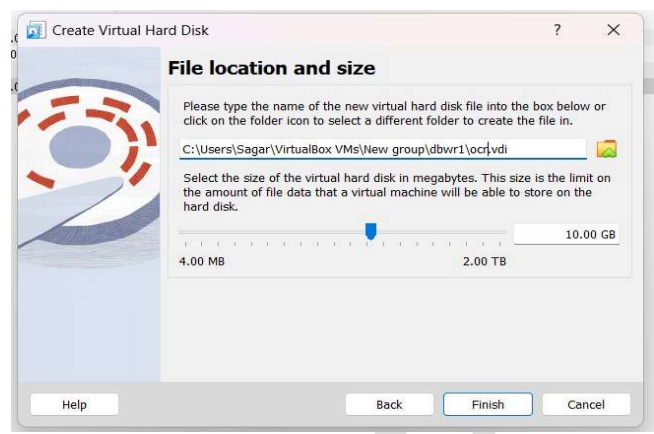
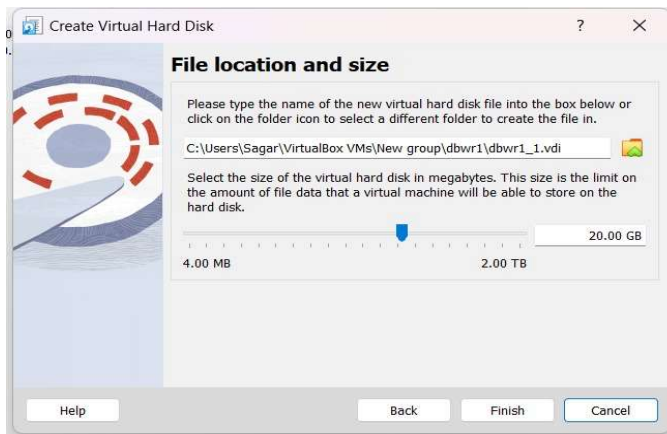


Click Create and select VDI



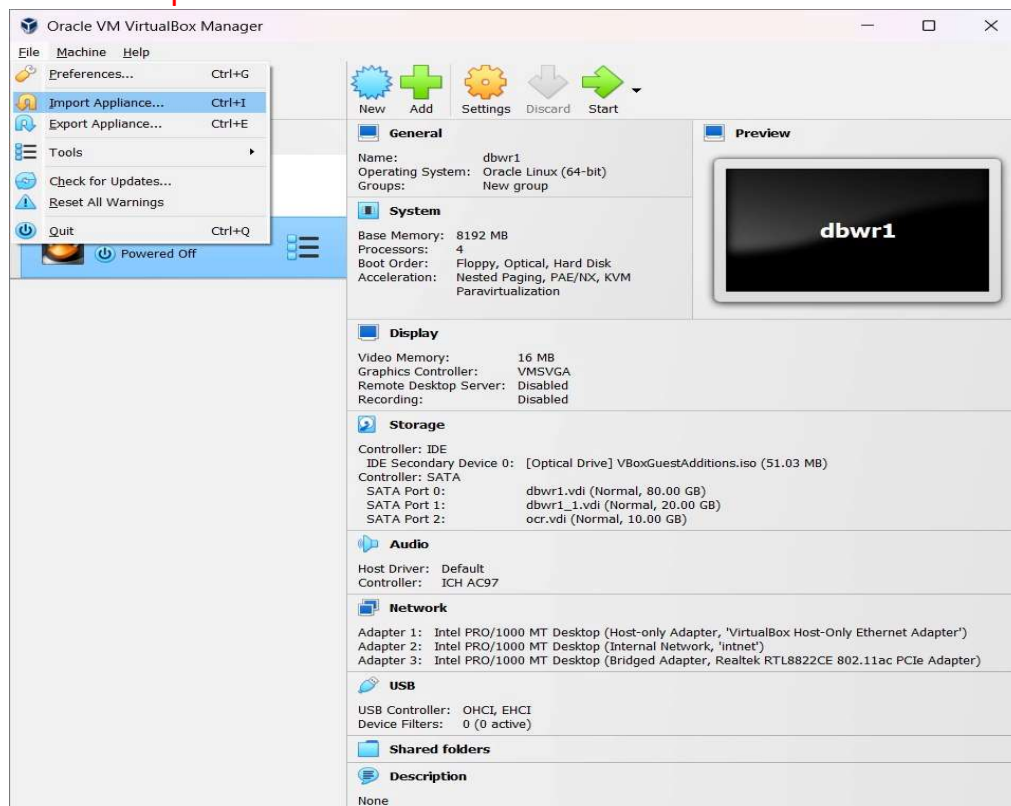
Select Storage Location

In My case example 3 folders i created like this Rac1,Rac2,Storage

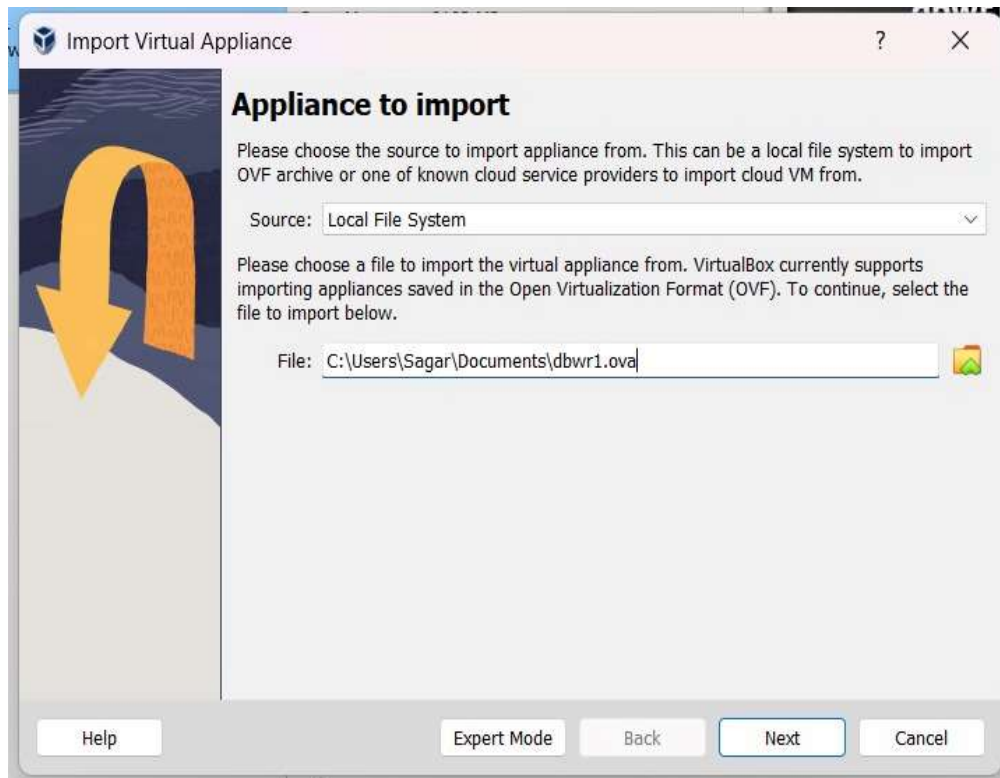


Copy the Virtual Box For Node:2 Preparation

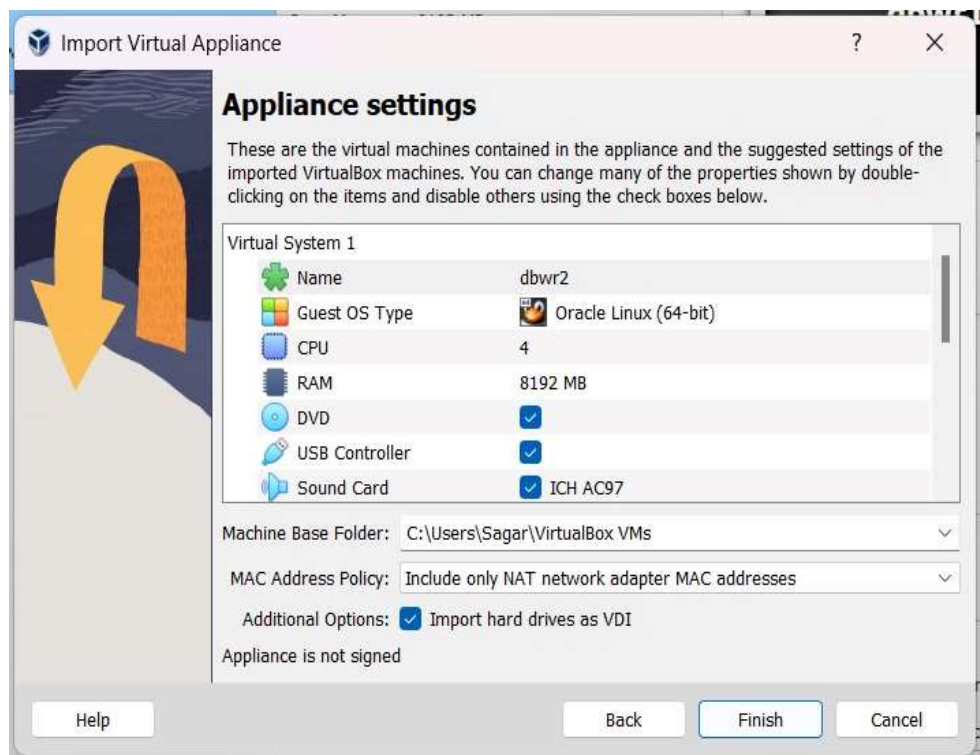
Select File and Click Import button



Now Restore the Backup that was taken earlier and change it as dbwr2



Change Name to DBWR2 and Change the location as below

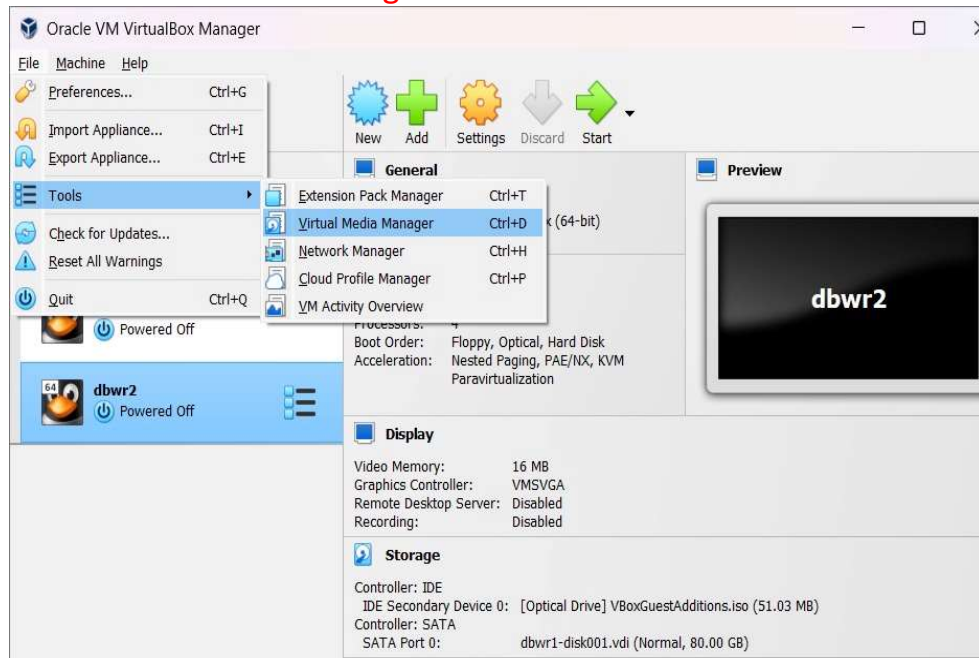


Import is completed successfully.

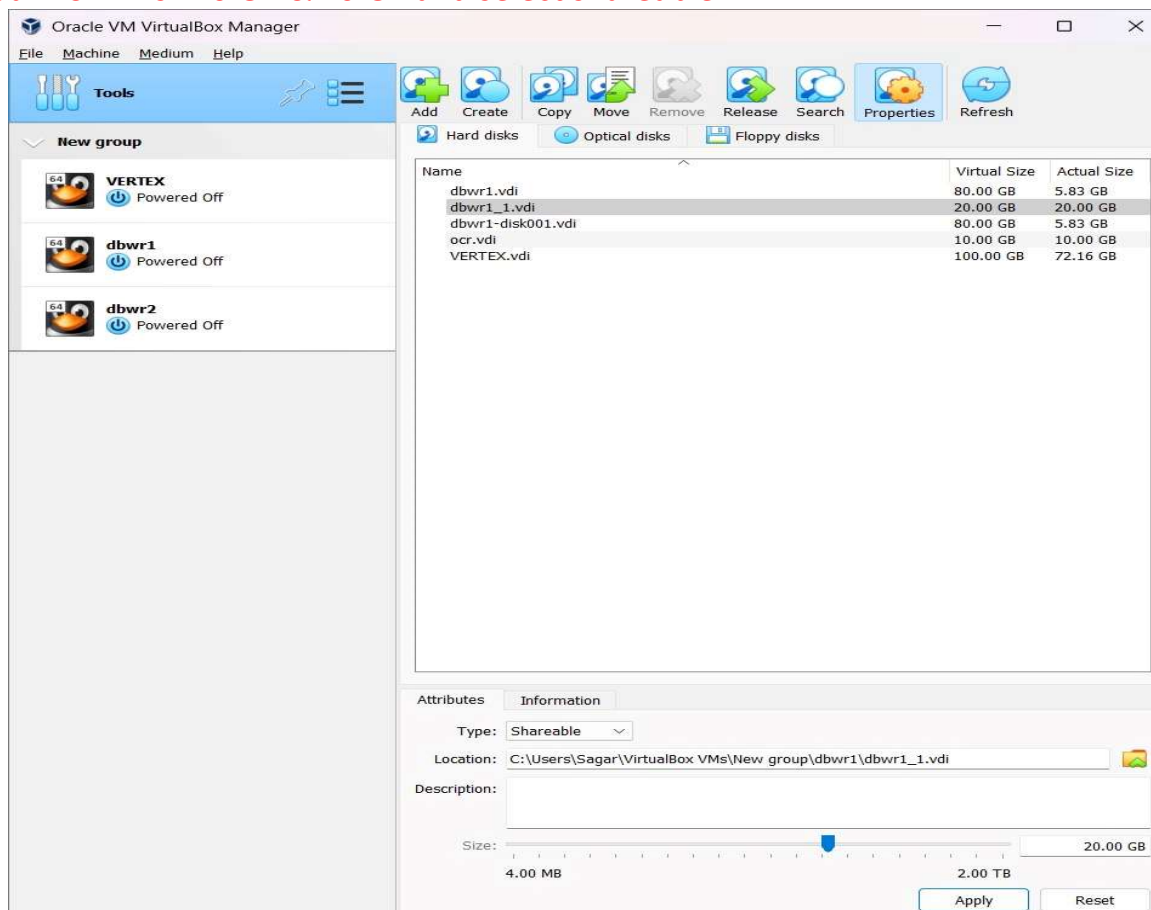
Creating a Shared Disk

After import is completed We need to Change the disks to shared that were created earlier for ASM

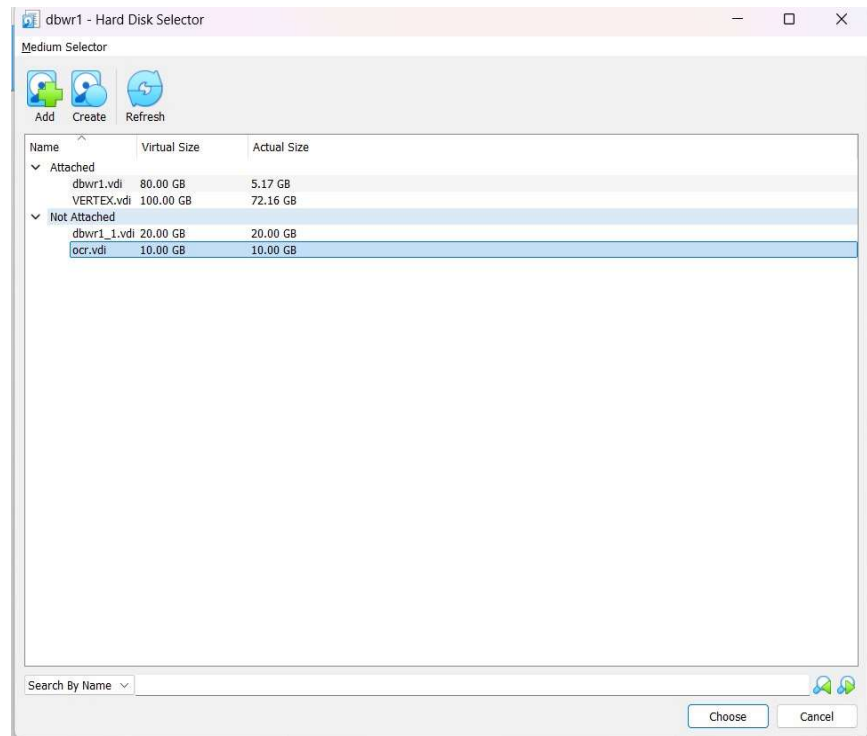
Select file and then Virtual media manager



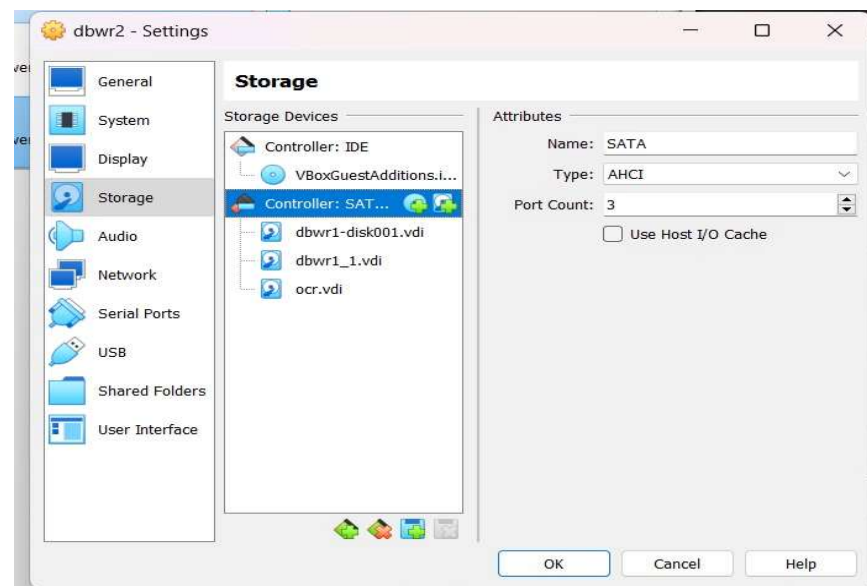
Select our ASM Disk 20 GB & 10 GB and select shareable:



Choose our disk and add (35 GB and 12 GB)



Both Asm Disk is added to 2nd node



Now start 2nd Node for changing network IP's (Only 2nd Node)

Change Host name

```
[root@dbwr1 ~]# nmcli general hostname
```

```
dbwr1.localdomain
```

```
[root@dbwr1 ~]# nmcli general hostname dbwr2.localdomain
```

```
[root@dbwr1 ~]# nmcli general hostname  
dbwr2.localdomain
```

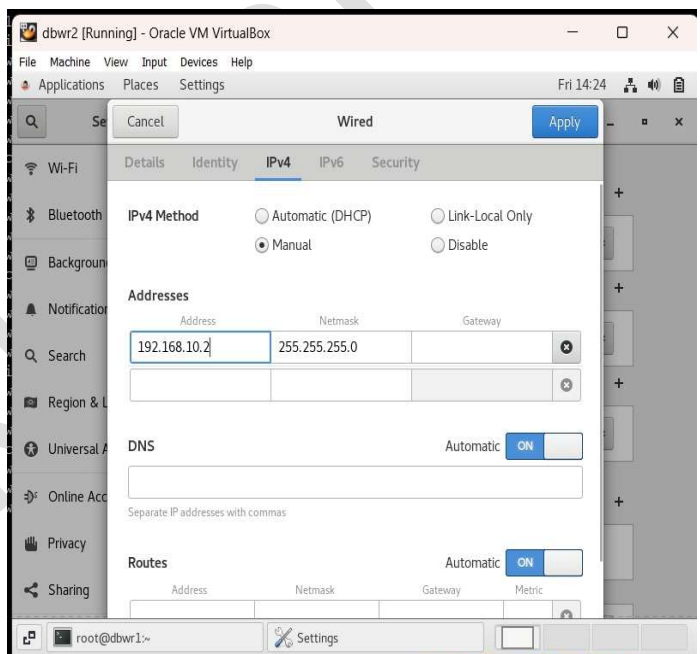
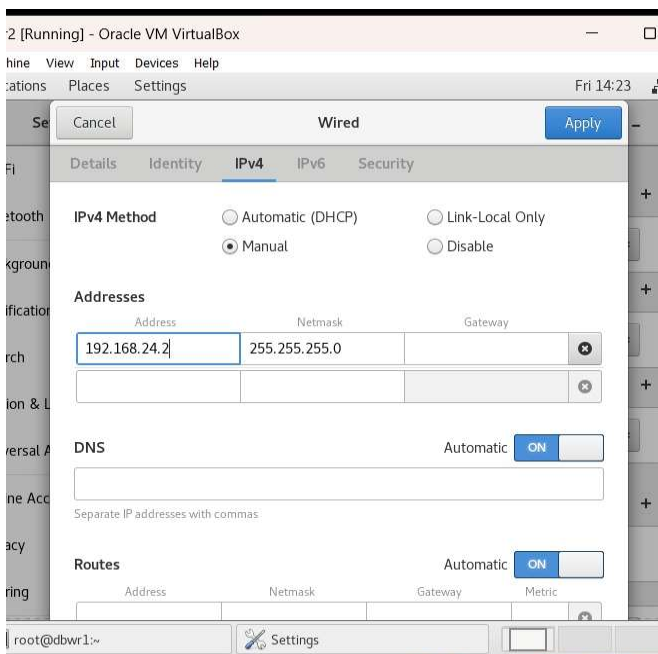
```
[root@dbwr1 ~]# service systemd-hostnamed restart
```

Redirecting to /bin/systemctl restart systemd-hostnamed.service

```
[root@dbwr1 ~]# hostname  
dbwr2.localdomain
```

Change Network

Select Applications and settings



```
[root@dbwr1 ~]# systemctl restart network
```

```
[root@dbwr1 ~]# ifconfig
```

```
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
```

```
inet 192.168.24.2 netmask 255.255.255.0 broadcast 192.168.24.255
```

```
inet6 fe80::a230:3ef0:a271:163c prefixlen 64 scopeid 0x20<link>
```

```
ether 08:00:27:92:99:bd txqueuelen 1000 (Ethernet)
```

```
RX packets 32 bytes 2930 (2.8 KiB)
```

```
RX errors 0 dropped 0 overruns 0 frame 0
```

```
TX packets 83 bytes 11287 (11.0 KiB)
```

```
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.10.2 netmask 255.255.255.0 broadcast 192.168.10.255
    inet6 fe80::454c:b14a:b50a:4eec prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:c1:c9:c6 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 81 bytes 10691 (10.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

To verify the network

```
ping -c 2 dbwr1
ping -c 2 dbwr2
[root@dbwr1 ~]# nslookup dbwr-scan
Server: 192.168.24.1
Address: 192.168.24.1#53
Name: dbwr-scan.localdomain
Address: 192.168.24.43
Name: dbwr-scan.localdomain
Address: 192.168.24.41
Name: dbwr-scan.localdomain
Address: 192.168.24.42
```

Copy the oracle 19c grid software using shared folder

```
[root@dbwr1 ~]# cd /media/sf_Grid_Sfw
[root@dbwr1 sf_Grid_Sfw]# ll
total 2821472
-rwxrwx---. 1 root vboxsf 2889184573 Oct 15 16:03 V982068-01.zip
[root@dbwr1 sf_Grid_Sfw]# cp V982068-01.zip /u01/app/19c/grid/
```

Install the package cvudisk from the grid home as the "root" user on all nodes.

1st Node

```
[root@dbwr1 grid]# cd /u01/app/19c/grid/cv/rpm
[root@dbwr1 rpm]# rpm -Uvh cvuqdisk*
Preparing... ##### [100%]
Using default group oinstall to install package
Updating / installing...
```

1:cvuqdisk-1.0.10-1

[100%]

2nd Node

[root@dbwr2 ~]# **ssh root@dbwr2 rpm -Uvh /tmp/cvuqdisk***

root@dbwr2's password:

Preparing...

#####

Using default group oinstall to install package

Updating / installing...

cvuqdisk-1.0.10-1

#####

Configure Oracle ASM

[root@dbwr1 ~]# **fdisk /dev/sdb**

Welcome to fdisk (util-linux 2.23.2).

Changes will remain in memory only, until you decide to write them.

Be careful before using the write command.

Device does not contain a recognized partition table

Building a new DOS disklabel with disk identifier 0xa0ccfdd4.

Command (m for help): n

Partition type:

p primary (0 primary, 0 extended, 4 free)

e extended

Select (default p): p

Partition number (1-4, default 1):

First sector (2048-41943039, default 2048):

Using default value 2048

Last sector, +sectors or +size{K,M,G} (2048-41943039, default 41943039):

Using default value 41943039

Partition 1 of type Linux and of size 20 GiB is set

Command (m for help): w

The partition table has been altered!

Calling ioctl() to re-read partition table.

Syncing disks.


```
[root@dbwr1 ~]# fdisk /dev/sdc
```

Welcome to fdisk (util-linux 2.23.2).

Changes will remain in memory only, until you decide to write them.

Be careful before using the write command.

Device does not contain a recognized partition table

Building a new DOS disklabel with disk identifier 0x6b457c4c.

Command (m for help): n

Partition type:

p primary (0 primary, 0 extended, 4 free)

e extended

Select (default p): p

Partition number (1-4, default 1):

First sector (2048-20971519, default 2048):

Using default value 2048

Last sector, +sectors or +size{K,M,G} (2048-20971519, default 20971519):

Using default value 20971519

Partition 1 of type Linux and of size 10 GiB is set

Command (m for help): w

The partition table has been altered!

Calling ioctl() to re-read partition table.

Syncing disks.

Configure oracleasm utility.

```
[root@dbwr1 grid]# /usr/sbin/oracleasm configure -i
```

Configuring the Oracle ASM library driver.

This will configure the on-boot properties of the Oracle ASM library driver. The following questions will determine whether the driver is loaded on boot and what permissions it will have. The current values will be shown in brackets ('[]'). Hitting <ENTER> without typing an answer will keep that current value. Ctrl-C will abort.

Default user to own the driver interface []: **oracle**

Default group to own the driver interface []: **oinstall**

Start Oracle ASM library driver on boot (y/n) [n]: y
Scan for Oracle ASM disks on boot (y/n) [y]: y
Writing Oracle ASM library driver configuration: done

This can be verified using:-

```
[root@dbwr1 grid]# /usr/sbin/oracleasm configure
ORACLEASM_ENABLED=true
ORACLEASM_UID=oracle
ORACLEASM_GID=oinstall
ORACLEASM_SCANBOOT=true
ORACLEASM_SCANORDER=""
ORACLEASM_SCANEXCLUDE=""
ORACLEASM_SCAN_DIRECTORIES=""
ORACLEASM_USE_LOGICAL_BLOCK_SIZE="false"
```

Initialize the asmlib with the oracleasm init command to load oracleasm module and mount the oracleasm filesystem:-

```
[root@dbwr1 grid]# /usr/sbin/oracleasm init
Creating /dev/oracleasm mount point: /dev/oracleasm
Loading module "oracleasm": oracleasm
Configuring "oracleasm" to use device physical block size
Mounting ASMLib driver filesystem: /dev/oracleasm
```

When ASMLIB is configured, a special file system is created and mounted as /dev/oracleasm, you should notice the following line from the df command:-

```
[root@dbwr1 grid]# df -ha | grep oracle
oracleasmfs          0      0      0  - /dev/oracleasm
[root@dbwr1 grid]#
[root@dbwr1 grid]# oracleasm createdisk ASMDISK_DATA /dev/sdb1
Writing disk header: done
Instantiating disk: done
[root@dbwr1 grid]# oracleasm createdisk ASMDISK_OCR /dev/sdc1
```

Writing disk header: done

Instantiating disk: done

```
[root@dbwr1 grid]# cd /dev/oracleasm/disks
```

```
[root@dbwr1 disks]# ll
```

total 0

brw-rw----. 1 oracle oinstall 8, 17 Oct 18 14:49 ASMDISK_DATA

brw-rw----. 1 oracle oinstall 8, 33 Oct 18 14:49 ASMDISK_OCR

```
[root@dbwr1 disks]# oracleasm scandisks
```

Reloading disk partitions: done

Cleaning any stale ASM disks...

Scanning system for ASM disks...

```
[root@dbwr1 disks]# oracleasm listdisks
```

ASMDISK_DATA

ASMDISK_OCR

Node 2:

```
[root@dbwr2 ~]# cd /u01/app/19c/grid
```

```
[root@dbwr2 grid]# /usr/sbin/oracleasm configure -i
```

Configuring the Oracle ASM library driver.

This will configure the on-boot properties of the Oracle ASM library driver. The following questions will determine whether the driver is loaded on boot and what permissions it will have. The current values will be shown in brackets ('[]'). Hitting <ENTER> without typing an answer will keep that current value. Ctrl-C will abort.

Default user to own the driver interface []: **oracle**

Default group to own the driver interface []: **oinstall**

Start Oracle ASM library driver on boot (y/n) [n]: **y**

Scan for Oracle ASM disks on boot (y/n) [y]: **y**

Writing Oracle ASM library driver configuration: done

```
[root@dbwr2 grid]# /usr/sbin/oracleasm init
```

Creating /dev/oracleasm mount point: /dev/oracleasm

Loading module "oracleasm": oracleasm

Configuring "oracleasm" to use device physical block size

Mounting ASMLib driver filesystem: /dev/oracleasm

```
[root@dbwr2 grid]# oracleasm scandisks
```

Reloading disk partitions: done

Cleaning any stale ASM disks...

Scanning system for ASM disks...

Instantiating disk "ASMDISK_OCR"

Instantiating disk "ASMDISK_DATA"

```
[root@dbwr2 grid]# oracleasm listdisks
```

ASMDISK_DATA

ASMDISK_OCR

Configure SSH Setup

```
[root@dbwr1 disks]# cd /u01/app/19c/grid/deinstall
```

```
[root@dbwr1 deinstall]# ./sshUserSetup.sh -user oracle -hosts "dbwr1 dbwr2" -noPromptPassphrase  
-confirm -advanced
```

The output of this script is also logged into /tmp/sshUserSetup_2024-10-18-14-52-08.log

Hosts are dbwr1 dbwr2

user is oracle

Platform:- Linux

Checking if the remote hosts are reachable

PING dbwr1.localdomain (192.168.24.1) 56(84) bytes of data.

64 bytes from dbwr1.localdomain (192.168.24.1): icmp_seq=1 ttl=64 time=12.9 ms

64 bytes from dbwr1.localdomain (192.168.24.1): icmp_seq=2 ttl=64 time=0.195 ms

64 bytes from dbwr1.localdomain (192.168.24.1): icmp_seq=3 ttl=64 time=0.086 ms

64 bytes from dbwr1.localdomain (192.168.24.1): icmp_seq=4 ttl=64 time=0.083 ms

64 bytes from dbwr1.localdomain (192.168.24.1): icmp_seq=5 ttl=64 time=0.083 ms

--- dbwr1.localdomain ping statistics ---

5 packets transmitted, 5 received, 0% packet loss, time 4071ms

rtt min/avg/max/mdev = 0.083/2.670/12.906/5.118 ms

PING dbwr2.localdomain (192.168.24.2) 56(84) bytes of data.

64 bytes from dbwr2.localdomain (192.168.24.2): icmp_seq=1 ttl=64 time=5.79 ms

64 bytes from dbwr2.localdomain (192.168.24.2): icmp_seq=2 ttl=64 time=1.50 ms

64 bytes from dbwr2.localdomain (192.168.24.2): icmp_seq=3 ttl=64 time=2.11 ms

64 bytes from dbwr2.localdomain (192.168.24.2): icmp_seq=4 ttl=64 time=1.24 ms

64 bytes from dbwr2.localdomain (192.168.24.2): icmp_seq=5 ttl=64 time=2.35 ms

--- dbwr2.localdomain ping statistics ---

5 packets transmitted, 5 received, 0% packet loss, time 4052ms

rtt min/avg/max/mdev = 1.243/2.602/5.796/1.647 ms

Remote host reachability check succeeded.

The following hosts are reachable: dbwr1 dbwr2.

The following hosts are not reachable: .

All hosts are reachable. Proceeding further...

firsthost dbwr1

numhosts 2

The script will setup SSH connectivity from the host dbwr1.localdomain to all the remote hosts. After the script is executed, the user can use SSH to run commands on the remote hosts or copy files between this host dbwr1.localdomain and the remote hosts without being prompted for passwords or confirmations.

NOTE 1:

As part of the setup procedure, this script will use ssh and scp to copy files between the local host and the remote hosts. Since the script does not store passwords, you may be prompted for the passwords during the execution of the script whenever ssh or scp is invoked.

NOTE 2:

AS PER SSH REQUIREMENTS, THIS SCRIPT WILL SECURE THE USER HOME DIRECTORY AND THE .ssh DIRECTORY BY REVOKING GROUP AND WORLD WRITE PRIVILEGES TO THESE directories.

Do you want to continue and let the script make the above mentioned changes (yes/no)?

Confirmation provided on the command line

The user chose yes

User chose to skip passphrase related questions.

Creating .ssh directory on local host, if not present already

Creating authorized_keys file on local host

Changing permissions on authorized_keys to 644 on local host

Creating known_hosts file on local host

Changing permissions on known_hosts to 644 on local host

Creating config file on local host

If a config file exists already at /root/.ssh/config, it would be backed up to /root/.ssh/config.backup.

Removing old private/public keys on local host

Running SSH keygen on local host with empty passphrase

Generating public/private rsa key pair.

Your identification has been saved in /root/.ssh/id_rsa.

Your public key has been saved in /root/.ssh/id_rsa.pub.

The key fingerprint is:

SHA256:0MU7PnhcwFLzIAB2aWeJCPxxxboD6plsRgi7L00Gvpo root@dbwr1.localdomain

The key's randomart image is:

+---[RSA 1024]-----+

| ...ooo*o== |

| ..o.=.*o++ |

| .+.+.o. |

|o o.. o. |

|o+...S+o |

|o.= o.= |

| O.o ... |

|ooO |

|E=. |

+---[SHA256]-----+

Creating .ssh directory and setting permissions on remote host dbwr1

THE SCRIPT WOULD ALSO BE REVOKING WRITE PERMISSIONS FOR group AND others ON THE HOME DIRECTORY FOR oracle. THIS IS AN SSH REQUIREMENT.

The script would create ~oracle/.ssh/config file on remote host dbwr1. If a config file exists already at ~oracle/.ssh/config, it would be backed up to ~oracle/.ssh/config.backup.

The user may be prompted for a password here since the script would be running SSH on host dbwr1.

Warning: Permanently added 'dbwr1,192.168.24.1' (ECDSA) to the list of known hosts.

oracle@dbwr1's password:

Permission denied, please try again.

oracle@dbwr1's password:

Done with creating .ssh directory and setting permissions on remote host dbwr1.

Creating .ssh directory and setting permissions on remote host dbwr2

THE SCRIPT WOULD ALSO BE REVOKING WRITE PERMISSIONS FOR group AND others ON THE HOME DIRECTORY FOR oracle. THIS IS AN SSH REQUIREMENT.

The script would create ~oracle/.ssh/config file on remote host dbwr2. If a config file exists already at ~oracle/.ssh/config, it would be backed up to ~oracle/.ssh/config.backup.

The user may be prompted for a password here since the script would be running SSH on host dbwr2.

Warning: Permanently added 'dbwr2,192.168.24.2' (ECDSA) to the list of known hosts.

oracle@dbwr2's password:

Done with creating .ssh directory and setting permissions on remote host dbwr2.

Copying local host public key to the remote host dbwr1

The user may be prompted for a password or passphrase here since the script would be using SCP for host dbwr1.

oracle@dbwr1's password:

Done copying local host public key to the remote host dbwr1

Copying local host public key to the remote host dbwr2

The user may be prompted for a password or passphrase here since the script would be using SCP for host dbwr2.

oracle@dbwr2's password:

Done copying local host public key to the remote host dbwr2

Creating keys on remote host dbwr1 if they do not exist already. This is required to setup SSH on host dbwr1.

Generating public/private rsa key pair.

Your identification has been saved in .ssh/id_rsa.

Your public key has been saved in .ssh/id_rsa.pub.

The key fingerprint is:

SHA256:lg31cdFjt/+7/Ll4AqVS+jve2xjZspb9NCJQx430N4 oracle@dbwr1.localdomain

The key's randomart image is:

```
+---[RSA 1024]-----+
|      ..o ++o |
|      . ..+ .+ |
|      .  ooo.. |
|      +o =.oE. |
|      S o* o  . |
|      o +..+   . |
|      . + +...  . |
|      o.o.O .o.o |
|      oBX ..o*= |
+-----[SHA256]-----+
```

Creating keys on remote host dbwr2 if they do not exist already. This is required to setup SSH on host dbwr2.

Generating public/private rsa key pair.

Your identification has been saved in .ssh/id_rsa.

Your public key has been saved in .ssh/id_rsa.pub.

The key fingerprint is:

SHA256:EwY8RTw6nKID28WN/W0//0TKFYIIYONb99xZ3hclEaY oracle@dbwr2.localdomain

The key's randomart image is:

```
+---[RSA 1024]-----+
|      ..+o  =o+.o |
|      o.o o + * . |
|      . = +oE  + =o |
|      .  = B. .  ooX |
|      + o . oS.  .X |
|      . +      ..o  .+. |
|      .      ..  o . |
|      o  . |
|      o... |
+-----[SHA256]-----+
```

Updating authorized_keys file on remote host dbwr1

Updating known_hosts file on remote host dbwr1

Updating authorized_keys file on remote host dbwr2

Updating known_hosts file on remote host dbwr2

SSH setup is complete.

Verifying SSH setup

=====

The script will now run the date command on the remote nodes using ssh to verify if ssh is setup correctly. IF THE SETUP IS CORRECTLY SETUP, THERE SHOULD BE NO OUTPUT OTHER THAN THE DATE AND SSH SHOULD NOT ASK FOR PASSWORDS. If you see any output other than date or are prompted for the password, ssh is not setup correctly and you will need to resolve the issue and set up ssh again.

The possible causes for failure could be:

1. The server settings in /etc/ssh/sshd_config file do not allow ssh for user oracle.
2. The server may have disabled public key based authentication.
3. The client public key on the server may be outdated.
4. ~oracle or ~oracle/.ssh on the remote host may not be owned by oracle.
5. User may not have passed -shared option for shared remote users or may be passing the -shared option for non-shared remote users.
6. If there is output in addition to the date, but no password is asked, it may be a security alert shown as part of company policy. Append the additional text to the <OMS HOME>/sysman/prov/resources/ignoreMessages.txt file.

--dbwr1:--

Running /usr/bin/ssh -x -l oracle dbwr1 date to verify SSH connectivity has been setup from local host to dbwr1.

IF YOU SEE ANY OTHER OUTPUT BESIDES THE OUTPUT OF THE DATE COMMAND OR IF YOU ARE PROMPTED FOR A PASSWORD HERE, IT MEANS SSH SETUP HAS NOT BEEN SUCCESSFUL. Please note that being prompted for a passphrase may be OK but being prompted for a password is ERROR.

Fri Oct 18 14:52:49 IST 2024

--dbwr2:--

Running /usr/bin/ssh -x -l oracle dbwr2 date to verify SSH connectivity has been setup from local host to dbwr2.

IF YOU SEE ANY OTHER OUTPUT BESIDES THE OUTPUT OF THE DATE COMMAND OR IF YOU ARE PROMPTED FOR A PASSWORD HERE, IT MEANS SSH SETUP HAS NOT BEEN SUCCESSFUL. Please note that being prompted for a passphrase may be OK but being prompted for a password is ERROR.

Fri Oct 18 14:52:49 IST 2024

Verifying SSH connectivity has been setup from dbwr1 to dbwr1

IF YOU SEE ANY OTHER OUTPUT BESIDES THE OUTPUT OF THE DATE COMMAND OR IF YOU ARE PROMPTED FOR A PASSWORD HERE, IT MEANS SSH SETUP HAS NOT BEEN SUCCESSFUL.

Fri Oct 18 14:52:50 IST 2024

Verifying SSH connectivity has been setup from dbwr1 to dbwr2

IF YOU SEE ANY OTHER OUTPUT BESIDES THE OUTPUT OF THE DATE COMMAND OR IF YOU ARE PROMPTED FOR A PASSWORD HERE, IT MEANS SSH SETUP HAS NOT BEEN SUCCESSFUL.

Fri Oct 18 14:52:51 IST 2024

-Verification from complete-

SSH verification complete.

[root@dbwr1 deinstall]#

Pre-check for RAC Setup

Pre-check for CRS installation using Cluvfy

[oracle@dbwr1 grid]\$./runcluvfy.sh stage -pre crsinst -n dbwr1,dbwr2 -verbose

Verifying Physical Memory ...

Node Name	Available	Required	Status
dbwr2	7.7666GB (8143852.0KB)	8GB (8388608.0KB)	passed
dbwr1	7.7666GB (8143852.0KB)	8GB (8388608.0KB)	passed

Verifying Physical Memory ...PASSED

Verifying Available Physical Memory ...

Node Name	Available	Required	Status
-----------	-----------	----------	--------

dbwr2	7.126GB (7472172.0KB)	50MB (51200.0KB)	passed
dbwr1	6.8511GB (7183856.0KB)	50MB (51200.0KB)	passed

Verifying Available Physical Memory ...PASSED

Verifying Swap Size ...

Node Name	Available	Required	Status
dbwr2	7.875GB (8257532.0KB)	7.7666GB (8143852.0KB)	passed
dbwr1	7.875GB (8257532.0KB)	7.7666GB (8143852.0KB)	passed

Verifying Swap Size ...PASSED

Verifying Free Space: dbwr2:/usr,dbwr2:/var,dbwr2:/etc,dbwr2:/sbin,dbwr2:/tmp ...

Path	Node Name	Mount point	Available	Required	Status
/usr	dbwr2	/	15.249GB	25MB	passed
/var	dbwr2	/	15.249GB	5MB	passed
/etc	dbwr2	/	15.249GB	25MB	passed
/sbin	dbwr2	/	15.249GB	10MB	passed
/tmp	dbwr2	/	15.249GB	1GB	passed

Verifying Free Space: dbwr2:/usr,dbwr2:/var,dbwr2:/etc,dbwr2:/sbin,dbwr2:/tmp ...PASSED

Verifying Free Space: dbwr1:/usr,dbwr1:/var,dbwr1:/etc,dbwr1:/sbin,dbwr1:/tmp ...

Path	Node Name	Mount point	Available	Required	Status
/usr	dbwr1	/	15.2432GB	25MB	passed
/var	dbwr1	/	15.2432GB	5MB	passed
/etc	dbwr1	/	15.2432GB	25MB	passed
/sbin	dbwr1	/	15.2432GB	10MB	passed
/tmp	dbwr1	/	15.2432GB	1GB	passed

Verifying Free Space: dbwr1:/usr,dbwr1:/var,dbwr1:/etc,dbwr1:/sbin,dbwr1:/tmp ...PASSED

Verifying User Existence: oracle ...

Node Name	Status	Comment
dbwr2	passed	exists(54321)

dbwr1 passed exists(54321)

Verifying Users With Same UID: 54321 ...PASSED

Verifying User Existence: oracle ...PASSED

Verifying Group Existence: asmadmin ...

Node Name	Status	Comment
-----------	--------	---------

dbwr2	passed	exists
-------	--------	--------

dbwr1	passed	exists
-------	--------	--------

Verifying Group Existence: asmadmin ...PASSED

Verifying Group Existence: asmdba ...

Node Name	Status	Comment
-----------	--------	---------

dbwr2	passed	exists
-------	--------	--------

dbwr1	passed	exists
-------	--------	--------

Verifying Group Existence: asmdba ...PASSED

Verifying Group Existence: oinstall ...

Node Name	Status	Comment
-----------	--------	---------

dbwr2	passed	exists
-------	--------	--------

dbwr1	passed	exists
-------	--------	--------

Verifying Group Existence: oinstall ...PASSED

Verifying Group Membership: asmadmin ...

Node Name	User Exists	Group Exists	User in Group	Status
-----------	-------------	--------------	---------------	--------

dbwr2	yes	yes	yes	passed
-------	-----	-----	-----	--------

dbwr1	yes	yes	yes	passed
-------	-----	-----	-----	--------

Verifying Group Membership: asmadmin ...PASSED

Verifying Group Membership: asmdba ...

Node Name	User Exists	Group Exists	User in Group	Status
-----------	-------------	--------------	---------------	--------

dbwr2	yes	yes	yes	passed
-------	-----	-----	-----	--------

dbwr1	yes	yes	yes	passed
-------	-----	-----	-----	--------

Verifying Group Membership: asmdba ...PASSED

Verifying Group Membership: oinstall(Primary) ...

Node Name	User Exists	Group Exists	User in Group	Primary	Status
dbwr2	yes	yes	yes	yes	passed
dbwr1	yes	yes	yes	yes	passed

Verifying Group Membership: oinstall(Primary) ...PASSED

Verifying Run Level ...

Node Name	run level	Required	Status
dbwr2	5	3,5	passed
dbwr1	5	3,5	passed

Verifying Run Level ...PASSED

Verifying Hard Limit: maximum open file descriptors ...

Node Name	Type	Available	Required	Status
dbwr2	hard	65536	65536	passed
dbwr1	hard	65536	65536	passed

Verifying Hard Limit: maximum open file descriptors ...PASSED

Verifying Soft Limit: maximum open file descriptors ...

Node Name	Type	Available	Required	Status
dbwr2	soft	1024	1024	passed
dbwr1	soft	1024	1024	passed

Verifying Soft Limit: maximum open file descriptors ...PASSED

Verifying Hard Limit: maximum user processes ...

Node Name	Type	Available	Required	Status
dbwr2	hard	16384	16384	passed
dbwr1	hard	16384	16384	passed

Verifying Hard Limit: maximum user processes ...PASSED

Verifying Soft Limit: maximum user processes ...

Node Name	Type	Available	Required	Status
dbwr2	soft	16384	2047	passed
dbwr1	soft	16384	2047	passed

Verifying Soft Limit: maximum user processes ...PASSED

Verifying Soft Limit: maximum stack size ...

Node Name	Type	Available	Required	Status
dbwr2	soft	10240	10240	passed
dbwr1	soft	10240	10240	passed

Verifying Soft Limit: maximum stack size ...PASSED

Verifying Architecture ...

Node Name	Available	Required	Status
dbwr2	x86_64	x86_64	passed
dbwr1	x86_64	x86_64	passed

Verifying Architecture ...PASSED

Verifying OS Kernel Version ...

Node Name	Available	Required	Status
dbwr2	5.4.17-2102.201.3.el7uek.x86_64	4.1.12	passed
dbwr1	5.4.17-2102.201.3.el7uek.x86_64	4.1.12	passed

Verifying OS Kernel Version ...PASSED

Verifying OS Kernel Parameter: semmsl ...

Node Name	Current	Configured	Required	Status	Comment
dbwr1	250	250	250	passed	
dbwr2	250	250	250	passed	

Verifying OS Kernel Parameter: semmsl ...PASSED

Verifying OS Kernel Parameter: semmns ...

Node Name	Current	Configured	Required	Status	Comment

dbwr1	32000	32000	32000	passed
dbwr2	32000	32000	32000	passed

Verifying OS Kernel Parameter: semmns ...PASSED

Verifying OS Kernel Parameter: semopm ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	100	100	100	passed	
dbwr2	100	100	100	passed	

Verifying OS Kernel Parameter: semopm ...PASSED

Verifying OS Kernel Parameter: semmni ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	128	128	128	passed	
dbwr2	128	128	128	passed	

Verifying OS Kernel Parameter: semmni ...PASSED

Verifying OS Kernel Parameter: shmmax ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	4398046511104	4398046511104	4169652224	passed	
dbwr2	4398046511104	4398046511104	4169652224	passed	

Verifying OS Kernel Parameter: shmmax ...PASSED

Verifying OS Kernel Parameter: shmmni ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	4096	4096	4096	passed	
dbwr2	4096	4096	4096	passed	

Verifying OS Kernel Parameter: shmmni ...PASSED

Verifying OS Kernel Parameter: shmall ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	1073741824	1073741824	1073741824	passed	
dbwr2	1073741824	1073741824	1073741824	passed	

Verifying OS Kernel Parameter: shmall ...PASSED

Verifying OS Kernel Parameter: file-max ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	6815744	6815744	6815744	passed	
dbwr2	6815744	6815744	6815744	passed	

Verifying OS Kernel Parameter: file-max ...PASSED

Verifying OS Kernel Parameter: ip_local_port_range ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	between 9000 & 65500	between 9000 & 65500	between 9000 & 65500	between 9000 & 65535	
passed					
dbwr2	between 9000 & 65500	between 9000 & 65500	between 9000 & 65500	between 9000 & 65535	
passed					

Verifying OS Kernel Parameter: ip_local_port_range ...PASSED

Verifying OS Kernel Parameter: rmem_default ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	262144	262144	262144	passed	
dbwr2	262144	262144	262144	passed	

Verifying OS Kernel Parameter: rmem_default ...PASSED

Verifying OS Kernel Parameter: rmem_max ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	4194304	4194304	4194304	passed	
dbwr2	4194304	4194304	4194304	passed	

Verifying OS Kernel Parameter: rmem_max ...PASSED

Verifying OS Kernel Parameter: wmem_default ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	262144	262144	262144	passed	
dbwr2	262144	262144	262144	passed	

Verifying OS Kernel Parameter: wmem_default ...PASSED

Verifying OS Kernel Parameter: wmem_max ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	1048576	1048576	1048576	passed	
dbwr2	1048576	1048576	1048576	passed	

Verifying OS Kernel Parameter: wmem_max ...PASSED

Verifying OS Kernel Parameter: aio-max-nr ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	1048576	1048576	1048576	passed	
dbwr2	1048576	1048576	1048576	passed	

Verifying OS Kernel Parameter: aio-max-nr ...PASSED

Verifying OS Kernel Parameter: panic_on_oops ...

Node Name	Current	Configured	Required	Status	Comment
-----	-----	-----	-----	-----	-----
dbwr1	1	1	1	passed	
dbwr2	1	1	1	passed	

Verifying OS Kernel Parameter: panic_on_oops ...PASSED

Verifying Package: kmod-20-21 (x86_64) ...

Node Name	Available	Required	Status
-----	-----	-----	-----
dbwr2	kmod(x86_64)-20-28.0.3.el7	kmod(x86_64)-20-21	passed
dbwr1	kmod(x86_64)-20-28.0.3.el7	kmod(x86_64)-20-21	passed

Verifying Package: kmod-20-21 (x86_64) ...PASSED

Verifying Package: kmod-libs-20-21 (x86_64) ...

Node Name	Available	Required	Status
-----	-----	-----	-----
dbwr2	kmod-libs(x86_64)-20-28.0.3.el7	kmod-libs(x86_64)-20-21	passed
dbwr1	kmod-libs(x86_64)-20-28.0.3.el7	kmod-libs(x86_64)-20-21	passed

Verifying Package: kmod-libs-20-21 (x86_64) ...PASSED

Verifying Package: binutils-2.23.52.0.1 ...

Node Name	Available	Required	Status
-----------	-----------	----------	--------

dbwr2	binutils-2.27-44.base.0.1.el7	binutils-2.23.52.0.1	passed
-------	-------------------------------	----------------------	--------

dbwr1	binutils-2.27-44.base.0.1.el7	binutils-2.23.52.0.1	passed
-------	-------------------------------	----------------------	--------

Verifying Package: binutils-2.23.52.0.1 ...PASSED

Verifying Package: compat-libcap1-1.10 ...

Node Name	Available	Required	Status
-----------	-----------	----------	--------

dbwr2	compat-libcap1-1.10-7.el7	compat-libcap1-1.10	passed
-------	---------------------------	---------------------	--------

dbwr1	compat-libcap1-1.10-7.el7	compat-libcap1-1.10	passed
-------	---------------------------	---------------------	--------

Verifying Package: compat-libcap1-1.10 ...PASSED

Verifying Package: libgcc-4.8.2 (x86_64) ...

Node Name	Available	Required	Status
-----------	-----------	----------	--------

dbwr2	libgcc(x86_64)-4.8.5-44.0.3.el7	libgcc(x86_64)-4.8.2	passed
-------	---------------------------------	----------------------	--------

dbwr1	libgcc(x86_64)-4.8.5-44.0.3.el7	libgcc(x86_64)-4.8.2	passed
-------	---------------------------------	----------------------	--------

Verifying Package: libgcc-4.8.2 (x86_64) ...PASSED

Verifying Package: libstdc++-4.8.2 (x86_64) ...

Node Name	Available	Required	Status
-----------	-----------	----------	--------

dbwr2	libstdc++(x86_64)-4.8.5-44.0.3.el7	libstdc++(x86_64)-4.8.2	passed
-------	------------------------------------	-------------------------	--------

dbwr1	libstdc++(x86_64)-4.8.5-44.0.3.el7	libstdc++(x86_64)-4.8.2	passed
-------	------------------------------------	-------------------------	--------

Verifying Package: libstdc++-4.8.2 (x86_64) ...PASSED

Verifying Package: libstdc++-devel-4.8.2 (x86_64) ...

Node Name	Available	Required	Status
-----------	-----------	----------	--------

dbwr2	libstdc++-devel(x86_64)-4.8.5-44.0.3.el7	libstdc++-devel(x86_64)-4.8.2	passed
-------	--	-------------------------------	--------

dbwr1	libstdc++-devel(x86_64)-4.8.5-44.0.3.el7	libstdc++-devel(x86_64)-4.8.2	passed
-------	--	-------------------------------	--------

Verifying Package: libstdc++-devel-4.8.2 (x86_64) ...PASSED

Verifying Package: sysstat-10.1.5 ...

Node Name	Available	Required	Status
-----------	-----------	----------	--------

--	--	--	--

dbwr2	sysstat-10.1.5-19.el7	sysstat-10.1.5	passed
dbwr1	sysstat-10.1.5-19.el7	sysstat-10.1.5	passed

Verifying Package: sysstat-10.1.5 ...PASSED

Verifying Package: ksh ...

Node Name	Available	Required	Status
-----	-----	-----	-----
dbwr2	ksh	ksh	passed
dbwr1	ksh	ksh	passed

Verifying Package: ksh ...PASSED

Verifying Package: make-3.82 ...

Node Name	Available	Required	Status
-----	-----	-----	-----
dbwr2	make-3.82-24.el7	make-3.82	passed
dbwr1	make-3.82-24.el7	make-3.82	passed

Verifying Package: make-3.82 ...PASSED

Verifying Package: glibc-2.17 (x86_64) ...

Node Name	Available	Required	Status
-----	-----	-----	-----
dbwr2	glibc(x86_64)-2.17-317.0.1.el7	glibc(x86_64)-2.17	passed
dbwr1	glibc(x86_64)-2.17-317.0.1.el7	glibc(x86_64)-2.17	passed

Verifying Package: glibc-2.17 (x86_64) ...PASSED

Verifying Package: glibc-devel-2.17 (x86_64) ...

Node Name	Available	Required	Status
-----	-----	-----	-----
dbwr2	glibc-devel(x86_64)-2.17-317.0.1.el7	glibc-devel(x86_64)-2.17	passed
dbwr1	glibc-devel(x86_64)-2.17-317.0.1.el7	glibc-devel(x86_64)-2.17	passed

Verifying Package: glibc-devel-2.17 (x86_64) ...PASSED

Verifying Package: libaio-0.3.109 (x86_64) ...

Node Name	Available	Required	Status
-----	-----	-----	-----
dbwr2	libaio(x86_64)-0.3.109-13.el7	libaio(x86_64)-0.3.109	passed
dbwr1	libaio(x86_64)-0.3.109-13.el7	libaio(x86_64)-0.3.109	passed

Verifying Package: libaio-0.3.109 (x86_64) ...PASSED

Verifying Package: libaio-devel-0.3.109 (x86_64) ...

Node Name	Available	Required	Status
dbwr2	libaio-devel(x86_64)-0.3.109-13.el7	libaio-devel(x86_64)-0.3.109	passed
dbwr1	libaio-devel(x86_64)-0.3.109-13.el7	libaio-devel(x86_64)-0.3.109	passed

Verifying Package: libaio-devel-0.3.109 (x86_64) ...PASSED

Verifying Package: nfs-utils-1.2.3-15 ...

Node Name	Available	Required	Status
dbwr2	nfs-utils-1.3.0-0.68.0.1.el7	nfs-utils-1.2.3-15	passed
dbwr1	nfs-utils-1.3.0-0.68.0.1.el7	nfs-utils-1.2.3-15	passed

Verifying Package: nfs-utils-1.2.3-15 ...PASSED

Verifying Package: smartmontools-6.2-4 ...

Node Name	Available	Required	Status
dbwr2	smartmontools-7.0-2.el7	smartmontools-6.2-4	passed
dbwr1	smartmontools-7.0-2.el7	smartmontools-6.2-4	passed

Verifying Package: smartmontools-6.2-4 ...PASSED

Verifying Package: net-tools-2.0-0.17 ...

Node Name	Available	Required	Status
dbwr2	net-tools-2.0-0.25.20131004git.el7	net-tools-2.0-0.17	passed
dbwr1	net-tools-2.0-0.25.20131004git.el7	net-tools-2.0-0.17	passed

Verifying Package: net-tools-2.0-0.17 ...PASSED

Verifying Port Availability for component "Oracle Notification Service (ONS)" ...

Node Name	Port Number	Protocol	Available	Status
dbwr2	6200	TCP	yes	successful
dbwr1	6200	TCP	yes	successful
dbwr2	6100	TCP	yes	successful
dbwr1	6100	TCP	yes	successful

Verifying Port Availability for component "Oracle Notification Service (ONS)" ...PASSED

Verifying Port Availability for component "Oracle Cluster Synchronization Services (CSSD)" ...

Node Name	Port Number	Protocol	Available	Status
-----	-----	-----	-----	-----
dbwr2	42424	TCP	yes	successful
dbwr1	42424	TCP	yes	successful

Verifying Port Availability for component "Oracle Cluster Synchronization Services (CSSD)" ...PASSED

Verifying Users With Same UID: 0 ...PASSED

Verifying Current Group ID ...PASSED

Verifying Root user consistency ...

Node Name	Status
-----	-----
dbwr2	passed
dbwr1	passed

Verifying Root user consistency ...PASSED

Verifying Package: cvuqdisk-1.0.10-1 ...

Node Name	Available	Required	Status
-----	-----	-----	-----
dbwr2	cvuqdisk-1.0.10-1	cvuqdisk-1.0.10-1	passed
dbwr1	cvuqdisk-1.0.10-1	cvuqdisk-1.0.10-1	passed

Verifying Package: cvuqdisk-1.0.10-1 ...PASSED

Verifying Host name ...PASSED

Verifying Node Connectivity ...

Verifying Hosts File ...

Node Name	Status
-----	-----
dbwr1	passed
dbwr2	passed

Verifying Hosts File ...PASSED

Interface information for node "dbwr2"

Name	IP Address	Subnet	Gateway	Def. Gateway	HW Address
MTU					


```

-----
enp0s3 192.168.24.2      192.168.24.0      0.0.0.0      192.168.119.36  08:00:27:92:99:BD
1500
enp0s8 192.168.10.2      192.168.10.0      0.0.0.0      192.168.119.36  08:00:27:C1:C9:C6 1500
enp0s9 192.168.119.223 192.168.119.0      0.0.0.0      192.168.119.36  08:00:27:E2:42:8C 1500
virbr0 192.168.122.1      192.168.122.0      0.0.0.0      192.168.119.36  52:54:00:E9:CF:D6 1500

```

Interface information for node "dbwr1"

Name	IP Address	Subnet	Gateway	Def. Gateway	HW Address	MTU
enp0s3	192.168.24.1	192.168.24.0	0.0.0.0	192.168.119.36	08:00:27:F4:7B:FF	1500
enp0s8	192.168.10.1	192.168.10.0	0.0.0.0	192.168.119.36	08:00:27:B5:1C:97	1500
enp0s9	192.168.119.25	192.168.119.0	0.0.0.0	192.168.119.36	08:00:27:EB:B3:4C	1500
virbr0	192.168.122.1	192.168.122.0	0.0.0.0	192.168.119.36	52:54:00:E9:CF:D6	1500

Check: MTU consistency of the subnet "192.168.119.0".

Node	Name	IP Address	Subnet	MTU
dbwr2	enp0s9	192.168.119.223	192.168.119.0	1500
dbwr1	enp0s9	192.168.119.25	192.168.119.0	1500

Check: MTU consistency of the subnet "192.168.10.0".

Node	Name	IP Address	Subnet	MTU
dbwr2	enp0s8	192.168.10.2	192.168.10.0	1500
dbwr1	enp0s8	192.168.10.1	192.168.10.0	1500

Check: MTU consistency of the subnet "192.168.122.0".

Node	Name	IP Address	Subnet	MTU
dbwr2	virbr0	192.168.122.1	192.168.122.0	1500

dbwr1	virbr0	192.168.122.1	192.168.122.0	1500
-------	--------	---------------	---------------	------

Check: MTU consistency of the subnet "192.168.24.0".

Node	Name	IP Address	Subnet	MTU
------	------	------------	--------	-----

dbwr2	enp0s3	192.168.24.2	192.168.24.0	1500
-------	--------	--------------	--------------	------

dbwr1	enp0s3	192.168.24.1	192.168.24.0	1500
-------	--------	--------------	--------------	------

Verifying Check that maximum (MTU) size packet goes through subnet ...PASSED

Source	Destination	Connected?
--------	-------------	------------

dbwr1[enp0s9:192.168.119.25]	dbwr2[enp0s9:192.168.119.223]	yes
------------------------------	-------------------------------	-----

Source	Destination	Connected?
--------	-------------	------------

dbwr1[enp0s8:192.168.10.1]	dbwr2[enp0s8:192.168.10.2]	yes
----------------------------	----------------------------	-----

Source	Destination	Connected?
--------	-------------	------------

dbwr1[virbr0:192.168.122.1]	dbwr2[virbr0:192.168.122.1]	yes
-----------------------------	-----------------------------	-----

Source	Destination	Connected?
--------	-------------	------------

dbwr1[enp0s3:192.168.24.1]	dbwr2[enp0s3:192.168.24.2]	yes
----------------------------	----------------------------	-----

Verifying subnet mask consistency for subnet "192.168.119.0" ...PASSED

Verifying subnet mask consistency for subnet "192.168.10.0" ...PASSED

Verifying subnet mask consistency for subnet "192.168.122.0" ...PASSED

Verifying subnet mask consistency for subnet "192.168.24.0" ...PASSED

Verifying Node Connectivity ...FAILED (PRVG-1172, PRVG-11067, PRVG-11095)

Verifying Multicast or broadcast check ...

Checking subnet "192.168.119.0" for multicast communication with multicast group "224.0.0.251"

Verifying Multicast or broadcast check ...PASSED

Verifying Network Time Protocol (NTP) ...

Verifying '/etc/chrony.conf' ...

Node Name	File exists?
-----------	--------------

```

dbwr2                                yes
dbwr1                                yes
Verifying '/etc/chrony.conf' ...PASSED
Verifying Daemon 'chronyd' ...
Node Name                            Running?
-----
dbwr2                                yes
dbwr1                                yes
Verifying Daemon 'chronyd' ...PASSED
Verifying NTP daemon or service using UDP port 123 ...
Node Name                            Port Open?
-----
dbwr2                                yes
dbwr1                                yes
Verifying NTP daemon or service using UDP port 123 ...PASSED
Verifying chrony daemon is synchronized with at least one external time source ...PASSED
Verifying Network Time Protocol (NTP) ...PASSED
Verifying Same core file name pattern ...PASSED
Verifying User Mask ...
Node Name    Available    Required    Comment
-----
dbwr2        0022            0022        passed
dbwr1        0022            0022        passed
Verifying User Mask ...PASSED
Verifying User Not In Group "root": oracle ...
Node Name    Status    Comment
-----
dbwr2        passed    does not exist
dbwr1        passed    does not exist
Verifying User Not In Group "root": oracle ...PASSED
Verifying Time zone consistency ...PASSED
Verifying Time offset between nodes ...PASSED

```

Verifying resolv.conf Integrity ...

Node Name	Status
-----	-----
dbwr1	passed
dbwr2	passed

checking response for name "dbwr2" from each of the name servers specified in
"/etc/resolv.conf"

Node Name	Source	Comment	Status
-----	-----	-----	-----
dbwr2	192.168.24.1	IPv4	passed

checking response for name "dbwr1" from each of the name servers specified in
"/etc/resolv.conf"

Node Name	Source	Comment	Status
-----	-----	-----	-----
dbwr1	192.168.24.1	IPv4	passed

Verifying resolv.conf Integrity ...PASSED

Verifying DNS/NIS name service ...PASSED

Verifying Domain Sockets ...PASSED

Verifying /boot mount ...PASSED

Verifying Daemon "avahi-daemon" not configured and running ...

Node Name	Configured	Status
-----	-----	-----
dbwr2	no	passed
dbwr1	no	passed
Node Name	Running?	Status
-----	-----	-----
dbwr2	no	passed
dbwr1	no	passed

Verifying Daemon "avahi-daemon" not configured and running ...PASSED

Verifying Daemon "proxyt" not configured and running ...

Node Name	Configured	Status
dbwr2	no	passed
dbwr1	no	passed
Node Name	Running?	Status
dbwr2	no	passed
dbwr1	no	passed

Verifying Daemon "proxyt" not configured and running ...PASSED

Verifying User Equivalence ...PASSED

Verifying RPM Package Manager database ...INFORMATION (PRVG-11250)

Verifying /dev/shm mounted as temporary file system ...PASSED

Verifying File system mount options for path /var ...PASSED

Verifying DefaultTasksMax parameter ...PASSED

Verifying zeroconf check ...PASSED

Verifying ASM Filter Driver configuration ...PASSED

Verifying Systemd login manager IPC parameter ...PASSED

Pre-check for cluster services setup was unsuccessful on all the nodes.

Failures were encountered during execution of CVU verification request "stage -pre crsinst".

Verifying Node Connectivity ...FAILED

PRVG-1172 : The IP address "192.168.122.1" is on multiple interfaces "virbr0"

on nodes "dbwr2,dbwr1"

dbwr2: PRVG-11067 : TCP connectivity from node "dbwr2": "192.168.122.1" to node

"dbwr2": "192.168.122.1" failed.

PRVG-11095 : The TCP system call "connect" failed with error "111" while
executing exectask on node "dbwr2"

Connection refused

dbwr2: PRVG-11067 : TCP connectivity from node "dbwr2": "192.168.122.1" to node

"dbwr1": "192.168.122.1" failed.

PRVG-11095 : The TCP system call "connect" failed with error "111" while
executing exectask on node "dbwr2"

Connection refused

Verifying RPM Package Manager database ...INFORMATION

PRVG-11250 : The check "RPM Package Manager database" was not performed because it needs 'root' user privileges.

CVU operation performed: stage -pre crsinst

Date: Oct 18, 2024 3:04:09 PM

CVU home: /u01/app/19c/grid/

User: oracle

[oracle@dbwr1 grid]\$

Display Config and Start Grid Installation

In root:

```
su -  
export DISPLAY=:0.0  
xhost +
```

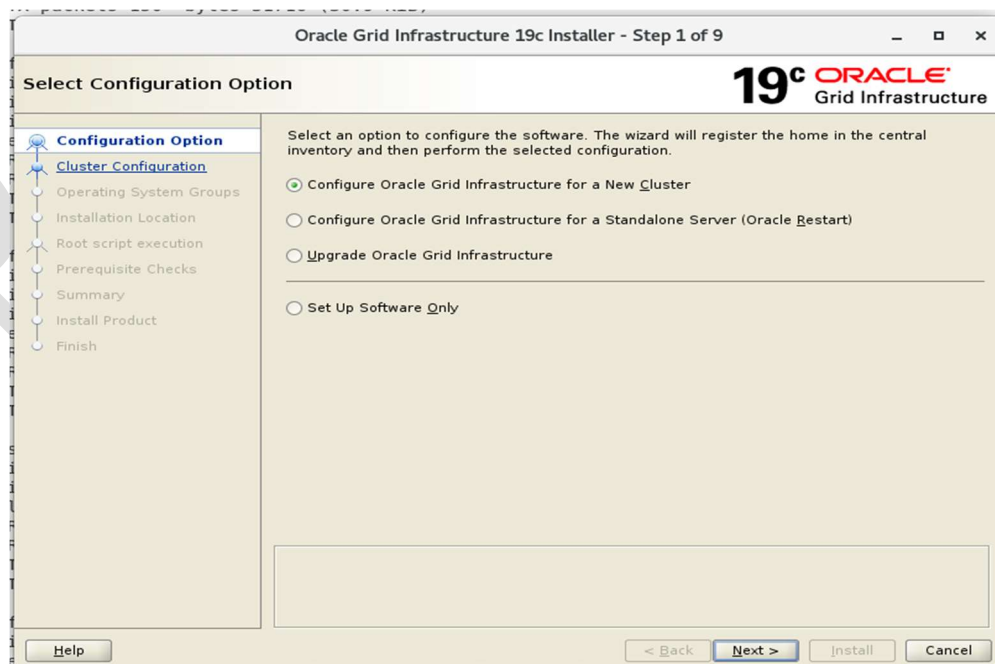
In oracle:

```
su - oracle  
[oracle@dbwr1 ~]$ DISPLAY=:0; export DISPLAY
```

Install and Configure Oracle 19c Grid Infrastructure for a Cluster

```
[oracle@dbwr1 ~]$ cd /u01/app/19c/grid/  
[oracle@dbwr1 grid]$ ./gridSetup.sh  
Launching Oracle Grid Infrastructure Setup Wizard...
```

Select the Configure Oracle Grid Infrastructure for a New Cluster option, then click the Next button.



Accept the Configure an Oracle Standalone Cluster option by clicking the Next button.

Oracle Grid Infrastructure 19c Installer - Step 2 of 9

19^c ORACLE[®]
Grid Infrastructure

Select Cluster Configuration

Configuration Option

Cluster Configuration

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Choose the required cluster configuration.

☒ Configure an Oracle Standalone Cluster

☐ Configure an Oracle Domain Services Cluster

☐ Configure an Oracle Member Cluster for Oracle Databases

☐ Configure an Oracle Member Cluster for Applications

Oracle Extended clusters are special purpose clusters that constitute nodes which span across multiple sites. Specify a minimum of 3 site names and a maximum of 5 (e.g., siteA, siteB, siteC).

☐ Configure as an Oracle Extended cluster

Site names:

Help < Back Next > Install Cancel

Enter the cluster name dbwr-cluster, SCAN name dbwr-scan and SCAN port 1521, then click the Next button.

Oracle Grid Infrastructure 19c Installer - Step 3 of 17

19^c ORACLE[®]
Grid Infrastructure

Grid Plug and Play Information

Configuration Option

Cluster Configuration

Grid Plug and Play

Cluster Node Information

Network Interface Usage

Storage Option

Create Grid Infrastructure Management Repository

Grid Infrastructure Management Repository

Create ASM Disk Group

ASM Password

Operating System Groups

Installation Location

Root script execution

Prerequisite Checks

Summary

Install Product

Finish

Single Client Access Name (SCAN) allows clients to use one name in connection strings to connect to the cluster as a whole. Client connect requests to the SCAN name can be handled by any

☒ Create Local SCAN

Cluster Name:

SCAN Name:

SCAN Port:

☐ Use Shared SCAN

SCAN Client Data: Browse...

☐ Configure GNS

☐ Configure nodes Virtual IPs as assigned by the Dynamic Networks

☒ Create a new GNS

GNS VIP Address:

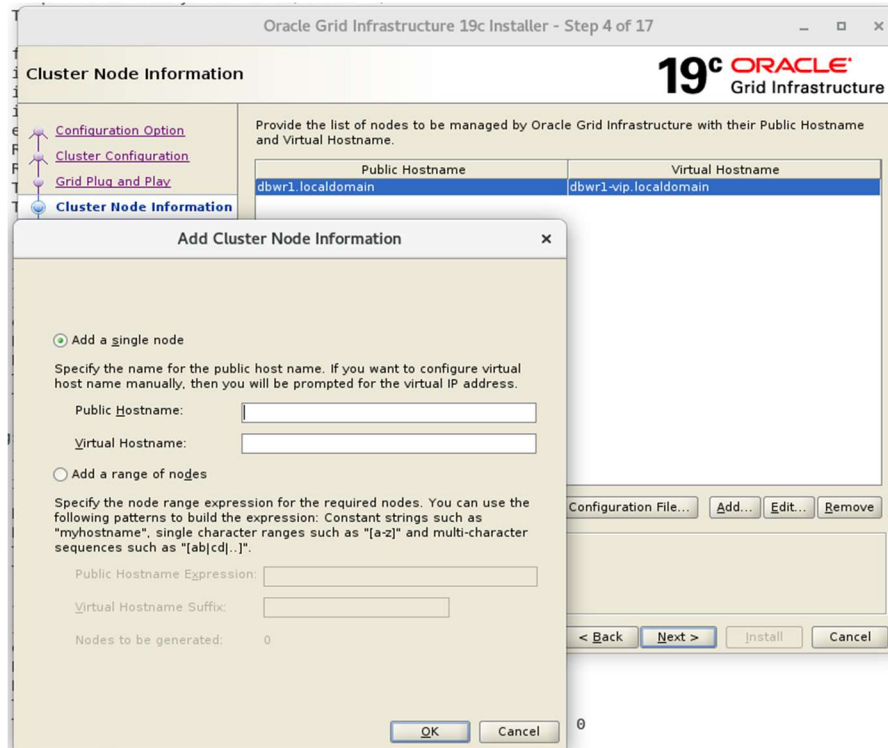
GNS Sub Domain:

☐ Use Shared GNS

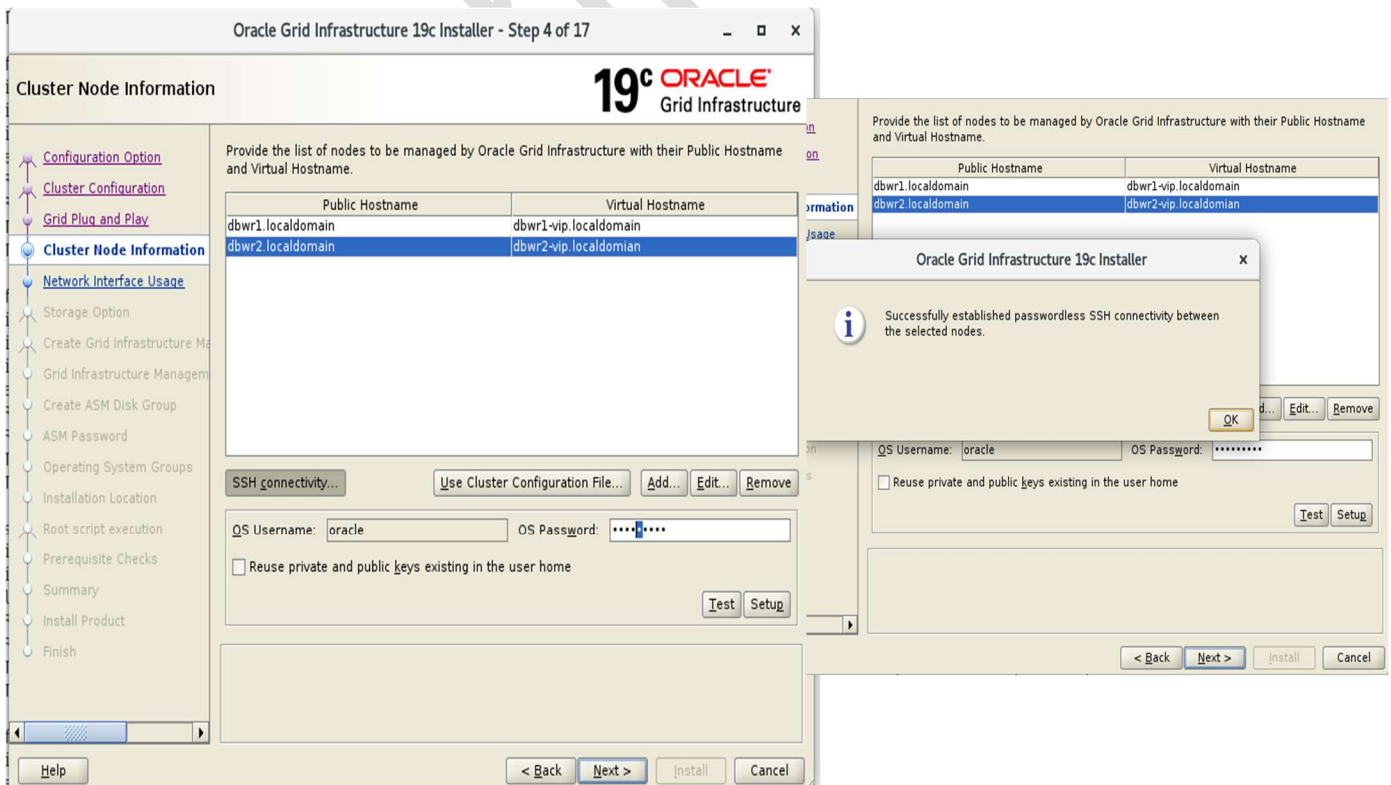
GNS Client Data: Browse...

Help < Back Next > Install Cancel

On the “Cluster Node Information” screen, click the Add button.
Enter the details of the second node in the cluster, then click the OK button.



Click the SSH connectivity button and enter the password for the oracle user. Click the Setup button to configure SSH connectivity, and the Test button to test it once it is complete. Once the test is complete, click the Next button.



Check the public and private networks are specified correctly as ASM & Private. Click the Next button.

Oracle Grid Infrastructure 19c Installer - Step 5 of 17

Specify Network Interface Usage

Private interfaces are used by Oracle Grid Infrastructure for internode traffic.

Interface Name	Subnet	Use for
enp0s3	192.168.24.0	Public
enp0s8	192.168.10.0	ASM & Private
enp0s9	192.168.214.0	Do Not Use
virbr0	192.168.122.0	Do Not Use

Navigation: Configuration Option, Cluster Configuration, Grid Plug and Play, Cluster Node Information, **Network Interface Usage**, Storage Option, Create Grid Infrastructure Management, Grid Infrastructure Management, Create ASM Disk Group, ASM Password, Operating System Groups, Installation Location, Root script execution, Prerequisite Checks, Summary, Install Product, Finish

Buttons: Help, < Back, Next >, Install, Cancel

Accept the Use Oracle Flex ASM for Storage option by clicking the Next button.

Oracle Grid Infrastructure 19c Installer - Step 6 of 17

Storage Option Information

You can place Oracle Cluster Registry (OCR) files and voting disk files on Oracle ASM storage, or on a file system.

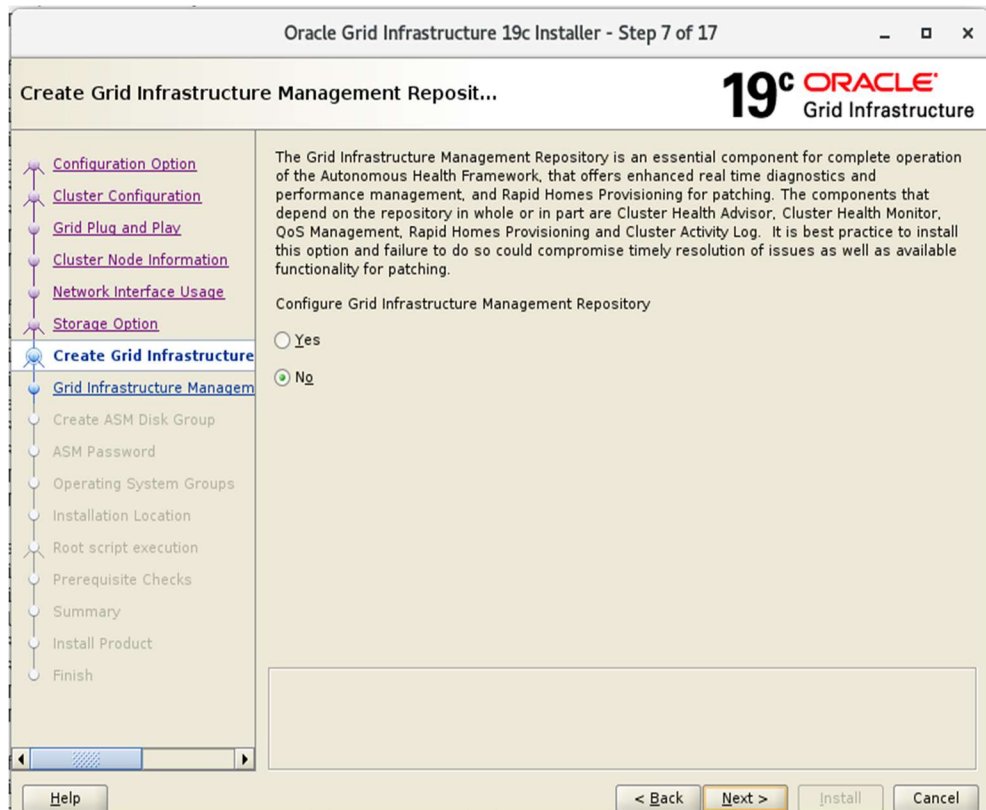
☒ Use Oracle Flex ASM for storage
Choose this option to configure OCR and voting disks on ASM storage. ASM instance will be configured on reduced number of cluster nodes.

☐ Use Shared File System
Choose this option to configure OCR and voting disk files on an existing shared file system.

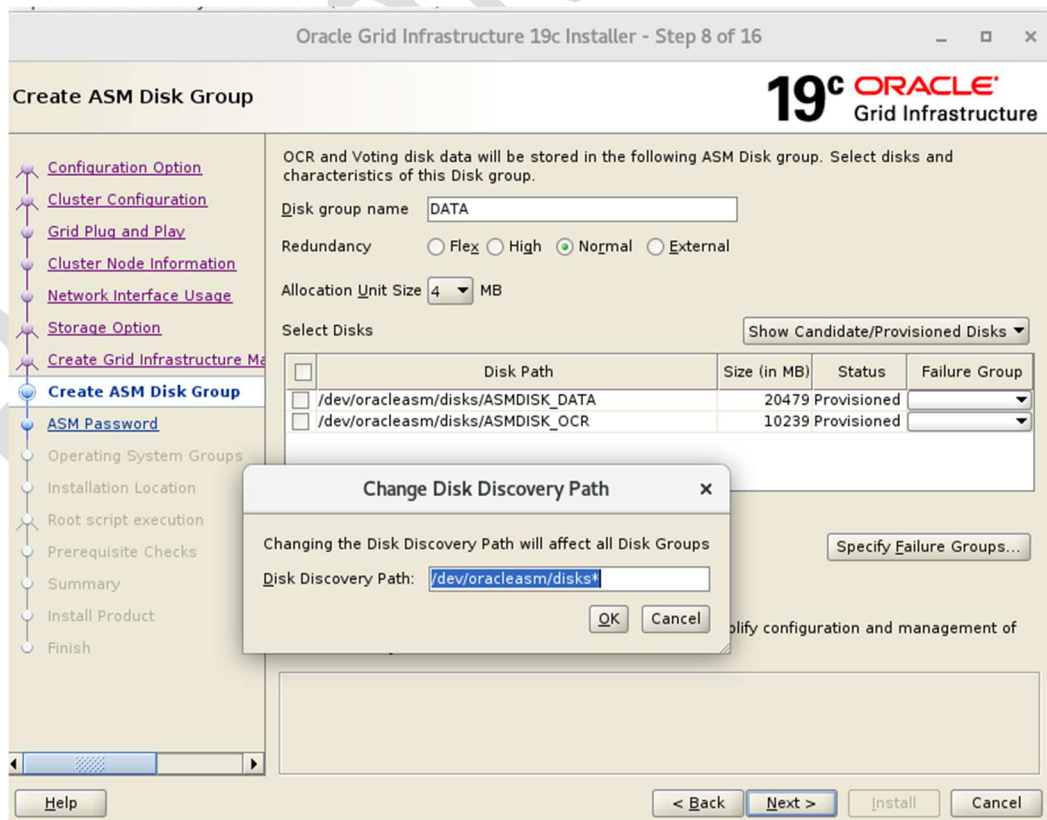
Navigation: Configuration Option, Cluster Configuration, Grid Plug and Play, Cluster Node Information, Network Interface Usage, **Storage Option**, Create Grid Infrastructure Management, Grid Infrastructure Management, Create ASM Disk Group, ASM Password, Operating System Groups, Installation Location, Root script execution, Prerequisite Checks, Summary, Install Product, Finish

Buttons: Help, < Back, Next >, Install, Cancel

Select the No option, as we don't want to create a separate disk group for the GIMR in this case. Click the Next button.



First Browse the path /dev/oracleasm/disks* using change discovery path



Set the redundancy to External, click the 10GB DISK for OCR configuration, then click the Next button.

Oracle Grid Infrastructure 19c Installer - Step 8 of 16

Create ASM Disk Group

OCR and Voting disk data will be stored in the following ASM Disk group. Select disks and characteristics of this Disk group.

Disk group name:

Redundancy: ☐ Flexible ☐ High ☐ Normal ☒ External

Allocation Unit Size: MB

Select Disks:

	Disk Path	Size (in MB)	Status
☒	/dev/oracleasm/disks/ASMDISK_DATA	20479	Provisioned
☐	/dev/oracleasm/disks/ASMDISK_OCR	10239	Provisioned

Disk Discovery Path: /dev/oracleasm/disks*

☐ Configure Oracle ASM Filter Driver
Select this option to configure ASM Filter Driver (AFD) to simplify configuration and management of disk devices by Oracle ASM.

Enter the credentials and click the Next button.

Oracle Grid Infrastructure 19c Installer - Step 9 of 16

Specify ASM Password

The new Oracle Automatic Storage Management (Oracle ASM) instance requires its own SYS user with SYSASM privileges for administration. Oracle recommends that you create a less privileged ASMSNMP user with SYSDBA privileges to monitor the ASM instance.

Specify the password for these user accounts.

☐ Use different passwords for these accounts

	Password	Confirm Password
SYS		
ASMSNMP		

☒ Use same passwords for these accounts

Specify Password: Confirm Password:

Accept the default IPMI option by clicking the Next button.

The screenshot shows the 'Failure Isolation Support' window in the Oracle Grid Infrastructure 19c Installer. The title bar indicates 'Step 10 of 18'. The left sidebar contains a list of configuration steps, with 'Failure Isolation' selected. The main area prompts the user to 'Choose one of the following Failure Isolation Support options.' There are two radio button options: 'Use Intelligent Platform Management Interface (IPMI)' and 'Do not use Intelligent Platform Management Interface (IPMI)'. The second option is selected. Below the first option, there are input fields for 'User Name' and 'Password'. At the bottom, there are buttons for '< Back', 'Next >', 'Install', and 'Cancel'.

Deselect EM. Click the Next button.

The screenshot shows the 'Specify Management Options' window in the Oracle Grid Infrastructure 19c Installer. The title bar indicates 'Step 11 of 18'. The left sidebar shows 'Management Options' selected. The main area explains that the user can configure the instance to be managed by Enterprise Manager Cloud Control. A checkbox labeled 'Register with Enterprise Manager (EM) Cloud Control' is present and is currently unchecked. Below this checkbox are input fields for 'OMS host', 'OMS port', 'EM Admin User Name', and 'EM Admin Password'. At the bottom, there are buttons for '< Back', 'Next >', 'Install', and 'Cancel'.

Set the groups to oinstall and click the Next button. Accept the warnings on the subsequent dialog by clicking the Yes button.

Oracle Grid Infrastructure 19c Installer - Step 12 of 18

Privileged Operating System Groups

Select the name of the operating system group, that you want to use for operating system authentication to Oracle Automatic Storage Management.

Oracle ASM Administrator (OSASM) Group:

Oracle ASM DBA (OSDBA for ASM) Group:

Oracle ASM Operator (OSOPER for ASM) Group (Optional):

Buttons: Help, < Back, Next >, Install, Cancel

Click the Next button.

Oracle Grid Infrastructure 19c Installer - Step 13 of 18

Specify Installation Location

Specify the Oracle base. The Oracle base directory for the Oracle Grid Infrastructure installation is the location where diagnostic and administrative logs, and other logs associated with Oracle ASM and Oracle Clusterware are stored. This location would also contain files pertaining to the configuration of Oracle Clusterware.

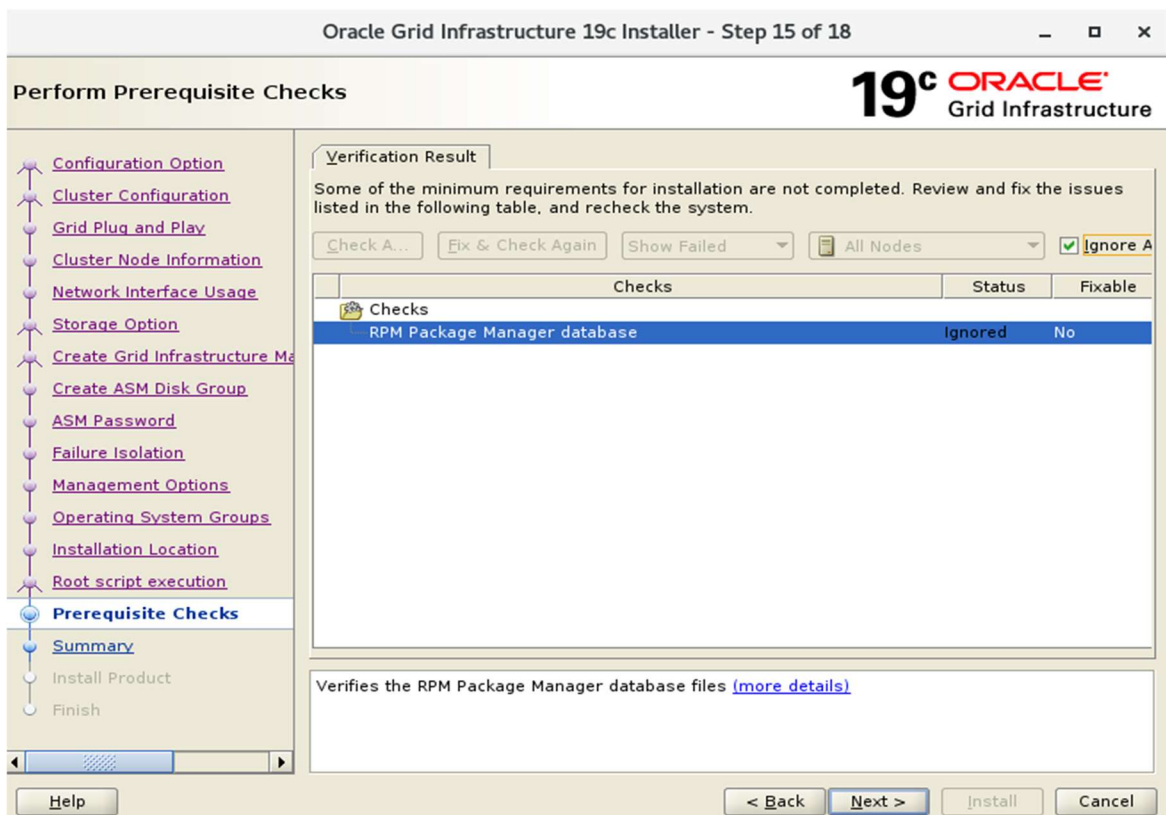
Oracle base: Browse...

This software directory is the Oracle Grid Infrastructure home directory.

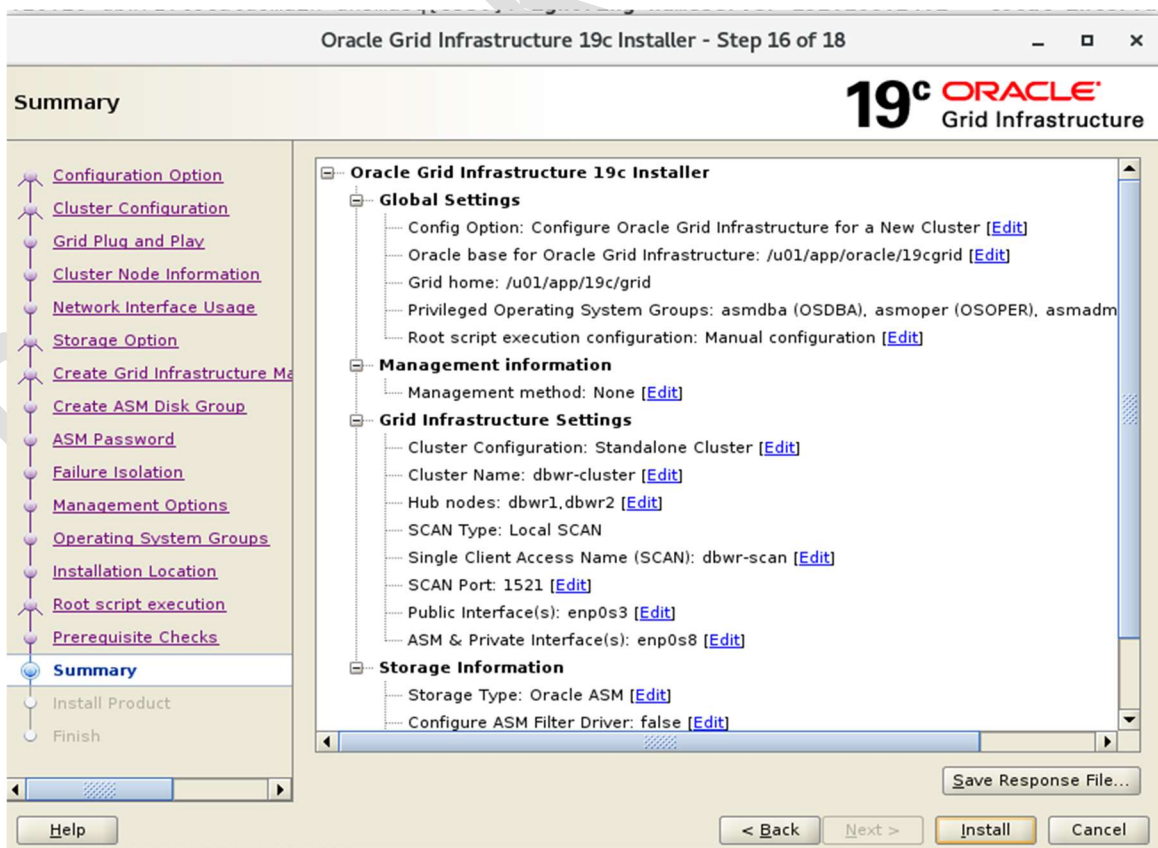
Software location: /u01/app/19c/grid

Buttons: Help, < Back, Next >, Install, Cancel

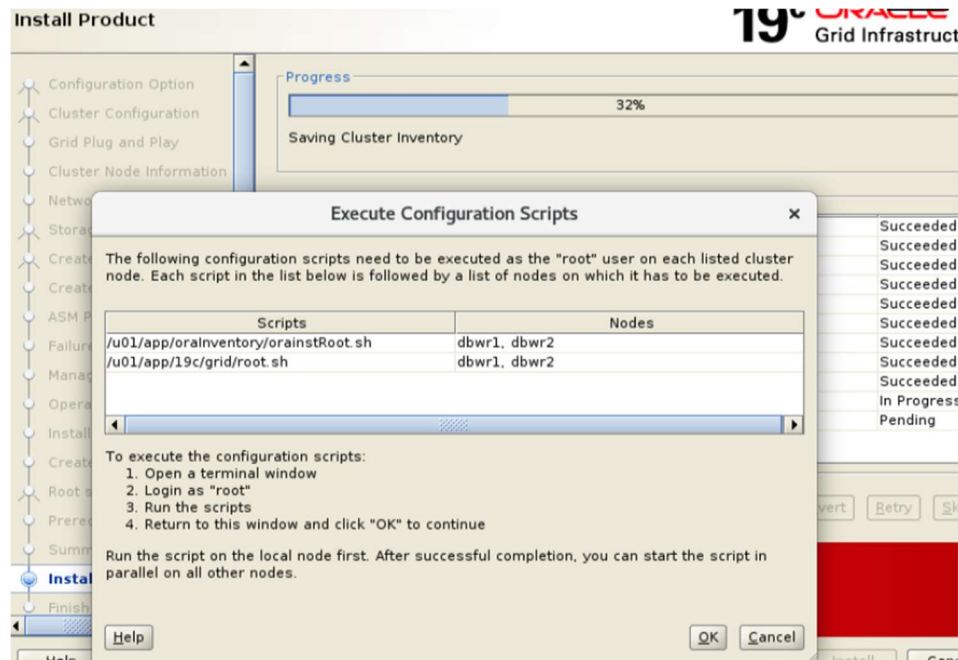
Accept the default inventory directory by click the Next button



Double check and click the Install button.



When prompted, run the configuration scripts on each node one by one



Node 1

```
[root@dbwr1 grid]# /u01/app/oraInventory/orainstRoot.sh
```

Changing permissions of /u01/app/oraInventory.

Adding read,write permissions for group.

Removing read,write,execute permissions for world.

Changing groupname of /u01/app/oraInventory to oinstall.

The execution of the script is complete.

Node 2

```
[root@dbwr2 oracle]#
```

```
[root@dbwr2 oracle]# /u01/app/oraInventory/orainstRoot.sh
```

Changing permissions of /u01/app/oraInventory.

Adding read,write permissions for group.

Removing read,write,execute permissions for world.

Changing groupname of /u01/app/oraInventory to oinstall.

The execution of the script is complete.

Node 1

```
dbwr1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Applications Places Terminal

root@dbwr1:~

File Edit View Search Terminal Help
[root@dbwr1 ~]# /u01/app/19c/grid/root.sh
Performing root user operation.

The following environment variables are set as:
    ORACLE_OWNER= oracle
    ORACLE_HOME= /u01/app/19c/grid

Enter the full pathname of the local bin directory: [/usr/local/bin]:
    Copying dbhome to /usr/local/bin ...
    Copying oraenv to /usr/local/bin ...
    Copying coraenv to /usr/local/bin ...

Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file: /u01/app/19c/grid/crs/install/crsconfig_params
The log of current session can be found at:
    /u01/app/oracle/19cgrid/crsdata/dbwr1/crsconfig/rootcrs_dbwr1_2024-10-16_04-56-09PM.log
2024/10/16 16:56:31 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.
2024/10/16 16:56:31 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.
2024/10/16 16:56:31 CLSRSC-363: User ignored prerequisites during installation
2024/10/16 16:56:32 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.
2024/10/16 16:56:35 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.
2024/10/16 16:56:37 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.
2024/10/16 16:56:37 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.
2024/10/16 16:56:38 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.
2024/10/16 16:56:57 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.
2024/10/16 16:57:04 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.
2024/10/16 16:57:04 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.
2024/10/16 16:57:23 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.
2024/10/16 16:57:23 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.
2024/10/16 16:57:31 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.
2024/10/16 16:57:31 CLSRSC-330: Adding Clusterware entries to file 'oracle-ohasd.service'
2024/10/16 16:58:05 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.
2024/10/16 16:58:13 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.
2024/10/16 16:58:21 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.
2024/10/16 16:58:29 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.

ASM has been created and started successfully.

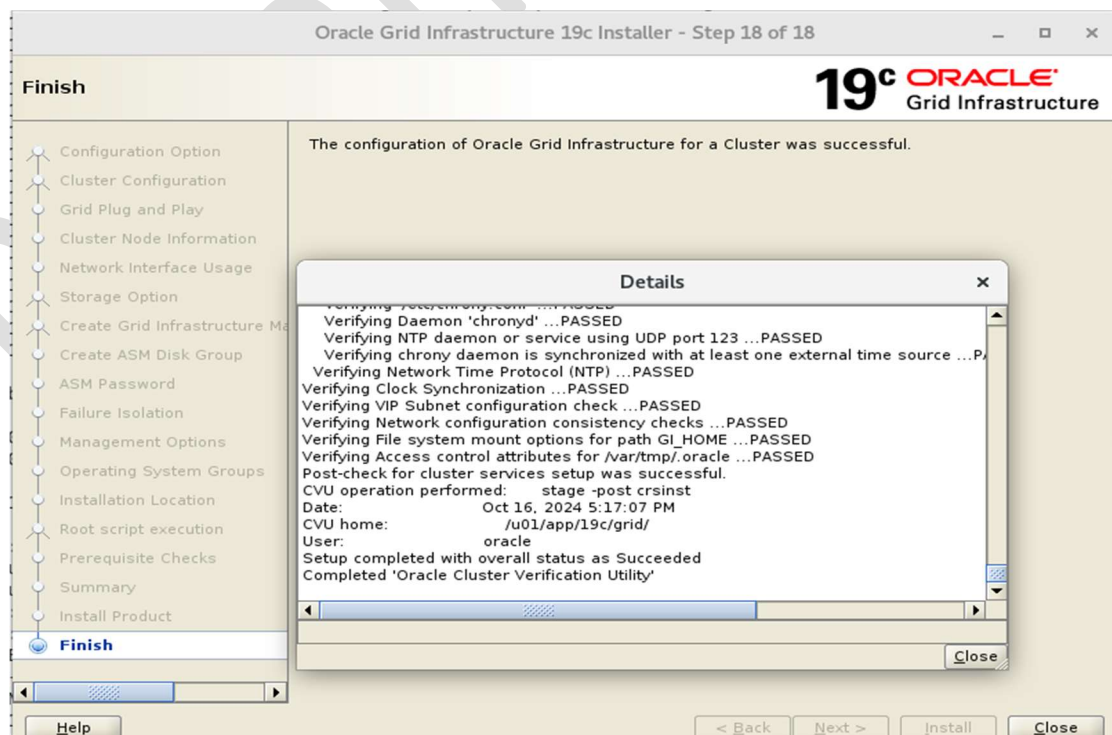
[DBT-30001] Disk groups created successfully. Check /u01/app/oracle/19cgrid/cfgtoollogs/asmca/asmca-241016PM045906.log for details.

2024/10/16 17:00:53 CLSRSC-482: Running command: '/u01/app/19c/grid/bin/ocrconfig -upgrade oracle oinstall'
CRS-4256: Updating the profile
Successful addition of voting disk 4ee44d9090b14f20bf684bd5f33d0fe8.
Successfully replaced voting disk group with +DATA.
CRS-4256: Updating the profile
CRS-4266: Voting file(s) successfully replaced
## STATE File Universal Id File Name Disk group
--  ----
1. ONLINE 4ee44d9090b14f20bf684bd5f33d0fe8 (/dev/oracleasm/disks/ASMDISK_DATA) [DATA]
Located 1 voting disk(s).
2024/10/16 17:03:03 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.
2024/10/16 17:04:29 CLSRSC-343: Successfully started Oracle Clusterware stack
2024/10/16 17:04:29 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.
2024/10/16 17:07:07 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.
2024/10/16 17:07:49 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded
[root@dbwr1 ~]#
[root@dbwr1 ~]#
```


Node 2

```
root@dbwr2:~  
[root@dbwr2 ~]# /u01/app/19c/grid/root.sh  
Performing root user operation.  
  
The following environment variables are set as:  
ORACLE_OWNER= oracle  
ORACLE_HOME= /u01/app/19c/grid  
  
Enter the full pathname of the local bin directory: [/usr/local/bin]:  
Copying dbhome to /usr/local/bin ...  
Copying oraenv to /usr/local/bin ...  
Copying coraenv to /usr/local/bin ...  
  
Creating /etc/oratab file...  
Entries will be added to the /etc/oratab file as needed by  
Database Configuration Assistant when a database is created  
Finished running generic part of root script.  
Now product-specific root actions will be performed.  
Relinking oracle with rac on option  
Using configuration parameter file: /u01/app/19c/grid/crs/install/crsconfig_params  
The log of current session can be found at:  
/u01/app/oracle/19cgrid/crsdata/dbwr2/crsconfig/rootcrs_dbwr2 2024-10-16 05-11-58PM.log  
2024/10/16 17:12:15 CLSRSC-594: Executing installation step 1 of 19: 'SetupTFA'.  
2024/10/16 17:12:15 CLSRSC-594: Executing installation step 2 of 19: 'ValidateEnv'.  
2024/10/16 17:12:15 CLSRSC-363: User ignored prerequisites during installation  
2024/10/16 17:12:15 CLSRSC-594: Executing installation step 3 of 19: 'CheckFirstNode'.  
2024/10/16 17:12:18 CLSRSC-594: Executing installation step 4 of 19: 'GenSiteGUIDs'.  
2024/10/16 17:12:18 CLSRSC-594: Executing installation step 5 of 19: 'SetupOSD'.  
2024/10/16 17:12:18 CLSRSC-594: Executing installation step 6 of 19: 'CheckCRSConfig'.  
2024/10/16 17:12:19 CLSRSC-594: Executing installation step 7 of 19: 'SetupLocalGPNP'.  
2024/10/16 17:12:21 CLSRSC-594: Executing installation step 8 of 19: 'CreateRootCert'.  
2024/10/16 17:12:22 CLSRSC-594: Executing installation step 9 of 19: 'ConfigOLR'.  
2024/10/16 17:12:32 CLSRSC-594: Executing installation step 10 of 19: 'ConfigCHMOS'.  
2024/10/16 17:12:32 CLSRSC-594: Executing installation step 11 of 19: 'CreateOHASD'.  
2024/10/16 17:12:34 CLSRSC-594: Executing installation step 12 of 19: 'ConfigOHASD'.  
2024/10/16 17:12:35 CLSRSC-330: Adding Clusterware entries to file 'oracle-ohasd.service'  
2024/10/16 17:12:53 CLSRSC-4002: Successfully installed Oracle Trace File Analyzer (TFA) Collector.  
2024/10/16 17:13:06 CLSRSC-594: Executing installation step 13 of 19: 'InstallAFD'.  
2024/10/16 17:13:09 CLSRSC-594: Executing installation step 14 of 19: 'InstallACFS'.  
2024/10/16 17:13:12 CLSRSC-594: Executing installation step 15 of 19: 'InstallKA'.  
2024/10/16 17:13:14 CLSRSC-594: Executing installation step 16 of 19: 'InitConfig'.  
2024/10/16 17:13:32 CLSRSC-594: Executing installation step 17 of 19: 'StartCluster'.  
2024/10/16 17:14:43 CLSRSC-343: Successfully started Oracle Clusterware stack  
2024/10/16 17:14:44 CLSRSC-594: Executing installation step 18 of 19: 'ConfigNode'.  
2024/10/16 17:15:13 CLSRSC-594: Executing installation step 19 of 19: 'PostConfig'.  
2024/10/16 17:15:28 CLSRSC-325: Configure Oracle Grid Infrastructure for a Cluster ... succeeded
```

Grid Installation is complete. Ignore the NTP error.



Install and Configure Oracle 19c Software binary

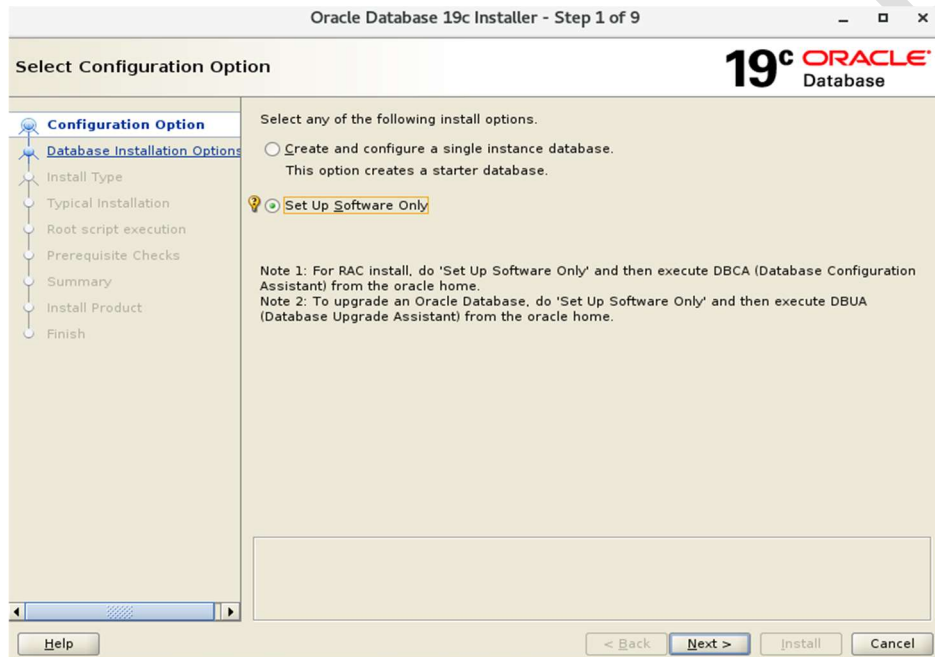
[oracle@dbwr1]\$. db.env

```
oracle@dbwr1:u01/app/oracle/product/19c/db_1
[oracle@dbwr1 db_1]$ ll
total 2988000
-rwxr-x---. 1 oracle oinstall 3059705302 Oct 17 11:15 LINUX.X64_193000_0
[oracle@dbwr1 db_1]$
[oracle@dbwr1 db_1]$ unzip LINUX.X64_193000_db_home.zip
```

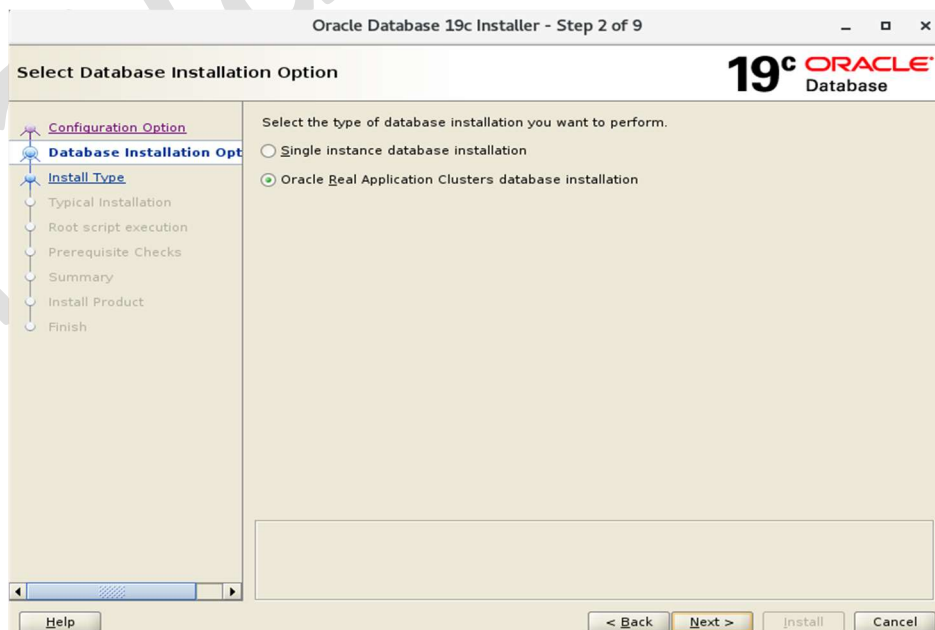
```
[oracle@dbwr1 db_1]$ export DISPLAY=:1;
[oracle@dbwr1 db_1]$ ./runInstaller
```

Launching Oracle Database Setup Wizard...

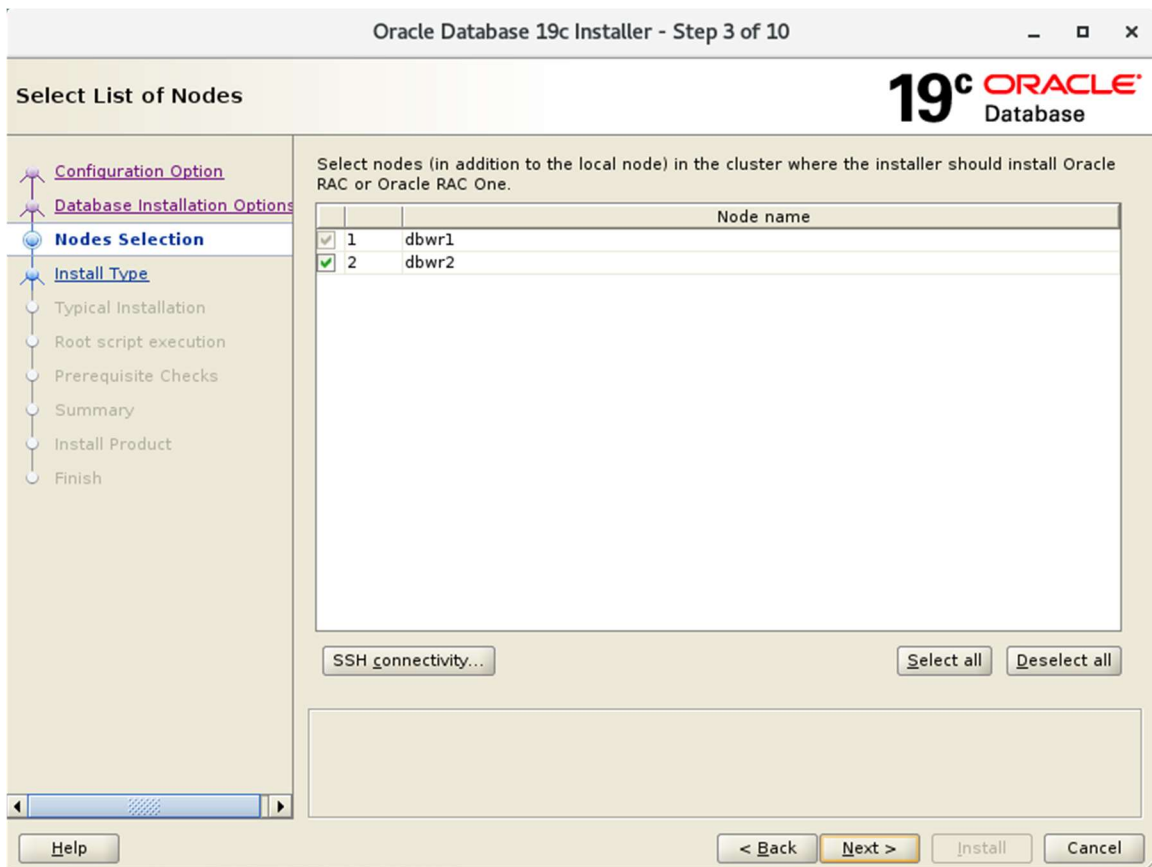
Select the setup software only option, then click the Next button.



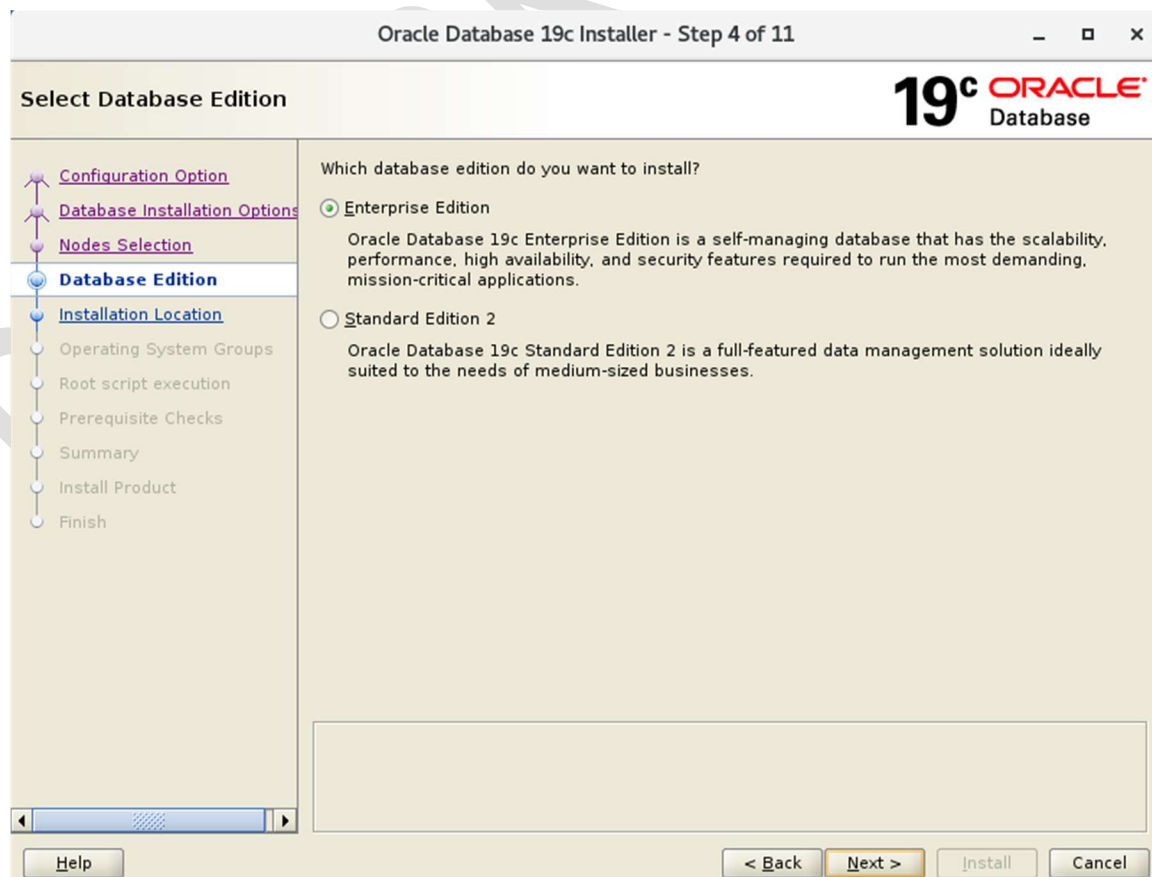
Accept the Oracle Real Application Clusters database installation option by clicking the Next button.



Make sure both nodes are selected, then click the Next button.



Select the Enterprise Edition option, then click the Next button.



Enter /u01/app/oracle as the Oracle base and /u01/app/oracle/product/19c/dbhome_1 as the software location, then click the Next button.

The screenshot shows the 'Specify Installation Location' window of the Oracle Database 19c Installer. The window title is 'Oracle Database 19c Installer - Step 5 of 11'. On the left is a navigation pane with links: Configuration Option, Database Installation Options, Nodes Selection, Database Edition, Installation Location (selected), Operating System Groups, Root script execution, Prerequisite Checks, Summary, Install Product, and Finish. The main area contains the following text: 'Specify a path to place all Oracle software and configuration-related files installed by this installation owner. This location is the Oracle base directory for the installation owner.' Below this, 'Oracle base:' is followed by a text box containing '/u01/app/oracle' and a 'Browse...' button. Further down, it states: 'This software directory is the Oracle Database home directory. Software location: /u01/app/oracle/product/19c/db_1'. At the bottom are buttons for '< Back', 'Next >', 'Install', and 'Cancel'. A 'Help' button is in the bottom left.

Click the Next button. Accept the warnings on the subsequent dialog by clicking the Yes button

The screenshot shows the 'Privileged Operating System groups' window of the Oracle Database 19c Installer. The window title is 'Oracle Database 19c Installer - Step 6 of 11'. The navigation pane on the left is the same as in Step 5, with 'Operating System Groups' selected. The main area contains the following text: 'SYS privileges are required to create a database using operating system (OS) authentication. Membership in OS Groups grants the corresponding SYS privilege, eg. membership in OSDBA grants the SYSDBA privilege.' Below this, several OS groups are listed with dropdown menus: 'Database Administrator (OSDBA) group:' (asmdba), 'Database Operator (OSOPER) group (Optional):' (oinstall), 'Database Backup and Recovery (OSBACKUPDBA) group:' (asmdba), 'Data Guard administrative (OSDGDBA) group:' (asmdba), 'Encryption Key Management administrative (OSKMDBA) group:' (asmdba), and 'Real Application Cluster administrative (OSRACDBA) group:' (asmdba). At the bottom are buttons for '< Back', 'Next >', 'Install', and 'Cancel'. A 'Help' button is in the bottom left.

Click the Next button.

Oracle Database 19c Installer - Step 7 of 11

19c ORACLE Database

Root script execution configuration

During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.

☐ Automatically run configuration scripts

☒ Use "root" user credential

Password :

☐ Use sudo

Program path :

User name :

Password :

Check the "Ignore All" checkbox and click the "Next" button.

Oracle Database 19c Installer - Step 7 of 11

19c ORACLE Database

Root script execution configuration

During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.

☒ Automatically run configuration scripts

☒ Use "root" user credential

Password :

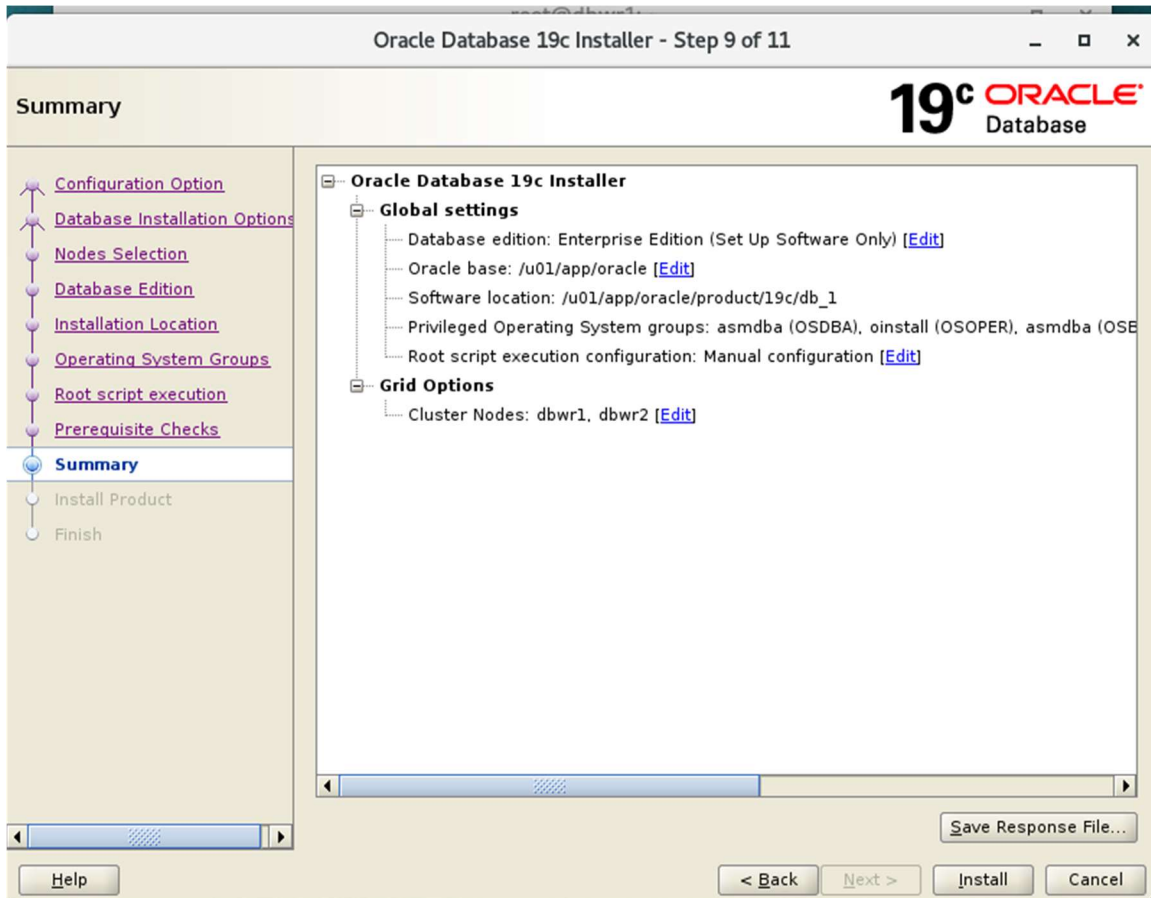
☐ Use sudo

Program path :

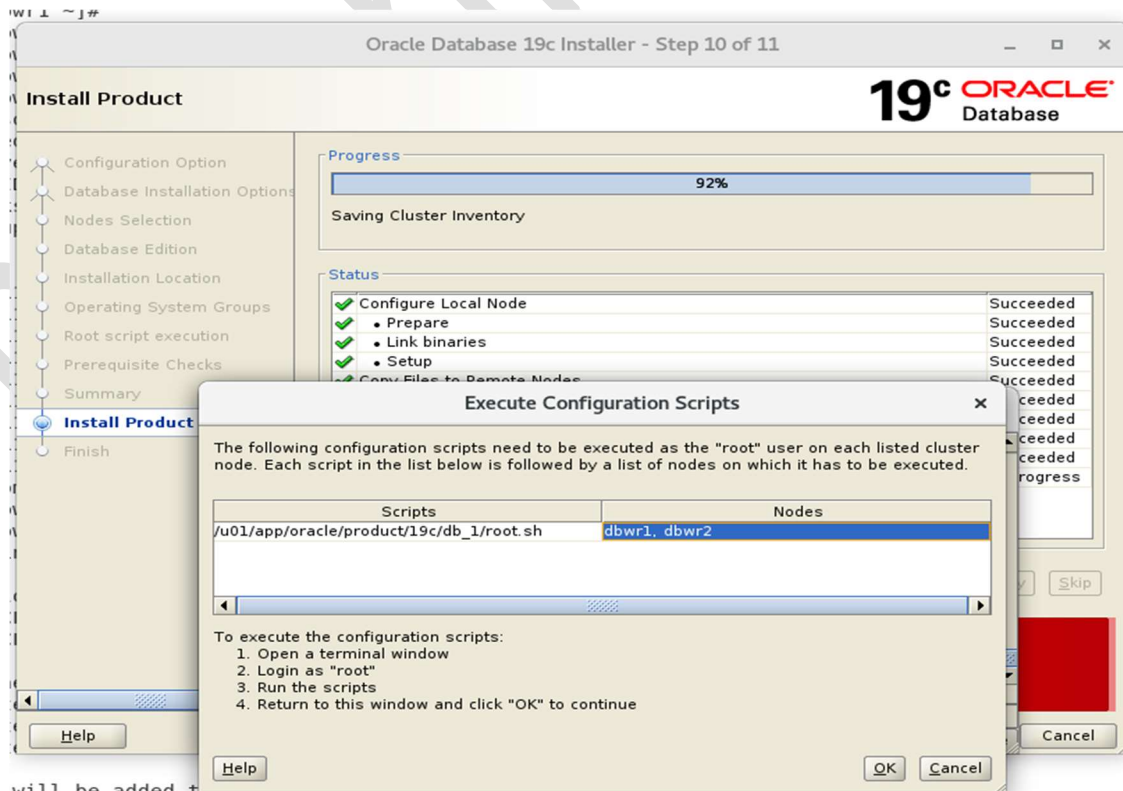
User name :

Password :

Click the Install button.



When prompted, run the configuration script on each node. When the scripts have been run on each node, click the OK button.



Node 1

```
[root@dbwr1 db_1]# /u01/app/oracle/product/19c/db_1/root.sh
```

Performing root user operation.

The following environment variables are set as:

```
ORACLE_OWNER= oracle
```

```
ORACLE_HOME= /u01/app/oracle/product/19c/db_1
```

Enter the full pathname of the local bin directory: [/usr/local/bin]:

The contents of "dbhome" have not changed. No need to overwrite.

The contents of "oraenv" have not changed. No need to overwrite.

The contents of "coraenv" have not changed. No need to overwrite.

Entries will be added to the /etc/oratab file as needed by

Database Configuration Assistant when a database is created

Finished running generic part of root script.

Now product-specific root actions will be performed.

Node 2

```
[root@dbwr2 db_1]# /u01/app/oracle/product/19c/db_1/root.sh
```

Performing root user operation.

The following environment variables are set as:

```
ORACLE_OWNER= oracle
```

```
ORACLE_HOME= /u01/app/oracle/product/19c/db_1
```

Enter the full pathname of the local bin directory: [/usr/local/bin]:

The contents of "dbhome" have not changed. No need to overwrite.

The contents of "oraenv" have not changed. No need to overwrite.

The contents of "coraenv" have not changed. No need to overwrite.

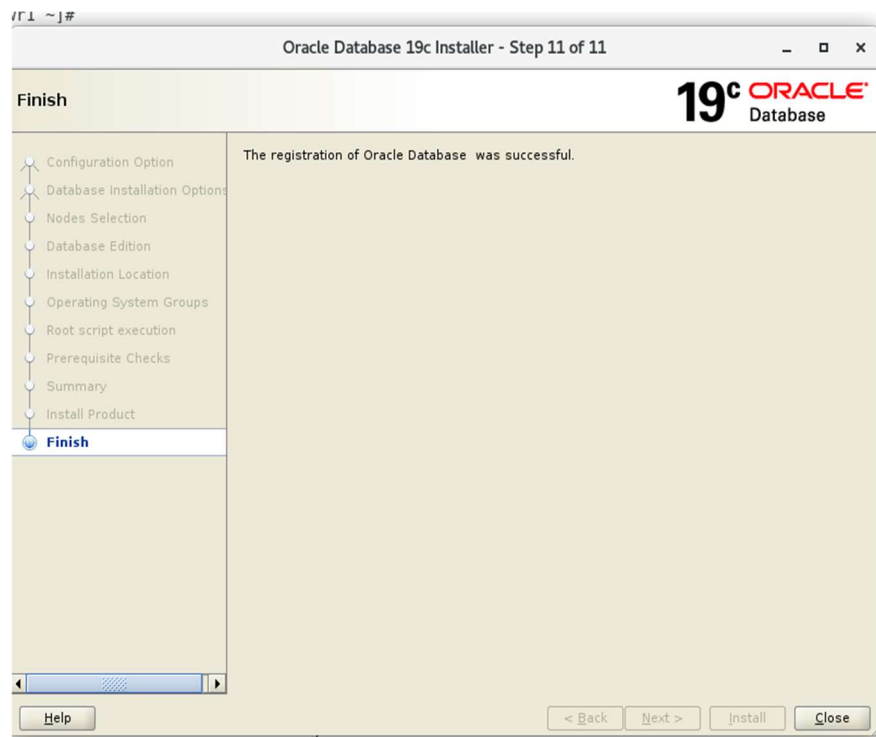
Entries will be added to the /etc/oratab file as needed by

Database Configuration Assistant when a database is created

Finished running generic part of root script.

Now product-specific root actions will be performed.

Oracle 19c Installation is completed.

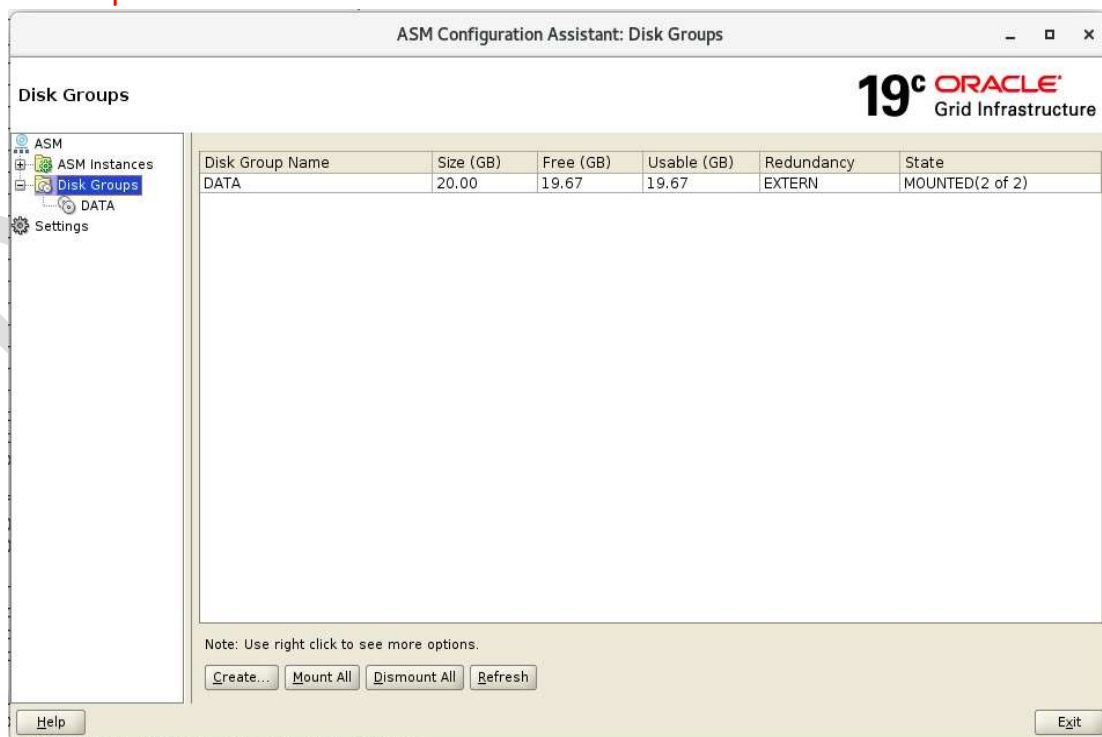


Creating ASM Diskgroup using ASMCA Tool for database creation

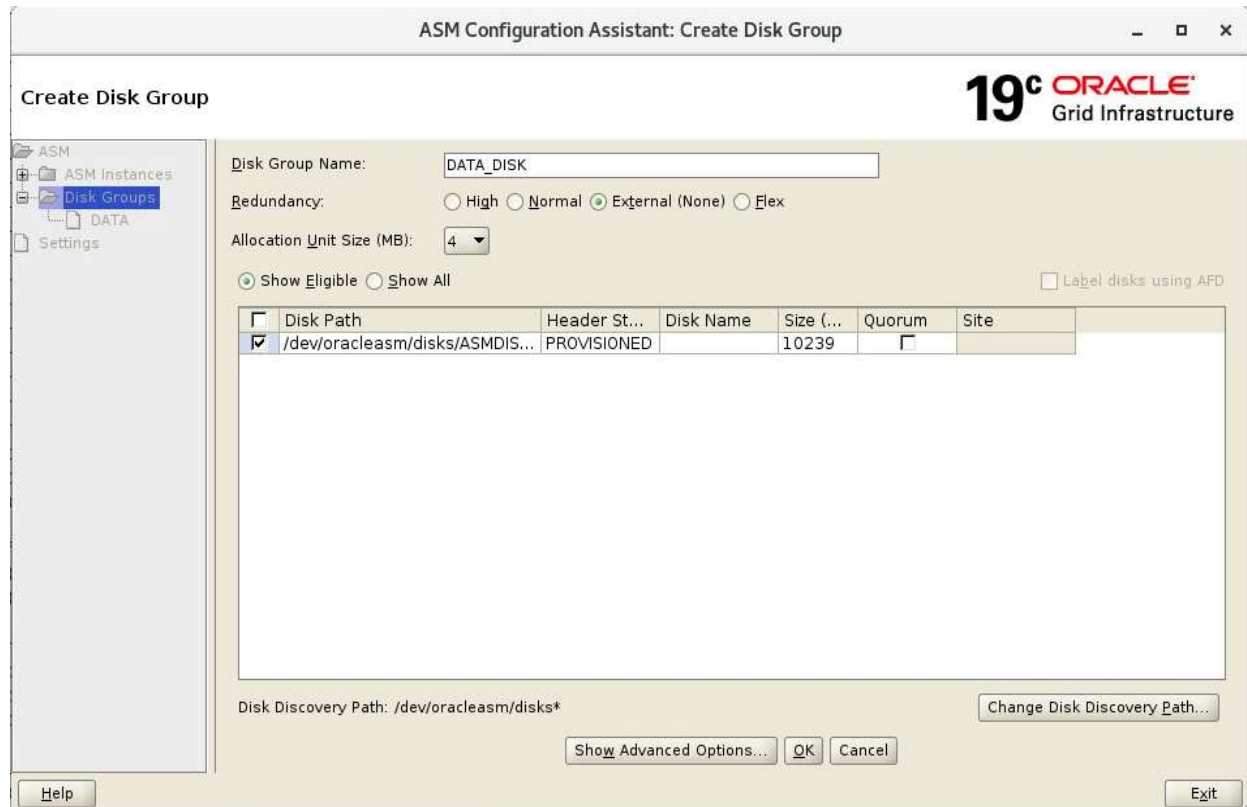
```
[oracle@dbwr1 ~]$ . grid.env
```

```
[oracle@dbwr1 ~]$ asmca
```

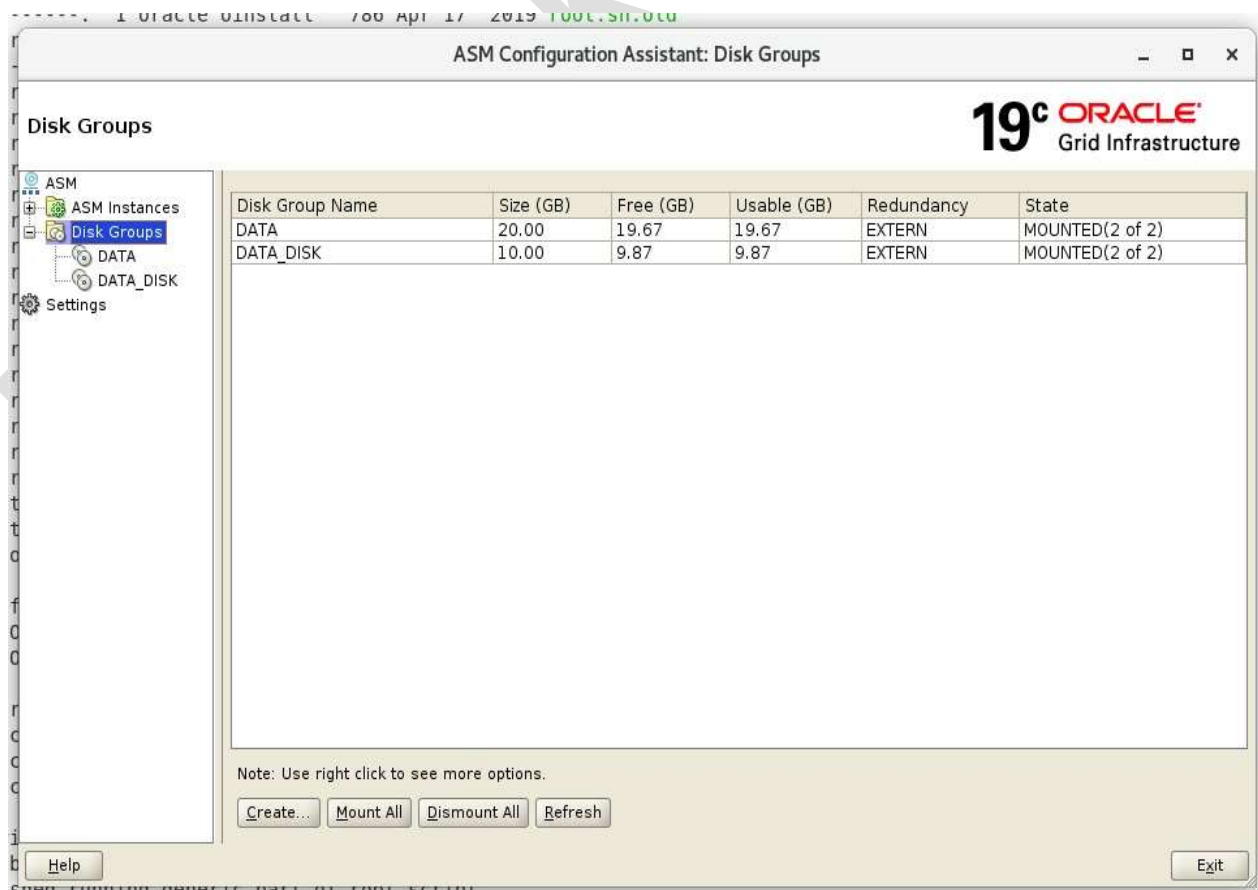
Select Disk Groups and create



Set the redundancy to External, click the 35GB DISK for Database configuration, then click the ok button.



Disk Group is ready for creating Database

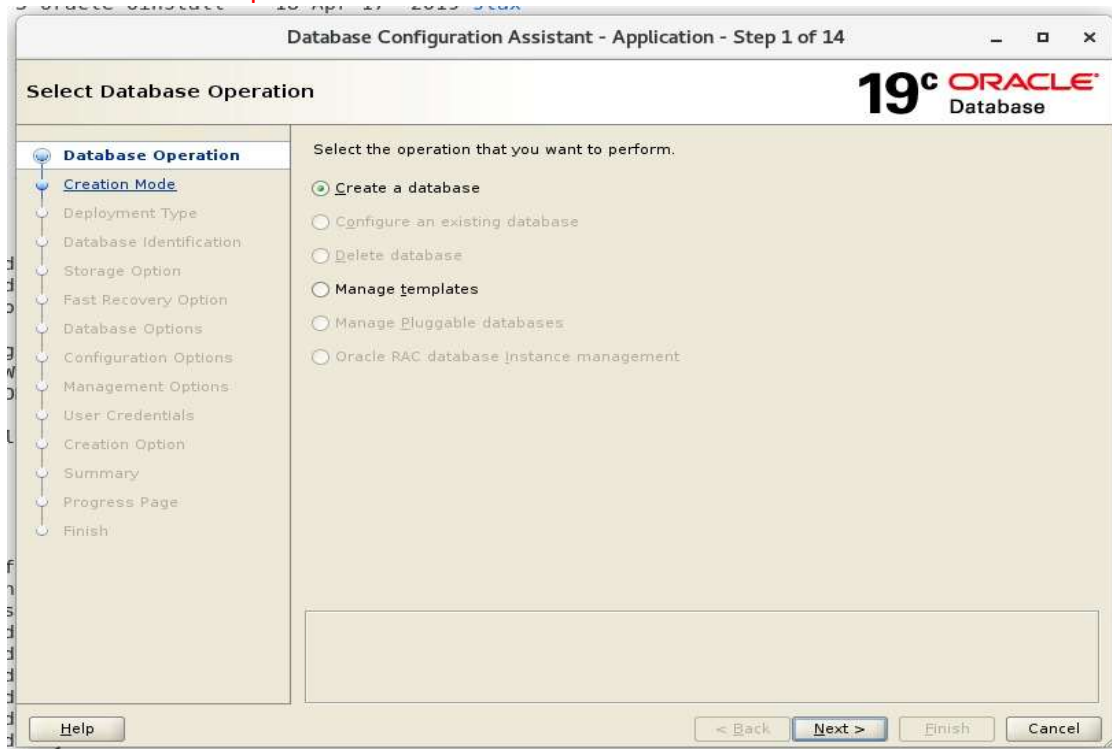


Database Creation

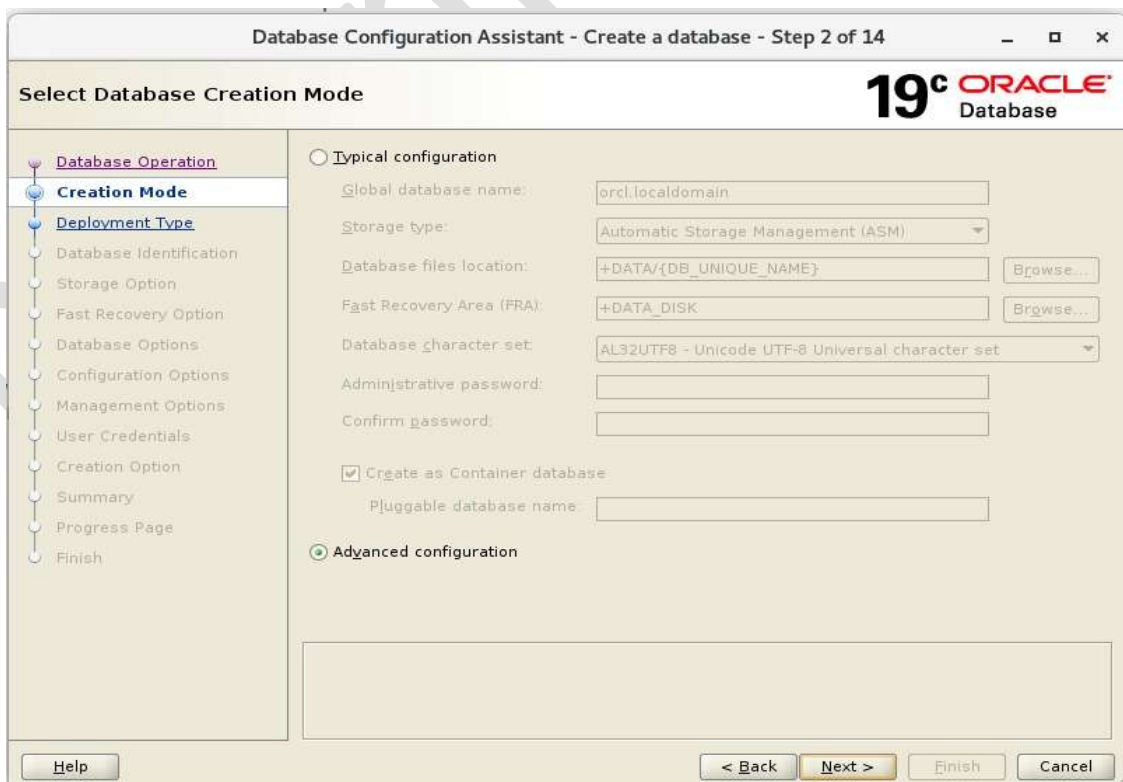
```
[oracle@dbwr1 ~]$ . db.env
```

```
[oracle@dbwr1 ~]$ dbca
```

Select the Create Database option and click the Next button.



Select the Advanced configuration option and click the Next button.



Select the General purpose and click the Next button.

Database Configuration Assistant - Create a database - Step 3 of 14

19^c ORACLE Database

Select Database Deployment Type

Select the type of database you want to create.

Database type: Oracle Real Application Cluster (RAC) datab...

Configuration type: Admin Managed

Select a template for your database.

Templates that include datafiles contain pre-created databases. They allow you to create a new database quickly. Use templates without datafiles only when necessary, such as when you need to change attributes like block size that cannot be altered after database creation.

	Template name	Include datafiles	Details
☐	Data Warehouse	Yes	View details
☐	Custom Database	No	View details
☒	General Purpose or Transaction Processing	Yes	View details

Template location: /u01/app/oracle/product/19c/db_1/assistants/dbca/templates [Change...](#)

Help < Back Next > Finish Cancel

Make sure both nodes are selected, then click the Next button.

Database Configuration Assistant - Create a database - Step 4 of 16

19^c ORACLE Database

Select List of Nodes

Select the nodes on which you want to create the cluster database. The local node "dbwr1" should always be selected.

	Node name
☒	1 dbwr1
☒	2 dbwr2

[Select all](#) [Deselect all](#)

Help < Back Next > Finish Cancel

Enter the Database Name and deselect CDB Database

Database Configuration Assistant - Create a database - Step 5 of 16

Specify Database Identification Details

19^c ORACLE Database

Database Operation
Creation Mode
Deployment Type
Nodes Selection
Database Identification
Storage Option
Fast Recovery Option
Database Options
Configuration Options
Management Options
User Credentials
Creation Option
Prerequisite Checks
Summary
Progress Page
Finish

Provide a unique database identifier information. An Oracle database is uniquely identified by a Global database name, typically of the form "name.domain".

Global database name: oradbwr

SID Prefix: oradbwr

Service name:

☐ Create as Container database

A Container database can be used for consolidating multiple databases into a single database, and it enables database virtualization. A Container database (CDB) can have zero or more pluggable databases (PDB).

☒ Use local Undo tablespace for PDBs

☐ Create an empty Container database

☒ Create a Container database with one or more PDBs

Number of PDBs: 1

PDB name: pdb

Help < Back Next > Finish Cancel

Default select as it is and Click next

Database Configuration Assistant - Create 'oradbwr' database - Step 6 of 16

Select Database Storage Option

19^c ORACLE Database

Database Operation
Creation Mode
Deployment Type
Nodes Selection
Database Identification
Storage Option
Fast Recovery Option
Database Options
Configuration Options
Management Options
User Credentials
Creation Option
Prerequisite Checks
Summary
Progress Page
Finish

☐ Use template file for database storage attributes

Storage type and location for database files will be picked up from the specified template (General Purpose or Transaction Processing).

☒ Use following for the database storage attributes

All the database files will be put at the specified location below. You can customize the name and location of each datafile in the subsequent screen.

Database files storage type: Automatic Storage Management (ASM)

Database files location: +DATA/{DB_UNIQUE_NAME} Browse...

Oracle Managed files option will enable Oracle to automatically generate the names of the datafiles for simplified database management.

☒ Use Oracle-Managed Files (OMF) Multiplex redo logs and control files...

File location variables...

Help < Back Next > Finish Cancel

Deselect FRA and Archivelog mode

Database Configuration Assistant - Create 'oradbwr' database - Step 7 of 16

19c ORACLE Database

Select Fast Recovery Option

Choose the recovery options for the database.

☐ Specify Fast Recovery Area

Recovery files storage type: Automatic Storage Management (ASM)

Fast Recovery Area: +DATA_DISK [Browse...](#)

Fast Recovery Area size: 8256 MB

☐ Enable archiving [Edit archive mode parameters...](#)

[Help](#) [Back](#) [Next >](#) [Finish](#) [Cancel](#)

Default as it is and Click next

Database Configuration Assistant - Create 'oradbwr' database - Step 8 of 16

19c ORACLE Database

Select Oracle Data Vault Config Option

☐ Configure Oracle Database Vault

Database Vault owner:

Password: Confirm password:

☐ Create a separate account manager

Account manager:

Password: Confirm password:

☐ Configure Oracle Label Security

☐ Configure Oracle Label Security with OLS

[Help](#) [Back](#) [Next >](#) [Finish](#) [Cancel](#)

Check ASMM Memory

Database Configuration Assistant - Create 'oradbwr' database - Step 9 of 16

Specify Configuration Options

19^c ORACLE Database

Database Operation
Creation Mode
Deployment Type
Nodes Selection
Database Identification
Storage Option
Fast Recovery Option
Data Vault Option
Configuration Options
Management Options
User Credentials
Creation Option
Prerequisite Checks
Summary
Progress Page
Finish

Memory | Sizing | Character sets | Connection mode | Sample schemas

☒ Use Automatic Shared Memory Management

SGA size: 2385 MB

PGA Size: 795 MB

☐ Use Manual Shared Memory Management

Shared pool size: 0 MB

Buffer cache size: 0 MB

Java pool size: 0 MB

Large pool size: 0 MB

PGA size: 0 MB

Total memory for database 0 MB

☐ Use Automatic Memory Management

Memory target: 3180 MB

< Back Next > Finish Cancel

Default as it is and Click next

Database Configuration Assistant - Create 'oradbwr' database - Step 10 of 16

Specify Management Options

19^c ORACLE Database

Database Operation
Creation Mode
Deployment Type
Nodes Selection
Database Identification
Storage Option
Fast Recovery Option
Data Vault Option
Configuration Options
Management Options
User Credentials
Creation Option
Prerequisite Checks
Summary
Progress Page
Finish

Specify the management options for the database.

☐ Run Cluster Verification Utility (CVU) checks periodically

☐ Configure Enterprise Manager (EM) database express

EM database express port: 5500

☐ Register with Enterprise Manager (EM) cloud control

OMS host:

OMS port:

EM admin username:

EM admin password:

ASMSNMP user password:

< Back Next > Finish Cancel

Enter oracle user Credential and click next

Database Configuration Assistant - Create 'oradbwr' database - Step 11 of 16

Specify Database User Credentials

You must specify passwords for the following user accounts in the new database for security reasons.

☐ Use different administrative passwords

Password Confirm password

SYS

SYSTEM

☒ Use the same administrative password for all accounts

Password: Confirm password:

< Back Next > Finish Cancel

Select Create Database and click finish

Database Configuration Assistant - Create 'oradbwr' database - Step 12 of 16

Select Database Creation Option

Select the database creation options.

☒ Create database

Specify the SQL scripts you want to run after the database is created. The scripts are run in the order listed below.

Post DB creation scripts: Browse...

☐ Save as a database template

Template name: dbca_template_2024-10-18_04-50-3

Template location: /u01/app/oracle/product/19c/db_1/assistants/dbca/templates Browse...

Description:

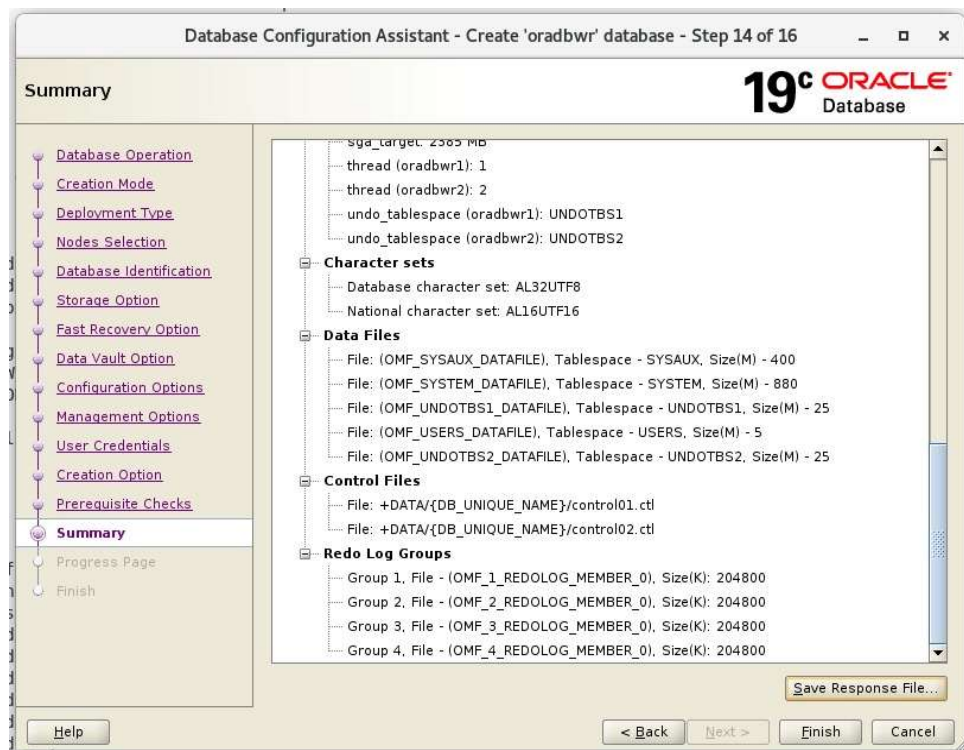
☐ Generate database creation scripts

Destination directory: {ORACLE_BASE}/admin/{DB_UNIQUE_NAME}/scripts Browse...

Following advanced configuration options can be used to configure initialization parameters and customize database storage locations.

All Initialization Parameters... Customize Storage Locations...

< Back Next > Finish Cancel



Oracle 19c Rac Database Creation is completed.

Post-Check For Rac Setup

Check the Status of the RAC

```
[oracle@dbwr1 ~]$
[oracle@dbwr1 ~]$ srvctl config database -d oradbwr
Database unique name: oradbwr
Database name: oradbwr
Oracle home: /u01/app/oracle/product/19c/db_1
Oracle user: oracle
Spfile: +DATA/ORADBWR/PARAMETERFILE/spfile.270.1182705515
Password file: +DATA/ORADBWR/PASSWORD/pwdoradbwr.258.1182704695
Domain:
Start options: open
Stop options: immediate
Database role: PRIMARY
Management policy: AUTOMATIC
Server pools:
Disk Groups: DATA
Mount point paths:
Services:
Type: RAC
Start concurrency:
Stop concurrency:
OSDBA group: oinstall
OSOPER group: oinstall
Database instances: oradbwr1,oradbwr2
Configured nodes: dbwr1,dbwr2
CSS critical: no
CPU count: 0
Memory target: 0
Maximum memory: 0
Default network number for database services:
Database is administrator managed
[oracle@dbwr1 ~]$
```


Database Level

```
oracle@dbwr1:~  
SQL*Plus: Release 19.0.0.0.0 - Production on Fri Oct 18 17:24:42 2024  
Version 19.3.0.0.0  
  
Copyright (c) 1982, 2019, Oracle. All rights reserved.  
  
Connected to:  
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0  
  
SQL> select INST_NUMBER, INST_NAME FROM v$active_instances;  
  
INST_NUMBER  
-----  
INST_NAME  
-----  
1  
dbwr1.localdomain:+ASM1  
2  
dbwr2.localdomain:+ASM2  
  
SQL> SELECT instance_name, host_name FROM gv$instance;  
  
INSTANCE_NAME  
-----  
HOST_NAME  
-----  
+ASM1  
dbwr1.localdomain  
+ASM2  
dbwr2.localdomain  
  
SQL>
```

```
oracle@dbwr1:~  
[oracle@dbwr1 ~]$ !sq  
sqlplus / as sysdba  
  
SQL*Plus: Release 19.0.0.0.0 - Production on Fri Oct 18 17:32:18 2024  
Version 19.3.0.0.0  
  
Copyright (c) 1982, 2019, Oracle. All rights reserved.  
  
Connected to:  
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0  
  
SQL> select name, open_mode from v$database;  
  
NAME          OPEN_MODE  
-----  
ORADBWR       READ WRITE  
  
SQL> select name, open_mode from gv$database;  
  
NAME          OPEN_MODE  
-----  
ORADBWR       READ WRITE  
ORADBWR       READ WRITE
```

Check the status of ASM using crsctl commands:

crsctl stat res -t

```
oracle@dbwr1:~  
[oracle@dbwr1 ~]$ . oraenv  
ORACLE_SID = [oradbwr] ? +ASM1  
ORACLE_HOME = [/home/oracle] ? /u01/app/oracle/19c/grid  
The Oracle base has been changed from /u01/app/oracle to /u01/app/oracle/19cgrid  
[oracle@dbwr1 ~]$  
[oracle@dbwr1 ~]$ crsctl stat res -t  
-----  
Name                Target  State        Server                State details  
-----  
Local Resources  
-----  
ora.LISTENER.lsnr  
      ONLINE  ONLINE      dbwr1                STABLE  
      ONLINE  ONLINE      dbwr2                STABLE  
ora.chad  
      ONLINE  ONLINE      dbwr1                STABLE  
      ONLINE  ONLINE      dbwr2                STABLE  
ora.net1.network  
      ONLINE  ONLINE      dbwr1                STABLE  
      ONLINE  ONLINE      dbwr2                STABLE  
ora.ons  
      ONLINE  ONLINE      dbwr1                STABLE  
      ONLINE  ONLINE      dbwr2                STABLE  
-----  
Cluster Resources  
-----  
ora.ASMNET1LSNR_ASM.lsnr(ora.asmgroup)  
      1      ONLINE  ONLINE      dbwr1                STABLE  
      2      ONLINE  ONLINE      dbwr2                STABLE  
      3      OFFLINE OFFLINE                        STABLE  
ora.DATA.dg(ora.asmgroup)  
      1      ONLINE  ONLINE      dbwr1                STABLE  
      2      ONLINE  ONLINE      dbwr2                STABLE  
      3      OFFLINE OFFLINE                        STABLE  
ora.LISTENER_SCAN1.lsnr  
      1      ONLINE  ONLINE      dbwr2                STABLE  
ora.LISTENER_SCAN2.lsnr  
      1      ONLINE  ONLINE      dbwr1                STABLE  
ora.LISTENER_SCAN3.lsnr  
      1      ONLINE  ONLINE      dbwr1                STABLE  
ora.asm(ora.asmgroup)  
      1      ONLINE  ONLINE      dbwr1                Started,STABLE  
      2      ONLINE  ONLINE      dbwr2                Started,STABLE  
      3      OFFLINE OFFLINE                        STABLE  
ora.asmnet1.asmnetwork(ora.asmgroup)  
      1      ONLINE  ONLINE      dbwr1                STABLE  
      2      ONLINE  ONLINE      dbwr2                STABLE  
      3      OFFLINE OFFLINE                        STABLE  
ora.cvu
```

Check the status of listener:

lsnctl status

```

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> show parameter name

NAME                                TYPE        VALUE
-----
db_unique_name                      string      +ASM
instance_name                      string      +ASM1
lock_name_space                    string
service_names                      string      +ASM
SQL> exit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
[oracle@dbwr1 ~]$ lsnrctl status

LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 16-OCT-2024 17:26:20

Copyright (c) 1991, 2019, Oracle. All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=IPC) (KEY=LISTENER)))
STATUS of the LISTENER
-----
Alias                                LISTENER
Version                            TNSLSNR for Linux: Version 19.0.0.0.0 - Production
Start Date                         16-OCT-2024 17:07:38
Uptime                             0 days 0 hr. 18 min. 42 sec
Trace Level                        off
Security                           ON: Local OS Authentication
SNMP                                OFF
Listener Parameter File            /u01/app/19c/grid/network/admin/listener.ora
Listener Log File                  /u01/app/oracle/19cgrid/diag/tnslsnr/dbwr1/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc) (KEY=LISTENER)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp) (HOST=192.168.24.1) (PORT=1521)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp) (HOST=192.168.24.31) (PORT=1521)))
Services Summary...
Service "+ASM" has 1 instance(s).
  Instance "+ASM1", status READY, has 1 handler(s) for this service...
Service "+ASM_DATA" has 1 instance(s).
  Instance "+ASM1", status READY, has 1 handler(s) for this service...
The command completed successfully
[oracle@dbwr1 ~]$

```

crsctl check crs

```

[oracle@dbwr2 ~]$ su - oracle
Last login: Wed Oct 16 17:20:43 IST 2024
[oracle@dbwr2 ~]$
[oracle@dbwr2 ~]$ . oraenv
ORACLE_SID = [oradbwr] ? +ASM2
ORACLE_HOME = [/home/oracle] ? /u01/app/19c/grid
The Oracle base has been changed from /u01/app/oracle to /u01/app/oracle/19cgrid
[oracle@dbwr2 ~]$
[oracle@dbwr2 ~]$
[oracle@dbwr2 ~]$
[oracle@dbwr2 ~]$
[oracle@dbwr2 ~]$ crsctl check crs
CRS-4638: Oracle High Availability Services is online
CRS-4537: Cluster Ready Services is online
CRS-4529: Cluster Synchronization Services is online
CRS-4533: Event Manager is online
[oracle@dbwr2 ~]$
[oracle@dbwr2 ~]$
[oracle@dbwr2 ~]$ crsctl stat res -t
-----
Name                                Target      State        Server                    State details
-----
Local Resources
-----
ora.LISTENER.lsnr
  ONLINE ONLINE dbwr1 STABLE
  ONLINE ONLINE dbwr2 STABLE
ora.chad
  ONLINE ONLINE dbwr1 STABLE
  ONLINE ONLINE dbwr2 STABLE
ora.net1.network
  ONLINE ONLINE dbwr1 STABLE
  ONLINE ONLINE dbwr2 STABLE
ora.ons
  ONLINE ONLINE dbwr1 STABLE
  ONLINE ONLINE dbwr2 STABLE
-----
Cluster Resources
-----
ora.ASMNET1LSNR_ASM.lsnr (ora.asmgroup)
  1 ONLINE ONLINE dbwr1 STABLE
  2 ONLINE ONLINE dbwr2 STABLE
  3 OFFLINE OFFLINE STABLE
ora.DATA.dg (ora.asmgroup)
  1 ONLINE ONLINE dbwr1 STABLE
  2 ONLINE ONLINE dbwr2 STABLE
  3 OFFLINE OFFLINE STABLE
ora.LISTENER_SCAN1.lsnr

```

crsctl check cluster -all

crsctl query crs activeversion

crsctl query crs releasepatch

crsctl query crs releaseversion

```
oracle@dbwr1:~  
[oracle@dbwr1 ~]$ crsctl check crs  
CRS-4638: Oracle High Availability Services is online  
CRS-4537: Cluster Ready Services is online  
CRS-4529: Cluster Synchronization Services is online  
CRS-4533: Event Manager is online  
[oracle@dbwr1 ~]$  
[oracle@dbwr1 ~]$ crsctl check cluster -all  
*****  
dbwr1:  
CRS-4537: Cluster Ready Services is online  
CRS-4529: Cluster Synchronization Services is online  
CRS-4533: Event Manager is online  
*****  
dbwr2:  
CRS-4537: Cluster Ready Services is online  
CRS-4529: Cluster Synchronization Services is online  
CRS-4533: Event Manager is online  
*****  
[oracle@dbwr1 ~]$  
[oracle@dbwr1 ~]$ ps -ef|grep smon  
root      8658      1  4 17:03 ?        00:01:16 /u01/app/19c/grid/bin/osysmond.bin  
oracle    9167      1  0 17:04 ?        00:00:00 asm_smon +ASML  
oracle    14274 26935  0 17:29 pts/17   00:00:00 grep --color=auto smon  
[oracle@dbwr1 ~]$  
[oracle@dbwr1 ~]$ crsctl query crs activeversion  
Oracle Clusterware active version on the cluster is [19.0.0.0.0]  
[oracle@dbwr1 ~]$  
[oracle@dbwr1 ~]$ crsctl query crs releasepatch  
Oracle Clusterware release patch level is [724960844] and the complete list of patches [29401763 29517242 29517247 29585399 ] have been applied on the local node. The release patch string i  
s [19.3.0.0.0].  
[oracle@dbwr1 ~]$ crsctl query crs releaseversion  
Oracle High Availability Services release version on the local node is [19.0.0.0.0]  
[oracle@dbwr1 ~]$
```

orachek

```
[oracle@dbwr1 ~]$  
[oracle@dbwr1 ~]$ ocrcheck  
Status of Oracle Cluster Registry is as follows :  
Version                :          4  
Total space (kbytes)    :       491684  
Used space (kbytes)     :       84192  
Available space (kbytes):       407492  
ID                      : 1920692064  
Device/File Name        :      +DATA  
                        Device/File integrity check succeeded  
  
                        Device/File not configured  
  
                        Device/File not configured  
  
                        Device/File not configured  
  
                        Device/File not configured  
  
Cluster registry integrity check succeeded  
  
Logical corruption check bypassed due to non-privileged user  
[oracle@dbwr1 ~]$
```